

KKS – GUIDELINES AND KEYS

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1. KKS-INTRODUCTION (IDENTIFICATION SYSTEM FOR POWER STATIONS)

1.1 Purpose and Area of Application

The power plant identification system is applied to clearly identify plants, systems, parts and components to their purpose, type and location.

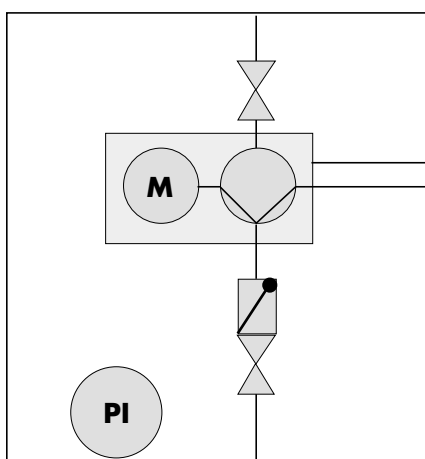
The content is in accordance with "KKS Identification Systems for Power Stations" issued by VGB PowerTech Service GmbH Essen, Germany.

1.2 Requirements

In order to perform the set tasks the identification system must be capable of satisfying the following requirements:

- Uniform identification for all types of power stations and any connected processes
- Sufficient capacity and detail for identification of all systems, components and structures
- Sufficient capacity for extension to accommodate new technologies
- Consistent identification for planning, licensing, construction, operation, maintenance and waste management
- Interdisciplinary applicability to mechanical engineering, civil engineering, electrical and instrument & control engineering combined with ability to identify according to process functions, points of installation and locations
- Consideration of national and international standards
- Non-language-based coding to ensure international usability
- Application in computer data processing.

Serial number of breakdown level	0	1	2	3
Name of breakdown level	Total plant	Function	Equipment unit	Component
Example	Unit 1	Feedwater system	Pump unit	Pump



Breakdown levels, referring to Process-Related Identification.

1.3 Structure and Application

The KKS consists of three types of identification:

- The **process-related code** identifies installations and equipment according to their assigned task in the power plant process
- The **point of installation code** identifies the points of installation within an installation unit (e.g. cubicles, consoles, panels)
- The **location code** identifies the rooms and floors, or other installation sites for installations and equipment in building structures.

A uniform identification structure, with a maximum of four breakdown levels, was created for all three types; the units referred to becoming smaller from left to right.

1.3.1 Identification and Type of Data Characters

Table 1 summarizes the identification and type of data characters of the breakdown levels for all three different types of identification appearing in the KKS.

1.3.2 Process-Related Identification

In this type of identification the entire system is subdivided according to the function or process, since, whether for mechanical, electrical, control or civil engineering, the equipment units and components must be identifiable in relation to the process.

The process-related identification is for many applications the most important identification, since it permits, for example, identification of electrical and control equipment, signals, and the identification in circuit

diagrams related to particular functions.

In the electrical and instrumentation & control (I&C) engineering sectors, the equipment for auxiliary services, power supply, open-loop-control, instrumentation, protection, etc., is treated as a process engineering function. The same applies to structures in civil engineering work.

The process-related identification corresponds to the identification block "Plant" in DIN 40719, part 2. This block has the prefix sign "=". According to the standard, the prefix sign can be omitted provided that the identification remains unambiguous.

Serial number of breakdown level	0	1	2	3
Name of break-down level	Total Plant	Function	Equipment Unit	Component
Designation of data character	G	F ₀ F ₁ F ₂ F ₃ F _n	A ₁ A ₂ A _N A ₃	B ₁ B ₂ B _N
Type of data character	(A) or (N)	(N) A A A NN	A A NNN (A)	A A NN

- A = Alphabetical symbols (letters, special symbols)
 N = Numerical symbols (digits)
 () = These data characters may be omitted

1.3.3 Point of Installation Identification

As with the process-related identification, the KKS is also used to identify locations, principally of electrical and I&C equipment, but also of mechanical equipment. Locations - for example, coordinates, racks and positions in cubicles etc. - are identified in the breakdown level **EQUIPMENT UNIT**.

The identification letters now used for the point of installation identification in the breakdown level **FUNCTION** may be the same as those for the

process-related identification. This improves recognition of the identification in the overall system. In order to prevent possible confusion between process-related identification and point of installation identification the prefix sign "+" must be added to the point of installation code (according to DIN 40719, part 2). Resp. the breakdown symbol "full stop" between breakdown Level 1 and 2 must be used. This prefix sign may be omitted only when there is absolutely no ambiguity – e.g. in layout documents.

1.3.4 Location Identification

In order to clearly identify the position of plants, sub-systems and equipment in the power station, the code of building structure and floor is entered at the breakdown level **FUNCTION** and the rooms on the various floors of the building structure at the breakdown level **EQUIPMENT UNIT**. The breakdown level **COMPONENT** is not used in location identification. Fire protection sections are identified according to the room identification.

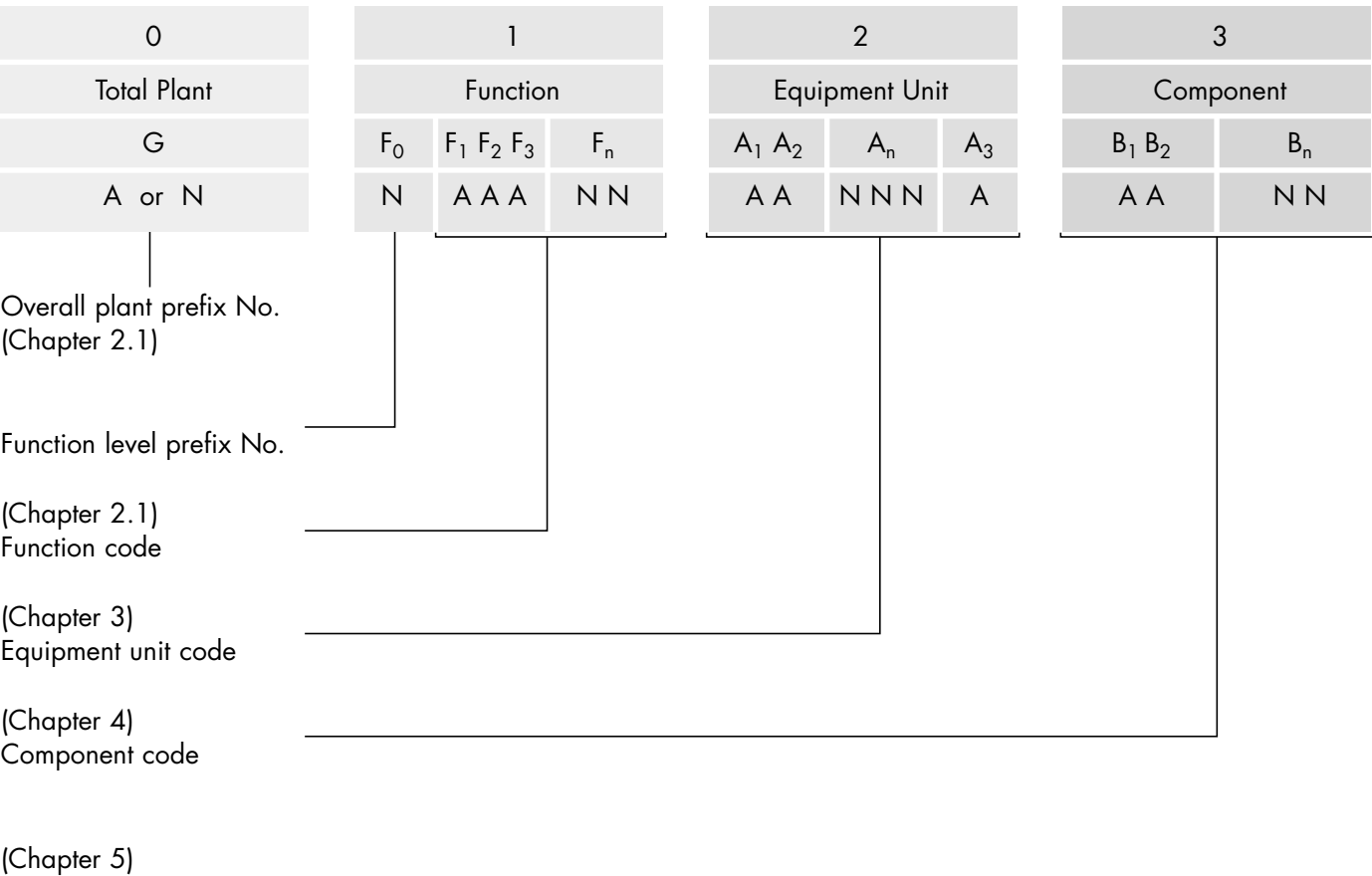
1.3.5 Title and Contents of the Breakdown Levels

Breakdown levels	0 Total plant	1 Function	2 Equipment unit	3 Component
Process-related code	Total plant	System code	Equipment unit code	Component code
Mechanical engineering	Unit	System	Pump unit	(Pump)
Civil engineering	Unit Unit	Structure, floor Structure, floor	Rolling door Fan unit	(Motor) (Fan)
Control and instrumentation (for mechanical and civil engineering)	Unit Unit Unit Unit	System Structure, floor System Unit coordinate level	Measuring circuit Measuring circuit Open-loop control Closed-loop control	Transducer Temp. sensor Push-button Controller
Electrical and control and instrumentation engineering	Unit Unit Unit Unit Unit Unit	Switchgear Transformer Transformer Transformer Elect. Equipment cab. Structure, floor	Switchgear assembly Fan unit Measuring circuit Terminal box Measuring circuit Junction box	Fuse Motor Indicator Terminal block Smoke detector Telephone jack
Point of installation	Total plant	Installation unit code	Installation space code	
Electrical and control & instrumentation engineering	Unit Unit Unit	Switchgear Elect. Equipment cab. Control console	Tier/Space Tier/Space Coordinates	
Location code	Total plant	Structure code	Room code	
Civil engineering	Unit Unit	Structure, floor Outdoor area	Room/coordinates Coordinates	

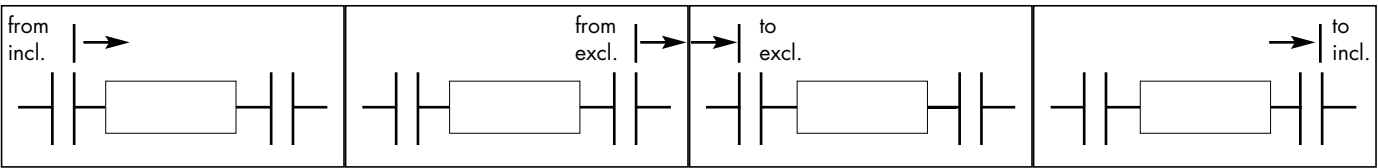
1.4 Structure and Contents of the Breakdown Levels

1.4.1 General

The KKS is divided into different **BREAKDOWN LEVELS** and codes from left to right in diminishing order of the units of a complete power plant:



Systems and sub-systems limit descriptions in the breakdown level Function are part of the function code list. Where they are not defined, the beginning of a new system is at branch-off or outlet of other system or at a point where unequivocal allocation of a component to a system is possible.

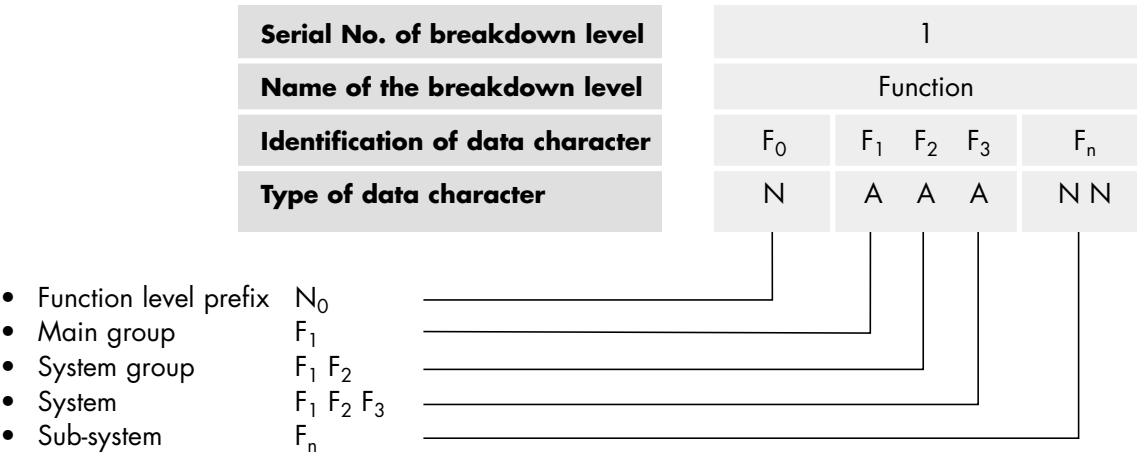


Legend for the limits:

- "from incl." means "including" the mentioned component
- "from excl." means "excluding" the mentioned component
- "to excl." means "excluding" the mentioned component
- "to incl." means "including" the mentioned component

1.4.2 Breakdown Level "Function"

Function identification:



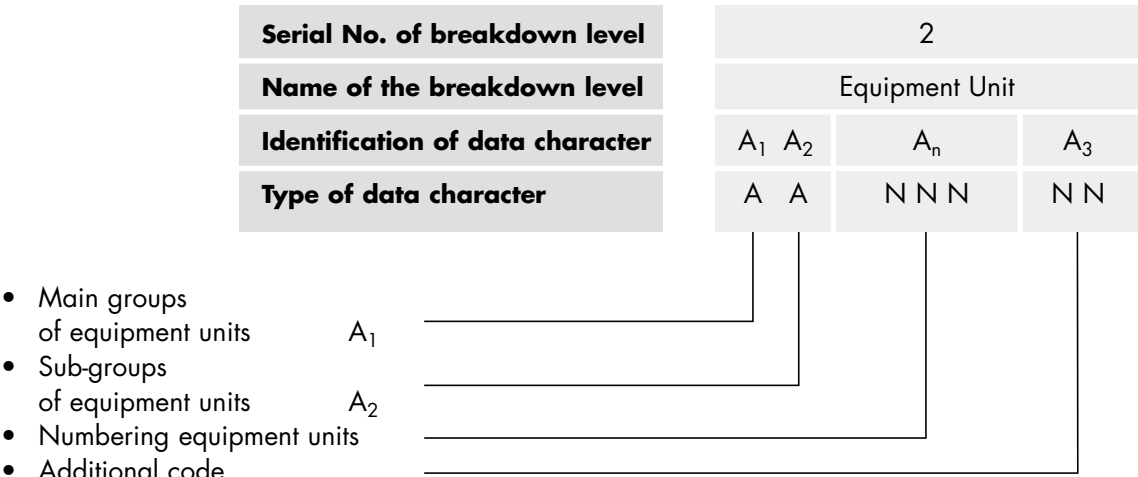
Function Key, main groups F₁: (Only the main groups for steam plants and combined cycle power plants are considered in this document)

Code	Identification
A	GRID AND DISTRIBUTION SYSTEMS
B	POWER TRANSMISSION AND AUXILIARY POWER SUPPLY
C	INSTRUMENTATION AND CONTROL EQUIPMENT
	(Identification on a priority basis according to main, instrumentation and control function also acceptable in composite structure hardware packaging systems.)
E	CONVENTIONAL FUEL SUPPLY AND RESIDUES DISPOSAL
G	WATER SUPPLY AND DISPOSAL
H	CONVENTIONAL HEAT GENERATION
L	STEAM, WATER, GAS CYCLES
M	MAIN MACHINE SETS
P	COOLING WATER SYSTEMS
Q	AUXILIARY SYSTEMS
S	ANCILLARY SYSTEMS
T	- blocked -
U	STRUCTURES
Z	WORKSHOP AND OFFICE EQUIPMENT

For complete function code list including system limits of function groups F₁F₂ and function subgroups F₁F₂F₃ see Function Key (chapter 3).

1.4.3 Breakdown Level "Equipment Unit"

Equipment unit identification:



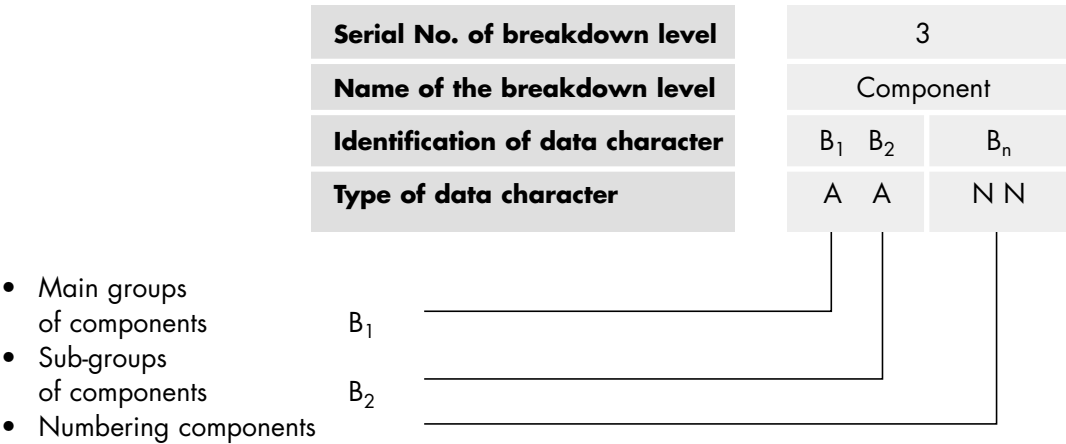
Equipment Unit Key, groups A₁:

Code	Identification
A	MECHANICAL EQUIPMENT
B	MECHANICAL EQUIPMENT
C	DIRECT MEASURING CIRCUITS
D	CLOSED-LOOP-CONTROL CIRCUITS
E	ANALOG AND BINARY SIGNAL CONDITIONING
F	INDIRECT MEASURING CIRCUITS
G	ELECTRICAL EQUIPMENT
H	SUB-ASSEMBLIES OF MAIN AND HEAVY MACHINERY

For complete equipment units code of the sub-groups A₁ A₂ see Equipment Unit Key (chapter 4)

1.4.4 Breakdown Level "Component"

Component identification:



Component code main groups B₁:

Code	Identification
K	MECHANICAL COMPONENTS
M	MECHANICAL COMPONENTS
Q	INSTRUMENTATION AND CONTROL COMPONENTS (Non electrical)
-	ELECTRICAL COMPONENTS
X	SIGNAL ORIGIN
Y	SIGNAL APPLICATION
Z	GATED SIGNALS

For complete component code of the sub-groups B₁ B₂ see Component Key (chapter 5). In P&IDs breakdown level 3 is NOT used. In other documents it may be used according to separate agreements.

1.4.5 Overview

Serial No. of breakdown level	0	1	2	3
Name of breakdown level	Total Plant	Function	Equipment Unit	Component
Identification of data character	G	F ₀ F ₁ F ₂ F ₃ F _n	A ₁ A ₂ A _n A ₃	B ₁ B ₂ B _n
Type of data character =	A or N	(N) AAA NN	AA NNN (A)	AA NN

- Prefix sign of process related identification - Identification of power station units and non-unit specific plants - Prefix No. of the function code Numbering of similar systems and plants in the parts of power station identified on breakdown level 0 - System-classifying Classifying sub-division of systems and plants of a unit as per Function key - System-numbering Numbering sub-division of systems and plants into sub-systems and sections of plants - Equipment-classifying Classifying sub-division of mechanical, electrical and I&C equipment as per Equipment unit key - Equipment-numbering Numbering of similar mechanical, electrical and I&C equipment - Additional identification of breakdown level 2 Identification of pilot valves, multiple drives/el. loads, measuring points with multiple output/operating points - Component-classifying Classifying sub-division of components, signals or signal-applications - Component-numbering Numbering of components, signals or signal-applications									
--	--	--	--	--	--	--	--	--	--

The data character marked () and the prefix sign can be omitted if the identification remains unequivocal.

1.5 KKS Relevant Symbols in P&ID's and Principle Sketches

In P&IDs the following symbols to indicate KKS code and limit functions and piping sections are recommended:



The symbol "pin with empty head" is used to identify the limits of functions and sub-functions.

(G F₀ F₁ F₂ F₃ F_N)



The symbol "pin with full head" is used to identify the limits of piping sections.

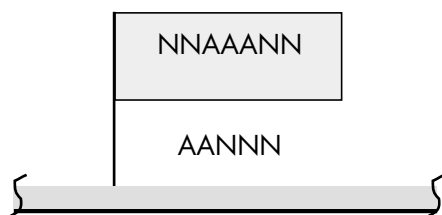
(A₁ A₂ A_N)



Reducers are assigned to the piping with the larger diameter.

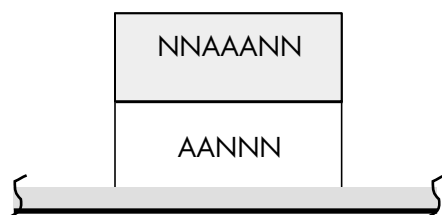
They need not be shown on flow diagrams.

(see KKS Application Commentaries Part B1, page 52)



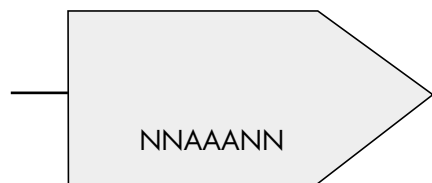
The symbol "flag" is used for the identification of pipings.

The direction of the flag shows the direction of the medium flow.



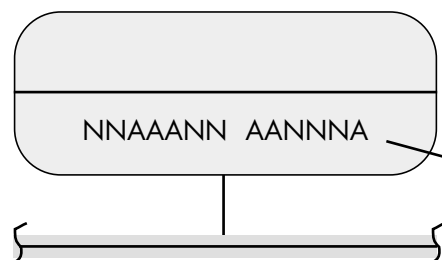
The extended "flag" symbol is used for the identification of a pipe

where medium flow direction alters according to the operating condition.



The symbol pointer between connected systems represented on different P&IDs.

Breakdown level Function of the connected system as a minimum information.



The symbol "I & C - circuits" (DIN 19227, sheet 1)

→ is used to represent measuring circuits.

KKS-code for measuring circuits.

2. PROJECT RELATED RULES, AGREEMENTS AND PROCEDURES

2.1 Agreement on Break-down Level "G" – Overall Plant and F_0 – Function Prefix Number

It may be necessary to identify units, non-unit specific plants or extension stages within a power station, such that a clear and unambiguous distinction exists between them. This is provided by breakdown level 0 – Overall plant. This identification must be agreed upon by all parties concerned, regarding the contents and and type of data character (A or N). Especially the identification of existing units has to be taken into account. The breakdown level "Overall plant" can be omitted when the designation remains unambiguous.

If in the overall plant two or more identical plants or systems are to be identified (e.g. 2 boiler plants, 2 gas turbines with attached el. generators and HRSGs, 2 structures with equal purpose) the prefix number F_0 is used to distinguish between them. Each F_0 number is valid only in the area of the function concerned. On no accounts it to be considered for the counting of system sections as parallel lines, data characters F_N being used for this purpose.

2.2 General Rules on Numbering Code Elements (F_N , A_N , B_N)

- Numbering starts anew when one of the preceding code elements changes.
- Numbering may be

consecutive or grouping.

- Numbering need not be continuous.
- Numbering conventions, once established, may not be altered, not even in the event of changes made in the progress of planning.
- Redundant zeros must be written.
- An application-specific scheme of numbering may be established. However, it has to be valid throughout all project disciplines in order of an unequivocal understanding of the code.
- Management systems (e.g. computer checking programs, allocation sheets) have to assure, that no **double KKS numbers** occur in the project and that no multiple KKS numbers are assigned to one item.
- If necessary, subdivisions of number ranges in F_N according to the division of work could be made. They have to be written down directly in the project-related Function key and must be respected by the partners involved.

2.3 General Rules Regarding Direction of Numbering

- Standard identifications have priority **in all cases**.
- Numbers are increasing in **media** or process **flow** direction.
- Project counting directives referred to local layouts are **related** to breakdown level "G", prefix number

for **system code "F₀"** and structures ($F_1 = U$) **only**.

- The counting is **not** related to **any geographical direction** (e.g. **not** from west to east, **not** from north to south)

2.4 Rules on Modifications and Cancellations of KKS Numbers

- It is **not** allowed to change allocated KKS numbers for a project, if these numbers have been **released** for the project.
- The original KKS number is still kept, even when another kind of equipment is used, provided that the function according to the KKS equipment unit code is not changed (e.g. substitution of a gate valve by a globe valve).

2.5 Writing Modus of KKS Code

Following structures of the KKS code are allowed:

- in one line with space between Function Level, Equipment Unit Level (and Component Level).

e.g.

19PAB10 AP001 (-M01)

- in two (three) lines

e.g. **19PAB10**

AP001

(-M01)

2.6 General Rules on Process-related Identification

- Piping & Instrument Diagrams (P&ID) are the basic

- documents (origins) for a process-related KKS coding.
- Single line diagrams are the basic documents (origins) for a process-related KKS coding in electrical engineering. However, KKS codes for electrical consumers, measuring loops and items of I & C represented in Function Diagrams are originated in P&IDs.
- For other engineering activities (e.g. component engineering), KKS code must always be taken from the above mentioned basic documents.
- In basic KKS documents (P&IDs, single line diagrams and site layouts) the break down level 3 (component code) is **never used**.
- The detailed KKS coding in process-related identification has to be made according to the KKS-Application Commentaries, Part B1, B3 and B4 by VGB PowerTech GmbH.

2.7 Rules on A_N Numbering for Piping and In-Pipe-Components

Serial No.	Application for
NNN	
001-199	Piping and in-pipe-components incl. instruments for main media/systems*)
201-299	Piping and in-pipe-components incl. instruments in discharge, drains and venting
301-399	1st isolation valves in instrument lines
401-499	Piping and isolation valves between technological and sampling & dosing systems
501-599	Piping and in-pipe-components incl. instruments for testing functions
601-699	Piping and in-pipe-components incl. instruments for heating and maintaining heat
701-799	Piping and in-pipe-components incl. instruments for special purposes:
701-719	– N ₂ purging / conservation connections
720-749	– Water / Steam purging (e.g. temporary blow-out)
750-779	– Compensators
780-789	– Hydro tests
790-799	– Free for project-specific allocations; documented agreement necessary
801-899	Piping and in-pipe-components incl. instruments for special purposes:
801-819	– Hydraulic control piping and in-pipe-components
820-829	– Temporary flushing piping
830-899	– Free for project-specific allocations; documented agreement necessary
901-999	Indirect measuring circuits (according to Application Commentaries B4, Chpt. 2.1.2.2)

*) Main media / system = in F₁, F₂, F₃ (F_N) identified system with its media; e.g. LAB(95) = High pressure feedwater piping system

Note: Special rules on point of installation identification and location identification have to be established within the relevant engineering disciplines (electrical, I&C and civil engineering) according to the KKS-Application Commentaries, Part B2, B3 and B4 by VGB PowerTech GmbH.

The KKS basic documents for these types of KKS code have to be identified as well.

3. KKS FUNCTION KEY (F₁ F₂ F₃)

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A

GRID AND DISTRIBUTION SYSTEMS

AB > 420 kV systems.

AC 380 (420) kV systems.

AD 220 - 359 kV systems.

AE 110 - 219 kV systems.

AF 60 - 109 kV systems.

AG 45 - 59 kV systems.

AH 30 - 44 kV systems.

AJ 20 - 29 kV systems.

AK 10 - 19 kV systems.

AL 6 - 9 kV systems.

AM 1 - 5 kV systems.

AN < 1 kV systems.

AP Control consoles.

AQ Measuring and metering equipment.

AR Protection equipment.

AS Decentralized panels and cabinets.

AT Transformer equipment.

AU Open-loop control, checkback and auxiliary equipment.

AV Marshalling racks.

AW Instrument panels.

AX Central equipment.

AY Communication equipment.



GRID AND DISTRIBUTION SYSTEMS

AB > 420 kV systems.

- ABA >420 kV systems.**
(diameter / feeder subdivision).
- ABB >420 kV systems.**
(diameter / feeder subdivision).
- ABC >420 kV systems.**
(diameter / feeder subdivision).
- ABD >420 kV systems.**
(free for use).
- ABM >420 kV systems.**
(busbar section).
- ABN >420 kV systems.**
(busbar section).
- ABZ >420 kV systems.**
(Line / transformer feeder).

AC 380 (420) kV systems.

- ACA 360 - 420 kV systems.**
(diameter / feeder subdivision).
- ACB 360 - 420 kV systems.**
(diameter / feeder subdivision).
- ACC 360 - 420 kV systems.**
(diameter / feeder subdivision).
- ACD 360 - 420 kV systems.**
(free for use).
- ACM 360 - 420 kV systems.**
(busbar section).
- ACN 360 - 420 kV systems.**
(busbar section).
- ACZ 360 - 420 kV systems.**
(Line / transformer feeder).

AD 220 - 359 kV systems.

- ADA 220 - 359 kV systems.**
(diameter / feeder subdivision).
- ADB 220 - 359 kV systems.**
(diameter / feeder subdivision).
- ADC 220 - 359 kV systems.**
(diameter / feeder subdivision).
- ADD 220 - 359 kV systems.**
(free for use).
- ADM 220 - 359 kV systems.**
(busbar section).
- ADN 220 - 359 kV systems.**
(busbar section).
- ADZ 220 - 359 kV systems.**
(Line / transformer feeder).

AE 110 - 219 kV systems.

- AEA 110 - 219 kV systems.**
(diameter / feeder subdivision).

- AEB 110 - 219 kV systems.**
(diameter / feeder subdivision).

- AEC 110 - 219 kV systems.**
(diameter / feeder subdivision).

- AED 110 - 219 kV systems.**
(free for use).

- AEM 110 - 219 kV systems.**
(busbar section).

- AEN 110 - 219 kV systems.**
(busbar section).

- AEZ 110 - 219 kV systems.**
(Line / transformer feeder).

AF 60 - 109 kV systems.

- AFA 60 - 109 kV systems.**
(diameter / feeder subdivision).

- AFB 60 - 109 kV systems.**
(diameter / feeder subdivision).

- AFC 60 - 109 kV systems.**
(diameter / feeder subdivision).

- AFD 60 - 109 kV systems.**
(free for use).

- AFM 60 - 109 kV systems.**
(busbar section).

- AFN 60 - 109 kV systems.**
(busbar section).

- AFZ 60 - 109 kV systems.**
(Line / transformer feeder).

AG 45 - 59 kV systems.

- AGA 45 - 59 kV systems.**
(diameter / feeder subdivision).

- AGB 45 - 59 kV systems.**
(diameter / feeder subdivision).

- AGC 45 - 59 kV systems.**
(diameter / feeder subdivision).

- AGD 45 - 59 kV systems.**
(free for use).

- AGM 45 - 59 kV systems.**
(busbar section).

- AGN 45 - 59 kV systems.**
(busbar section).

- AGZ 45 - 59 kV systems.**
(Line / transformer feeder).

AH 30 - 44 kV systems.

- AHA 30 - 44 kV systems.**
(diameter / feeder subdivision).

- AHB 30 - 44 kV systems.**
(diameter / feeder subdivision).

- AHC 30 - 44 kV systems.**
(diameter / feeder subdivision).

GRID AND DISTRIBUTION SYSTEMS

AHD 30 - 44 kV systems.
(free for use).

AHM 30 - 44 kV systems.
(busbar section).

AHN 30 - 44 kV systems.
(busbar section).

AHZ 30 - 44 kV systems.
(Line / transformer feeder).

AJ 20 - 29 kV systems.

AJA 20 - 29 kV systems.
(diameter / feeder subdivision).

AJB 20 - 29 kV systems.
(diameter / feeder subdivision).

AJC 20 - 29 kV systems.
(diameter / feeder subdivision).

AJD 20 - 29 kV systems.
(free for use).

AJM 20 - 29 kV systems.
(busbar section).

AJN 20 - 29 kV systems.
(busbar section).

AJZ 20 - 29 kV systems.
(Line / transformer feeder).

AK 10 - 19 kV systems.

AKA 10 - 19 kV systems.
(diameter / feeder subdivision).

AKB 10 - 19 kV systems.
(diameter / feeder subdivision).

AKC 10 - 19 kV systems.
(diameter / feeder subdivision).

AKD 10 - 19 kV systems.
(free for use).

AKM 10 - 19 kV systems.
(busbar section).

AKN 10 - 19 kV systems.
(busbar section).

AKZ 10 - 19 kV systems.
(Line / transformer feeder).

AL 6 - 9 kV systems.

ALA 6 - 9 kV systems.
(diameter / feeder subdivision).

ALB 6 - 9 kV systems.
(diameter / feeder subdivision).

ALC 6 - 9 kV systems.
(diameter / feeder subdivision).

ALD 6 - 9 kV systems.
(free for use).

ALM 6 - 9 kV systems.
(busbar section).

ALN 6 - 9 kV systems.
(busbar section).

ALZ 6 - 9 kV systems.
(Line / transformer feeder).

AM 1 - 5 kV systems.

AMA 1 - 5 kV systems.
(diameter / feeder subdivision).

AMB 1 - 5 kV systems.
(diameter / feeder subdivision).

AMC 1 - 5 kV systems.
(diameter / feeder subdivision).

AMD 1 - 5 kV systems.
(free for use).

AMM 1 - 5 kV systems.
(busbar section).

AMN 1 - 5 kV systems.
(busbar section).

AMZ 1 - 5 kV systems.
(Line / transformer feeder).

AN < 1 kV systems.

**ANA Low voltage switchgear 500-1000 V,
Three-phase/single phase alter-
nating current.**
(free for use).

**ANB Low voltage switchgear 500-1000 V,
Three-phase/single phase alter-
nating current.**
(free for use).

**ANC Low voltage switchgear 500-1000 V,
Three-phase/single phase alter-
nating current.**
(free for use).

AND < 1 kV systems.
(free for use).

**ANE Low voltage switchgear < 500 V,
Three-phase/single phase alter-
nating current.**
(free for use).

**ANF Low voltage switchgear < 500 V,
Three-phase/single phase alter-
nating current.**
(free for use).

**ANG Low voltage switchgear < 500 V,
Three-phase/single phase alter-
nating current.**
(free for use).

**ANH Low voltage switchgear < 500 V,
Three-phase/single phase alter-
nating current.**
(free for use).

GRID AND DISTRIBUTION SYSTEMS

ANK Direct voltage switchgear 220/110 V.
(free for use).

ANL Direct current switchgear 220/125 V.
(free for use).

ANM Direct current switchgear 220/125 V.
(free for use).

ANN Direct current switchgear 220/125 V.
(free for use).

ANQ Direct current switchgear 60/48 V.
(free for use).

ANR Direct current switchgear 60/48 V.
(free for use).

ANS Direct current switchgear 60/48 V.
(free for use).

ANU Direct current switchgear 24/12 V.
(free for use).

ANV Direct current switchgear 24/12 V.
(free for use).

ANW Direct current switchgear 24/12 V.
(free for use).

ANZ < 1 kV systems.
(line).

AP Control consoles.

APA Control consoles.
(free for use).

APB Control consoles.
(free for use).

APC Control consoles.
(free for use).

APD Control consoles.
(free for use).

AQ Measuring and metering equipment.

AQA Measuring and metering equipment.
(free for use).

AQB Measuring and metering equipment.
(free for use).

AQC Measuring and metering equipment.
(free for use).

AQD Measuring and metering equipment.
(free for use).

AQZ Measuring and metering equipment.
(line).

AR Protection equipment.

ARA Protection equipment.
(free for use).

ARB Protection equipment.
(free for use).

ARC Protection equipment.
(free for use).

ARD Protection equipment.
(free for use).

ARZ Protection equipment
(line).

AS Decentralized panels and cabinets.

ASA Circuit breaker appurtenances.

ASB Multiplication, conversion,
decoupling.

ASC Transducer appurtenances.

ASD Compressed air, hydraulics.

ASJ Automated controls, closed-loop
control.

ASL Grid simulation, voltage group
selection.

ASM Measuring equipment.

ASN Auxiliary power supply.

ASP Recording.

ASQ Metering.

ASR Protection.

ASS Synchronization.

AST Transformation.

ASU Panels and cabinets for auxiliary
equipment.

ASV Group, intermediate and general
terminal blocks.

ASW Indication, manual operation,
monitoring.

ASX Alarm annunciation.

AT Transformer equipment.

ATA Transformer equipment.
(free for use).

ATB Transformer equipment.
(free for use).

ATC Transformer equipment.
(free for use).

ATD Transformer equipment.
(free for use).

ATZ Transformer equipment.
(free for use).

AU Open-loop control, checkback and auxiliary equipment.

AUA Open-loop control, checkback and
auxiliary equipment.
(free for use).

GRID AND DISTRIBUTION SYSTEMS

AUB Open-loop control, checkback and auxiliary equipment.

(free for use).

AUC Open-loop control, checkback and auxiliary equipment.

(free for use).

AUD Open-loop control, checkback and auxiliary equipment.

(free for use).

AUE Open-loop control, checkback and auxiliary equipment.

(free for use).

AUF Open-loop control, checkback and auxiliary equipment.

(free for use).

AUG Open-loop control, checkback and auxiliary equipment.

(free for use).

AUH Open-loop control, checkback and auxiliary equipment.

(free for use).

AV Marshalling racks.

AVA Marshalling racks.

(free for use).

AVB Marshalling racks.

(free for use).

AVC Marshalling racks.

(free for use).

AVD Marshalling racks.

(free for use).

AVE Marshalling racks.

(free for use).

AVF Marshalling racks.

(free for use).

AVG Marshalling racks.

(free for use).

AVH Marshalling racks.

(free for use).

AW Instrument panels.

AWA Instrument panels.

(free for use).

AWB Instrument panels.

(free for use).

AWC Instrument panels.

(free for use).

AWD Instrument panels.

(free for use).

AX Central equipment.

AXA Battery chargers.

AXB Battery plant.

AXC Uninterruptible power supply system.

(UPS).

AXD Low voltage distribution.

AXE Compressed air generation system.

AXF Central equipment.

(free for use).

AXG Central equipment.

(free for use).

AXH Central equipment.

(free for use).

AXJ Central equipment.

(free for use).

AXK Central equipment.

(free for use).

AXL Central equipment.

(free for use).

AXM Central equipment.

(free for use).

AXN Central equipment.

(free for use).

AXP Central equipment.

(free for use).

AXR Central equipment.

(free for use).

AXS Central equipment.

(free for use).

AXT Auxiliary power transformer.

AXU Central equipment.

(free for use).

AXV Central equipment.

(free for use).

AXW Central equipment.

(free for use).

AXX Central equipment.

(free for use).

AXY Central equipment.

(free for use).

AXZ Central equipment.

(free for use).

AY Communication equipment.

AYA Telephone system

(PABX).

AYB Control console telephone system.

AYC Alarm system

(acoustic).

AYD Alarm system

(optical).

AYE Fire alarm system.

AYF Clock system.

AYG Remote control system.

AYH Telemetry system.

GRID AND DISTRIBUTION SYSTEMS

AYJ	Remote metering system.
AYK	HF carrier telephone system.
AYL	Staff paging system, wireless.
AYM	Staff paging system, inductive.
AYN	Staff paging system, hardwired.
AYP	Optical monitoring system.
AYQ	General gas alarm system. (if not assigned to specific system).
AYS	Radiotelephone system.



B

POWER TRANSMISSION AND AUXILIARY POWER SUPPLY

- BA** Power transmission.
- BB** Medium-voltage distribution boards and transformers, normal system.
- BC** Medium voltage distribution boards and transformers, general-purpose.
- BD** Medium voltage emergency distribution boards and transformers, (diesel) emergency power system.
- BF** Low voltage main distribution boards and transformers, normal system.
- BH** Low voltage main distribution boards and transformers, general-purpose.
- BJ** Low voltage subdistribution boards and transformers, normal system.
- BL** Low voltage subdistribution boards and transformers, general-purpose.
- BM** Low voltage distribution boards and transformers, (diesel) emergency power system 1.
- BP** Power installations for large variable-speed drives (e.g. for feedwater pump, excitation equipment, GT starting equipment, not power adjusters in switchgear).
- BR** Low voltage distribution boards, converter, uninterruptible power system (UPS) and stand-by diesel power system.
- BT** Battery and charger systems.
- BU** Direct current distribution boards.
- BV** Direct current distribution boards.
- BY** Control and protection equipment.



POWER TRANSMISSION AND AUXILIARY POWER SUPPLY

BA Power transmission.

BAA Generator leads.

from excl. generator bushings, incl. current and voltage transformers, cooling and ventilation systems to excl. generator transformer low side bushings or to excl. auxiliary power transformer high side bushings.

BAB Foundation cabinets.

BAC Generator circuit breaker, also commutating pole circuit breaker, incl. cooling system.

BAT Generator transformers, including cooling system.

BAU Earthing and lightning protection systems.

BAX Fluid supply system for control and protection equipment.

BAY Control and protection equipment.

BB Medium-voltage distribution boards and transformers, normal system.

BBA Medium voltage distribution boards, normal system. (free for use).

BBB Medium voltage distribution boards, normal system. (free for use).

BBC Medium voltage distribution boards, normal system. (free for use).

BBD Medium voltage distribution boards, normal system. (free for use).

BBE Medium voltage distribution boards, normal system. (free for use).

BBF Medium voltage distribution boards, normal system. (free for use).

BBG Medium voltage distribution boards, normal system. (free for use).

BBH Medium voltage distribution boards, normal system. (free for use).

BBT Medium voltage auxiliary power transformers.

BBX Fluid supply system for control and protection equipment.

BBY Control and protection equipment.

BC Medium voltage distribution boards and transformers, general-purpose.

BCA Medium voltage distribution boards, general-purpose. (free for use).

BCB Medium voltage distribution boards, general-purpose. (free for use).

BCC Medium voltage distribution boards, general-purpose. (free for use).

BCD Medium voltage distribution boards, general-purpose. (free for use).

BCE Medium voltage distribution boards, general-purpose. (free for use).

BCF Medium voltage distribution boards, general-purpose. (free for use).

BCG Medium voltage distribution boards, general-purpose. (free for use).

BCH Medium voltage distribution boards, general-purpose. (free for use).

BCT Start-up, offsite, general-purpose transformers.

BCX Fluid supply system for control and protection equipment.

BCY Control and protection equipment.

BD Medium voltage emergency distribution boards and transformers, (diesel) emergency power system.

BDA Medium voltage distribution boards, blackstart-purpose. (free for use).

BDB Medium voltage distribution boards, blackstart-purpose. (free for use).

BDC Medium voltage distribution boards, blackstart-purpose. (free for use).

BDD Medium voltage distribution boards, blackstart-purpose. (free for use).

BDT Start-up, off site, blackstart-purpose transformer.

BDX Fluid supply system for control and protection equipment.

BDY Control and protection equipment BDZ -blocked-.

POWER TRANSMISSION AND AUXILIARY POWER SUPPLY

BF Low voltage main distribution boards and transformers, normal system.

BFA Low voltage main distribution boards, normal system.
(free for use).

BFB Low voltage main distribution boards, normal system.
(free for use).

BFC Low voltage main distribution boards, normal system (ACC).

BFD Low voltage main distribution boards, normal system.
(free for use).

BFE Low voltage main distribution boards, normal system.
(free for use).

BFF Low voltage main distribution boards, normal system (option e.g. for heating & freezing protection).

BFG Low voltage main distribution boards, normal system.
(free for use).

BFH Low voltage main distribution boards, normal system.
(free for use).

BFT Low voltage auxiliary power transformer.

BH Low voltage main distribution boards and transformers, general-purpose.

BHA Low voltage main distribution boards, general purpose.
(free for use).

BHB Low voltage main distribution boards, general purpose.
(free for use).

BHC Low voltage main distribution boards, general purpose.
(free for use).

BHD Low voltage main distribution boards, general purpose.
(free for use).

BHE Low voltage main distribution boards, general purpose.
(free for use).

BHF Low voltage main distribution boards, general purpose.
(free for use).

BHG Low voltage main distribution boards, general purpose.
(free for use).

BHH Low voltage main distribution boards, general purpose.
(free for use).

BHR Low voltage auxiliary distribution boards, general-purpose.
Lighting, ventilation, air conditioning.

BHT Low voltage auxiliary power transformers.
(free for use according to voltage level).

BHU Low voltage auxiliary power transformers.
(free for use according to voltage level).

BHV Low voltage auxiliary power transformers.
(free for use according to voltage level).

BHW Low voltage auxiliary power transformers.
(free for use according to voltage level).

BHX Fluid supply system for control and protection equipment.

BHY Control and protection equipment.

BJ Low voltage subdistribution boards and transformers, normal system.

BJA Low voltage subdistribution boards and transformers, normal system.
(free for use).

BJB Low voltage subdistribution boards and transformers, normal system.
(free for use).

BJC Low voltage subdistribution boards, normal system. Water treatment.

BJD Low voltage subdistribution boards, normal system.

BJH Low voltage subdistribution boards, normal system.
(free for use).

BJT Low voltage auxiliary power transformers.

BL Low voltage subdistribution boards and transformers, general-purpose.

BLA Low voltage subdistribution boards, general-purpose.
(free for use).

BLB Low voltage subdistribution boards, general-purpose.
(free for use).

BLD Low voltage subdistribution boards, general-purpose.
(free for use).

POWER TRANSMISSION AND AUXILIARY POWER SUPPLY

BLE	Low voltage subdistribution boards, general-purpose. (free for use).	BMW	Low voltage auxiliary power transformers. (free for use according to voltage level).
BLF	Low voltage subdistribution boards, general-purpose. (free for use).	BMX	Fluid supply system for control and protection equipment.
BLG	Low voltage subdistribution boards, general-purpose. (free for use).	BMY	Control and protection equipment.
BLH	Low voltage subdistribution boards, general-purpose. (free for use).	BP	Power installations for large variable-speed drives (e.g. for feedwater pump, excitation equipment, gt starting equipment, not power adjusters in switchgear). (e.g. for feedwater pump, excitation equipment, not power adjusters in switchgear).
BLT	Low voltage auxiliary power transformer.	BPA	Power installations for large variable-speed drives. (free for use).
BLX	Fluid supply system for control and protection equipment.	BPC	Power installations for large variable-speed drives. (free for use).
BLY	Control and protection equipment.	BPD	Power installations for large variable-speed drives. (free for use).
BM	Low voltage distribution boards and transformers, (diesel) emergency power system 1.	BPT	Transformer for large variable-speed drives. (free for use).
BMA	Low voltage emergency distribution boards. (free for use).	BPX	Fluid supply system for control and protection equipment.
BMB	Low voltage emergency distribution boards. (free for use).	BPY	Control and protection equipment.
BMC	Low voltage emergency distribution boards. (free for use).	BR	Low voltage distribution boards, converter, uninterruptible power system (UPS) and stand-by diesel power system.
BMD	Low voltage emergency distribution boards. (free for use).	BRA	Low voltage (UPS) distribution boards. (free for use).
BME	Low voltage emergency distribution boards. (free for use).	BRB	Low voltage (UPS) distribution boards. (free for use).
BMF	Low voltage emergency distribution boards. (free for use).	BRC	Low voltage emergency distribution boards. (free for use).
BMG	Low voltage emergency distribution boards. (free for use).	BRD	Low voltage emergency distribution boards. (free for use).
BMH	Low voltage emergency distribution boards. (free for use).	BRF	Low voltage emergency distribution boards. (free for use).
BMT	Low voltage auxiliary power transformers. (free for use according to voltage level).	BRG	Low voltage emergency distribution boards. (free for use).
BMU	Low voltage auxiliary power transformers. (free for use according to voltage level).	BRH	Low voltage emergency distribution boards. (free for use).
BMV	Low voltage auxiliary power transformers. (free for use according to voltage level).		

POWER TRANSMISSION AND AUXILIARY POWER SUPPLY

BRT Converter (rotary).
BRU Converter (static), inverter.
BRV Stand-by power generation equipment.
BRX Fluid supply system for control and protection equipment.
BRY Control and protection equipment.

BT Battery and charger systems.

BTA Batteries 125 V or 220 V DC.
BTB Batteries 125 V or 220 V DC.
BTC Batteries.
BTD Batteries.
BTE Batteries 24 V DC.
BTf Batteries 24 V DC.
BTL Battery chargers 125 V or 220 V DC.
BTM Battery chargers 125 V or 220 V DC.
BTN Battery chargers.
BTP Battery chargers.
BTQ Battery chargers 24V DC.
BTR Battery chargers 24 V DC.
BTS DC/DC converter 125 or 220/48 V DC.
BTt DC/DC converter 125 or 220/48 V DC.
BTW Common equipment.
BTX Common equipment.
BTY Common equipment.
BTZ Common equipment.

BU Direct current distribution boards.

BUA Direct current distribution boards 125 V or 220 V.
 (free for use).
BUB Direct current distribution boards 125 V or 220 V.
 (free for use).
BUC Direct current distribution boards 125 V or 220 V.
 (free for use).
BUD Direct current distribution boards
 (free for use).
BUE Direct current distribution boards 24 V.
 (free for use).
BUF Direct current distribution boards 24 V.
 (free for use).
BUG Direct current distribution boards 24 V.
 (free for use).

BUH Direct current distribution boards.
 (free for use).

BUL Direct current distribution boards 48 V.
 (free for use).

BUM Direct current distribution boards 48 V.
 (free for use).

BUX Fluid supply system for control and protection equipment.

BV Direct current distribution boards.

BVA Direct current distribution boards 125 V or 220 V.
 (free for use).

BVB Direct current distribution boards 125 V or 220 V.
 (free for use).

BVC Direct current distribution boards 125 V or 220 V.
 (free for use).

BVD Direct current distribution boards.
 (free for use).

BVE Direct current distribution boards 24 V.
 (free for use).

BVF Direct current distribution boards 24 V.
 (free for use).

BVG Direct current distribution boards 24 V.
 (free for use).

BVH Direct current distribution boards.
 (free for use).

BVL Direct current distribution boards 48 V.
 (free for use).

BVM Direct current distribution boards 48 V.
 (free for use).

BVX Fluid supply system for control and protection equipment.

BVY Control and protection equipment.

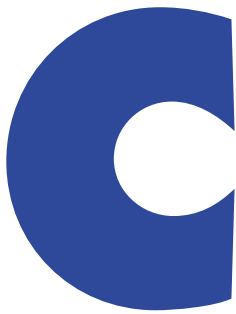
BY Control and protection equipment.

BYA Control and protection equipment.
 (free for use).

BYB Control and protection equipment.
 (free for use).

BYC Control and protection equipment.
 (free for use).

BYD Control and protection equipment.
 (free for use).



INSTRUMENTATION AND CONTROL EQUIPMENT

(Identification on a priority basis according to main instrumentation and control function also acceptable in composite structure hardware packaging systems).

- | | |
|--|--|
| CA Protective interlocks. | CS Instrumentation and control equipment. |
| CB Functional group control, subloop control. | CU Closed-loop control (power section). |
| CC Binary signal conditioning. | CV Marshalling racks. |
| CD Drive control interface. | CW Control rooms. |
| CE Annunciation. | CX Local control stations (e.g. for coal handling plants, ash handling plants, cooling water systems, diesel units, supervision of generator cooling, remote shutdown station). |
| CF Measurement, recording. | CY Communication equipment. |
| CG Closed-loop control (excl. power section). | |
| CH Protection (excl. reactor protection). | |
| CJ Unit coordination level. | |
| CK Process computer system. | |
| CM Instrumentation and control equipment. | |
| CN Instrumentation and control equipment. | |
| CP Instrumentation and control equipment. | |
| CQ Instrumentation and control equipment. | |
| CR Instrumentation and control equipment. | |



INSTRUMENTATION AND CONTROL EQUIPMENT

CA Protective interlocks.

- CAA** Cabinets for protective interlocks.
(free for use).
- CAB** Cabinets for protective interlocks.
- CAC** Cabinets for protective interlocks.
- CAD** Cabinets for protective interlocks.

CB Functional group control, subloop control.

- CBA** Cabinets for DCS cubicles.
(free for use).
- CBP** Cabinets for synchronization.
(free for use)
- CBQ** Cabinets for auxiliary power
changeover.
(free for use)

CC Binary signal conditioning.

- CCA** Cabinets for binary signal
conditioning.
(free for use).
- CCB** Cabinets for binary signal
conditioning.
(free for use).
- CCC** Cabinets for binary signal
conditioning.
(free for use).
- CCD** Cabinets for binary signal
conditioning.
(free for use).

CD Drive control interface.

- CDA** Cabinets for drive control interface.
(free for use).
- CDB** Cabinets for drive control interface.
(free for use).
- CDC** Cabinets for drive control interface.
(free for use).
- CDD** Cabinets for drive control interface.
(free for use).

CE Annunciation.

- CEK** Fault recording.

CF Measurement, recording.

- CFA** Cabinets for measurement.
(free for use).
- CFB** Cabinets for measurement.
(free for use).
- CFC** Cabinets for measurement.
(free for use).

- CFD** Cabinets for measurement.
(free for use).

- CFQ** Cabinets for recording (meters,
pen recorders).

CG Closed-loop control (excl. power section).

- CGA** Cabinets for closed-loop control.
(free for use).
- CGB** Cabinets for closed-loop control.
(free for use).
- CGC** Cabinets for closed-loop control.
(free for use).
- CGD** Cabinets for closed-loop control.
(free for use).

CH Protection (excl. reactor protection).

- CHA** Cabinets for generator and
transformer protection.
(free for use).
- CHB** Cabinets for generator and
transformer protection.
(free for use).
- CHC** Cabinets for generator and
transformer protection.
(free for use).
- CHD** Cabinets for generator and
transformer protection.
(free for use).
- CHE** Protection (excl. reactor protection).
(free for use).
- CHF** Protection (excl. reactor protection).
(free for use).

CJ Unit coordination level.

- CJA** Unit control system (including
cabinets).
- CJD** Start-up control, setpoint control
(unit) (incl. cabinets).
- CJF** Boiler control system (incl.
cabinets).
- CJJ** Instrumentation and control
cabinets for steam turbine set.
(free for use).
- CJK** Instrumentation and control
cabinets for steam turbine set.
(free for use).
- CJP** Instrumentation and control
cabinets for gas turbine set.
- CJQ** Instrumentation and control
cabinets for gas turbine set.
(free for use).

INSTRUMENTATION AND CONTROL EQUIPMENT

CJU	Instrumentation and control cabinets for other main and heavy machinery. (free for use).	CNC	Instrumentation and control equipment. (free for use for system combination).
CK	Process computer system.	CND	Instrumentation and control equipment. (free for use for system combination).
CKA	Online supervisory and diagnostic computer. (free for use).	CNT	Instrumentation and control equipment (network cubicles).
CKB	Online supervisory and diagnostic computer. (free for use).	CP	Instrumentation and control equipment.
CKC	Online supervisory and diagnostic computer. (free for use).	CPA	Instrumentation and control equipment. (free for use for system combination).
CKD	Online supervisory and diagnostic computer. (free for use).	CPB	Instrumentation and control equipment. (free for use for system combination).
CKE	Online supervisory and diagnostic computer. (free for use).	CPC	Instrumentation and control equipment. (free for use for system combination).
CKJ	Access control computer. (free for use).	CPD	Instrumentation and control equipment. (free for use for system combination).
CKK	Access control computer. (free for use).	CQ	Instrumentation and control equipment.
CKL	Access control computer. (free for use).	CQA	Instrumentation and control equipment. (free for use for system combination).
CKM	Access control computer. (free for use).	CQB	Instrumentation and control equipment. (free for use for system combination).
CKN	Process computer system. (free for use).	CQC	Instrumentation and control equipment. (free for use for system combination).
CM	Instrumentation and control equipment.	CQD	Instrumentation and control equipment. (free for use for system combination).
CMA	Instrumentation and control equipment. (free for use for system combination).	CR	Instrumentation and control equipment.
CMB	Instrumentation and control equipment. (free for use for system combination).	CRA	Instrumentation and control equipment. (free for use for system combination).
CMC	Instrumentation and control equipment. (free for use for system combination).	CRB	Instrumentation and control equipment. (free for use for system combination).
CMD	Instrumentation and control equipment. (free for use for system combination).	CRC	Instrumentation and control equipment. (free for use for system combination).
CN	Instrumentation and control equipment.	CRD	Instrumentation and control equipment. (free for use for system combination).
CNA	Instrumentation and control equipment. (free for use for system combination).		
CNB	Instrumentation and control equipment. (free for use for system combination).		

INSTRUMENTATION AND CONTROL EQUIPMENT

CS Instrumentation and control equipment.

- CSA Instrumentation and control equipment.**
(free for use for system combination).
- CSB Instrumentation and control equipment.**
(free for use for system combination).
- CSC Instrumentation and control equipment.**
(free for use for system combination).
- CSD Instrumentation and control equipment.**
(free for use for system combination).

CU Closed-loop control (power section).

- CUA Static excitation system or generator voltage regulator.**
(free for use).
- CUB Closed-loop control (power section).**
- CUC Closed-loop control (power section).**
- CUG Closed-loop control (power section).**

CV Marshalling racks.

- CVA Marshalling racks.**
(free for use).
- CVB Marshalling racks.**
(free for use).
- CVC Marshalling racks.**
(free for use).
- CVD Marshalling racks.**
(free for use).
- CVE Marshalling racks.**
(free for use).
- CVF Marshalling racks.**
(free for use).
- CVG Marshalling racks.**
(free for use).
- CVH Marshalling racks.**
(free for use).

CW Control rooms.

- CWA Main control consoles.**
(free for use).
- CWB Main control consoles.**
(free for use).
- CWC Main control consoles.**
(free for use).
- CWD Main control consoles.**
(free for use).
- CWE Main control consoles.**
(free for use).

CWF Main control panels.
(free for use).

CWG Main control panels.
(free for use).

CWH Main control panels.
(free for use).

CWJ Main control panels.
(free for use).

CWP Main control panels.
(free for use).

CWQ Control rooms.
(free for use).

CWR Control rooms.
(free for use).

CWS Control rooms.
(free for use).

CWT Control rooms.
(free for use).

CWU Control rooms.
(free for use).

CX Local control stations (e.g. for coal handling plants, ash handling plants, cooling water systems, diesel units, supervision of generator cooling, remote shutdown station).

CXA Local control stations.
(free for use).

CXB Local control stations.
(free for use).

CXC Local control stations.
(free for use).

CXD Local control stations.
(free for use).

CXE Local control stations.
(free for use).

CXF Local control stations.
(free for use).

CXG Local control stations.
(free for use).

CXH Local control stations.
(free for use).

CXQ Local control stations.
(free for use).

CY Communication equipment.

CYA Telephone system (PABX).

CYB Control console telephone system.

CYC Alarm system (acoustic).

CYD Alarm system (optical).

CYE Fire alarm system.

CYF Clock system.

CYG Remote control system.

INSTRUMENTATION AND CONTROL EQUIPMENT

CYH Telemetry system.

CYH80 Plant emission monitoring system,

CYH90 Weather station.

CYH8 Plant emission monitoring system

CYH9 Weather station.

CYJ Remote metering system.

CYK HF carrier telephone system.

CYL Staff paging system, wireless.

CYM Staff paging system, inductive.

CYN Staff paging system, hardwired.

CYP Optical monitoring system.

CYQ General gas alarm system (if not assignet to a specific system).

CYR Pneumatic tube conveyor.

CYS Radiotelephone system.



E

CONVENTIONAL FUEL SUPPLY AND RESIDUES DISPOSAL

- EA** Unloading and storage of solid fuels.
- EB** Mechanical treatment of solid fuels (also for gas generation and treatment) (Crushing, mixing, drying, etc.).
- EC** Distribution of solid fuels.
- ED** Chemical treatment of solid fuels incl. residues removal (e.g. desulfurization plant).
- EE** Conversion of solid fuels (Gas generation and treatment see function key *R*).
- EG** Supply of liquid fuels.
- EH** Chemical treatment of liquid fuels incl. residues removal.
- EK** Supply of gaseous fuels.
- EL** Chemical treatment of gaseous fuels incl. residues removal.
- EM** Supply and treatment of additive fuels.
- EN** Supply of other fuels (use only for combinations of different fuel types).
- EP** Treatment of other fuels (use only for combinations of different fuel types).
- EQ** Conversion of other fuels (use only for combinations of different fuel types).
- ER** Ignition fuel supply.
- ET** Ash and slag removal system (from excl. removal equipment).
- EU** Treatment and transport system for combustion, fuel treatment, fuel conversion, flue gas cleaning, gas generation residues.
- EV** Lubricant supply system.
- EW** Sealing/loosening medium supply.
- EX** Fluid supply system for control and protection equipment.
- EY** Control and protection equipment.



CONVENTIONAL FUEL SUPPLY AND RESIDUES DISPOSAL

EA Unloading and storage of solid fuels.

- EAA Ship unloading system.**
from incl. unloading equipment
to excl. transport or storage plant.
- EAB Rail wagon and truck unloading bay.**
from incl. unloading equipment
to excl. transport or storage plant.
- EAC Transport system.**
from incl. receiving point
to excl. storage incl. internal transport
system within unloading and storage plant
to excl. stacker.
- EAD Stacking system.**
from excl. transport system
to excl. storage plant.
- EAE Bunker system, storage area (stockyard).**
from excl. unloading equipment or
transport system
to excl. transport or bucket wheel or
reclaimer or stacking system.
- EAF Bucket wheel system, reclaimer system.**
from excl. bunker system or stockyard
to excl. transport system.
- EAT Weighing equipment.**
- EAU Sampling equipment.**
- EAV Lubricant supply system.**
- EAX Fluid supply system for control and protection equipment.**
- EAY Control and protection equipment.**

EB Mechanical treatment of solid fuels (also for gas generation and treatment) (Crushing, mixing, drying, etc.).

- EBA Transport system.**
from incl. receiving point
to excl. discharge into treatment or mixing
system.
- EBB Mixing system.**
from incl. receiving point
to excl. discharge into other system.
- EBC Crushing system, pulverizing system.**
from incl. receiving point
to excl. discharge into other system.
- EBD Screening system.**
from incl. receiving point
to excl. discharge into other system.

EBE Separator system and discharge equipment thereof - unless not structural part of one of the former systems (*EBA* to *EBD*).

- from incl. receiving point
to excl. discharge into other system.
- EBF Temporary storage system for milled raw coal.**
from incl. receiving point
to excl. discharge into other system.
- EBG Predrying system.**
from incl. receiving point
to excl. discharge into other system.
- EBH Main drying system.**
from incl. receiving point
to excl. discharge into other system.
- EBJ Dried coal transport system incl. aftercooling.**
from incl. receiving point
to excl. discharge into other system.
- EBK Dried coal temporary storage system.**
from incl. receiving point
to excl. discharge into other system.
- EBL Vapor compressor system.**
from incl. receiving point
to excl. discharge into other system.
- EBM Exhaust system.**
from incl. receiving point
to excl. discharge into other system.
- EBR Residues removal system.**
- EBT Weighing equipment.**
- EBU Sampling equipment.**
- EBV Lubricant supply system.**
- EBX Fluid supply system for control and protection equipment.**
- EBY Control and protection equipment.**

EC Distribution of solid fuels.

- ECA Distribution of solid fuels.**
(free for use according to transport systems)
from incl. receiving point
to excl. discharge into other systems.
- ECB Distribution of solid fuels.**
(free for use according to transport systems)
from incl. receiving point
to excl. discharge into other systems.
- ECC Distribution of solid fuels.**
(free for use according to transport systems)
from incl. receiving point
to excl. discharge into other systems.

CONVENTIONAL FUEL SUPPLY AND RESIDUES DISPOSAL

ECD Distribution of solid fuels.
(free for use according to transport systems)
from incl. receiving point
to excl. discharge into other systems.

ECE Distribution of solid fuels.
(free for use according to transport systems)
from incl. receiving point
to excl. discharge into other systems.

ECF Distribution of solid fuels.
(free for use according to transport systems)
from incl. receiving point
to excl. discharge into other systems.

ECG Distribution of solid fuels.
(free for use according to transport systems)
from incl. receiving point
to excl. discharge into other systems.

ECH Distribution of solid fuels.
(free for use according to transport systems)
from incl. receiving point
to excl. discharge into other systems.

ECJ Distribution of solid fuels.
(free for use according to transport systems)
from incl. receiving point
to excl. discharge into other systems.

ECK Distribution of solid fuels.
(free for use according to transport systems)
from incl. receiving point
to excl. discharge into other systems.

ECL Distribution of solid fuels.
(free for use according to transport systems)
from incl. receiving point
to excl. discharge into other systems.

ECM Distribution of solid fuels.
(free for use according to transport systems)
from incl. receiving point
to excl. discharge into other systems.

ECN Distribution of solid fuels.
(free for use according to transport systems)
from incl. receiving point
to excl. discharge into other systems.

ECP Distribution of solid fuels.
(free for use according to transport systems)
from incl. receiving point
to excl. discharge into other systems.

ECQ Distribution of solid fuels.
(free for use according to transport systems)
from incl. receiving point
to excl. discharge into other systems.

ECR Distribution of solid fuels.
(free for use according to transport systems)
from incl. receiving point
to excl. discharge into other systems.

ECS Distribution of solid fuels.
(free for use according to transport systems)
from incl. receiving point
to excl. discharge into other systems.

ECT Weighing equipment.

ECU Sampling equipment.

ECX Fluid supply system for control and protection equipment.

ECY Control and protection equipment.

ED Chemical treatment of solid fuels incl. residues removal (e.g. desulfurization plant).

EDA Chemical treatment of solid fuels incl. residues removal (e.g. desulfurization plant).
(free for use).

EDB Chemical treatment of solid fuels incl. residues removal (e.g. desulfurization plant).
(free for use).

EDC Chemical treatment of solid fuels incl. residues removal (e.g. desulfurization plant).
(free for use).

EDD Chemical treatment of solid fuels incl. residues removal (e.g. desulfurization plant).
(free for use).

EDE Chemical treatment of solid fuels incl. residues removal (e.g. desulfurization plant).
(free for use).

EDF Chemical treatment of solid fuels incl. residues removal (e.g. desulfurization plant).
(free for use).

EDG Chemical treatment of solid fuels incl. residues removal (e.g. desulfurization plant).
(free for use).

EDH Chemical treatment of solid fuels incl. residues removal (e.g. desulfurization plant).
(free for use).

EDJ Chemical treatment of solid fuels incl. residues removal (e.g. desulfurization plant).
(free for use).

EDK Chemical treatment of solid fuels incl. residues removal (e.g. desulfurization plant).
(free for use).

EDL Chemical treatment of solid fuels incl. residues removal (e.g. desulfurization plant).
(free for use).

CONVENTIONAL FUEL SUPPLY AND RESIDUES DISPOSAL

EDM	Chemical treatment of solid fuels incl. residues removal (e.g. desulfurization plant). (free for use).	EEJ	Conversion of solid fuels. (free for use).
EDN	Chemical treatment of solid fuels incl. residues removal (e.g. desulfurization plant). (free for use).	EEK	Conversion of solid fuels. (free for use).
EDP	Chemical treatment of solid fuels incl. residues removal (e.g. desulfurization plant). (free for use).	EEL	Conversion of solid fuels. (free for use).
EDQ	Chemical treatment of solid fuels incl. residues removal (e.g. desulfurization plant). (free for use).	EEM	Conversion of solid fuels. (free for use).
EDR	Chemical treatment of solid fuels incl. residues removal (e.g. desulfurization plant). (free for use).	EEN	Conversion of solid fuels. (free for use).
EDS	Chemical treatment of solid fuels incl. residues removal (e.g. desulfurization plant). (free for use).	EEP	Conversion of solid fuels. (free for use).
EDT	Chemical treatment of solid fuels incl. residues removal (e.g. desulfurization plant). (free for use).	EEQ	Conversion of solid fuels. (free for use).
EDU	Chemical treatment of solid fuels incl. residues removal (e.g. desulfurization plant). (free for use).	EER	Conversion of solid fuels. (free for use).
EDV	Lubricant supply system.	EES	Conversion of solid fuels. (free for use).
EDX	Fluid supply system for control and protection equipment.	EET	Conversion of solid fuels. (free for use).
EDY	Control and protection equipment.	EEU	Conversion of solid fuels. (free for use).
EE	Conversion of solid fuels (Gas generation and treatment see function key *R*).	EEV	Lubricant supply system.
EEA	Conversion of solid fuels. (free for use).	EEY	Fluid supply system for control and protection equipment.
EEB	Conversion of solid fuels. (free for use).	EEY	Control and protection equipment.
EEC	Conversion of solid fuels. (free for use).	EG	Supply of liquid fuels.
EED	Conversion of solid fuels. (free for use).	EGA	Receiving equipment incl. pipeline. from excl. receiving point to excl. tank, incl. pump.
EEE	Conversion of solid fuels. (free for use).	EGB	Tank farm. from incl. tank inlet to incl. tank outlet incl. drains, vents, heating equipment etc.
EEF	Conversion of solid fuels. (free for use).	EGC	Pump system. from incl. pump system suction nozzles to incl. pump system discharge nozzles.
EEG	Conversion of solid fuels. (free for use).	EGD	Piping system. from excl. outlet another system to excl. inlet another system.
EEH	Conversion of solid fuels. (free for use).	EGE	Mechanical cleaning, scrubbing. from incl. inlet into mechanical cleaning / scrubbing to incl. outlet mechanical cleaning / scrubbing.
		EGF	Intermediate storage. from incl. daily tank inlet to incl. daily tank outlet incl. drains, vents, heating equipment etc.
		EGG	Preheating. from incl. inlet preheating system to incl. outlet preheating system incl. drains, vents.

CONVENTIONAL FUEL SUPPLY AND RESIDUES DISPOSAL

EGR	Residues removal system. from incl. inlet collecting tank or collecting pipe to incl. inlet into another system incl. pumping equipment.	EHM	Chemical treatment of liquid fuels incl. residues removal. (free for use).
EGT	Heating medium system. from excl. branch off supply system to excl. user (*EGG*), and from excl. user (*EGG*) to excl. inlet another system.	EHN	Chemical treatment of liquid fuels incl. residues removal. (free for use).
EGU	Custody transfer metering station. from incl. inlet metering station to incl. outlet metering station.	EHP	Chemical treatment of liquid fuels incl. residues removal. (free for use).
EGV	Lubricant supply system.	EHQ	Chemical treatment of liquid fuels incl. residues removal. (free for use).
EGX	Fluid supply system for control and protection equipment.	EHR	Chemical treatment of liquid fuels incl. residues removal. (free for use).
EGY	Control and protection equipment.	EHS	Chemical treatment of liquid fuels incl. residues removal. (free for use).
EH	Chemical treatment of liquid fuels incl. residues removal.	EHT	Chemical treatment of liquid fuels incl. residues removal. (free for use).
EHA	Chemical treatment of liquid fuels incl. residues removal. (free for use).	EHU	Chemical treatment of liquid fuels incl. residues removal. (free for use).
EHB	Chemical treatment of liquid fuels incl. residues removal. (free for use).	EHV	Lubricant supply system.
EHC	Chemical treatment of liquid fuels incl. residues removal. (free for use).	EHX	Fluid supply system for control and protection equipment.
EHD	Chemical treatment of liquid fuels incl. residues removal. (free for use).	EHY	Control and protection equipment.
EHE	Chemical treatment of liquid fuels incl. residues removal. (free for use).	EK	Supply of gaseous fuels.
EHF	Chemical treatment of liquid fuels incl. residues removal. (free for use).	EKA	Receiving equipment incl. pipeline. from excl. receiving point to incl. inlet to other system.
EHG	Chemical treatment of liquid fuels incl. residues removal. (free for use).	EKB	Preliminary cleaning (moisture and dust separation). from incl. inlet moisture separation to incl. outlet dust separation.
EHH	Chemical treatment of liquid fuels incl. residues removal. (free for use).	EKC	Heating system. from incl. inlet heating system to incl. outlet heating system incl. drains, vents, gas leakage monitoring equipment.
EHJ	Chemical treatment of liquid fuels incl. residues removal. (free for use).	EKD	Main reducing station, expansion turbine. from incl. main reducing station inlet, expansion turbine inlet to incl. main reducing station outlet, expansion turbine outlet.
EHK	Chemical treatment of liquid fuels incl. residues removal. (free for use).	EKE	Mechanical filtering, cleaning, scrubbing. from incl. inlet of mechanical cleaning, scrubbing system to incl. outlet mechanical cleaning, scrubbing system.
EHL	Chemical treatment of liquid fuels incl. residues removal. (free for use).		

CONVENTIONAL FUEL SUPPLY AND RESIDUES DISPOSAL

EKF	Storage system. from incl. inlet storage system to incl. outlet storage system incl. drains, vents, gas leakage monitoring equipment.	ELH	Chemical treatment of gaseous fuels incl. residues removal. (free for use).
EKG	Piping system. from excl. outlet another system to excl. inlet another system.	ELJ	Chemical treatment of gaseous fuels incl. residues removal. (free for use).
EKH	Main pressure boosting system. from incl. inlet compressor system to incl. discharge compressor system incl. drains, vents, gas leakage monitoring equipment, sealing medium connections.	ELK	Chemical treatment of gaseous fuels incl. residues removal. (free for use).
EKR	Residues removal system. from incl. inlet collecting tank or collecting pipe to incl. inlet into another system incl. pumping equipment.	ELL	Chemical treatment of gaseous fuels incl. residues removal. (free for use).
EKT	Heating medium system. from excl. branch off supply system to excl. user (*EKC*) and from excl. user (*EKC*) to excl. inlet to other system.	ELM	Chemical treatment of gaseous fuels incl. residues removal. (free for use).
EKU	Custody transfer metering station. from incl. inlet metering station to incl. outlet metering station.	ELN	Chemical treatment of gaseous fuels incl. residues removal. (free for use).
EKV	Lubricant supply system.	ELP	Chemical treatment of gaseous fuels incl. residues removal. (free for use).
EKW	Sealing fluid supply system.	ELQ	Chemical treatment of gaseous fuels incl. residues removal. (free for use).
EKX	Fluid supply system for control and protection equipment.	ELR	Chemical treatment of gaseous fuels incl. residues removal. (free for use).
EKY	Control and protection equipment.	ELS	Chemical treatment of gaseous fuels incl. residues removal. (free for use).
EL	Chemical treatment of gaseous fuels incl. residues removal.	ELT	Chemical treatment of gaseous fuels incl. residues removal. (free for use).
ELA	Chemical treatment of gaseous fuels incl. residues removal. (free for use).	ELU	Chemical treatment of gaseous fuels incl. residues removal. (free for use).
ELB	Chemical treatment of gaseous fuels incl. residues removal. (free for use).	ELV	Lubricant supply system.
ELC	Chemical treatment of gaseous fuels incl. residues removal. (free for use).	ELW	Sealing fluid supply system.
ELD	Chemical treatment of gaseous fuels incl. residues removal. (free for use).	ELX	Fluid supply system for control and protection equipment.
ELE	Chemical treatment of gaseous fuels incl. residues removal. (free for use).	ELY	Control and protection equipment.
ELF	Chemical treatment of gaseous fuels incl. residues removal. (free for use).	EM	Supply and treatment of additive fuels.
ELG	Chemical treatment of gaseous fuels incl. residues removal. (free for use).	EMA	Supply and treatment of additive fuels. (free for use).
		EMB	Supply and treatment of additive fuels. (free for use).
		EMC	Supply and treatment of additive fuels. (free for use).

CONVENTIONAL FUEL SUPPLY AND RESIDUES DISPOSAL

EMD Supply and treatment of additive fuels.
(free for use).

EME Supply and treatment of additive fuels.
(free for use).

EMF Supply and treatment of additive fuels.
(free for use).

EMG Supply and treatment of additive fuels.
(free for use).

EMH Supply and treatment of additive fuels.
(free for use).

EMJ Supply and treatment of additive fuels.
(free for use).

EMK Supply and treatment of additive fuels.
(free for use).

EML Supply and treatment of additive fuels.
(free for use).

EMM Supply and treatment of additive fuels.
(free for use).

EMN Supply and treatment of additive fuels.
(free for use).

EMP Supply and treatment of additive fuels.
(free for use).

EMQ Supply and treatment of additive fuels.
(free for use).

EMR Supply and treatment of additive fuels.
(free for use).

EMS Supply and treatment of additive fuels.
(free for use).

EMT Supply and treatment of additive fuels.
(free for use).

EMU Supply and treatment of additive fuels.
(free for use).

EMV Lubricant supply system.

EMX Fluid supply system for control and protection equipment.

EMY Control and protection equipment.

EN Supply of other fuels (use only for combinations of different fuel types).

ENA Supply of other fuels (use only for combination of different fuel types).
(free for use).

ENB Supply of other fuels (use only for combination of different fuel types).
(free for use).

ENC Supply of other fuels (use only for combination of different fuel types).
(free for use).

END Supply of other fuels (use only for combination of different fuel types).
(free for use).

ENE Supply of other fuels (use only for combination of different fuel types).
(free for use).

ENF Supply of other fuels (use only for combination of different fuel types).
(free for use).

ENG Supply of other fuels (use only for combination of different fuel types).
(free for use).

ENH Supply of other fuels (use only for combination of different fuel types).
(free for use).

ENJ Supply of other fuels (use only for combination of different fuel types).
(free for use).

ENK Supply of other fuels (use only for combination of different fuel types).
(free for use).

ENL Supply of other fuels (use only for combination of different fuel types).
(free for use).

ENM Supply of other fuels (use only for combination of different fuel types).
(free for use).

ENN Supply of other fuels (use only for combination of different fuel types).
(free for use).

ENP Supply of other fuels (use only for combination of different fuel types).
(free for use).

ENQ Supply of other fuels (use only for combination of different fuel types).
(free for use).

ENR Supply of other fuels (use only for combination of different fuel types).
(free for use).

ENS Supply of other fuels (use only for combination of different fuel types).
(free for use).

ENT Supply of other fuels (use only for combination of different fuel types).
(free for use).

CONVENTIONAL FUEL SUPPLY AND RESIDUES DISPOSAL

ENU	Supply of other fuels (use only for combination of different fuel types). (free for use).	EPM	Treatment of other fuels (use only for combinations of different fuel types). (free for use).
ENV	Lubricant supply system.	EPN	Treatment of other fuels (use only for combinations of different fuel types). (free for use).
ENX	Fluid supply system for control and protection equipment.	EPP	Treatment of other fuels (use only for combinations of different fuel types). (free for use).
ENY	Control and protection equipment.	EPQ	Treatment of other fuels (use only for combinations of different fuel types). (free for use).
EP	Treatment of other fuels (use only for combinations of different fuel types).	EPR	Treatment of other fuels (use only for combinations of different fuel types). (free for use).
EPA	Treatment of other fuels (use only for combinations of different fuel types). (free for use).	EPS	Treatment of other fuels (use only for combinations of different fuel types). (free for use).
EPB	Treatment of other fuels (use only for combinations of different fuel types). (free for use).	EPT	Treatment of other fuels (use only for combinations of different fuel types). (free for use).
EPC	Treatment of other fuels (use only for combinations of different fuel types). (free for use).	EPU	Treatment of other fuels (use only for combinations of different fuel types). (free for use).
EPD	Treatment of other fuels (use only for combinations of different fuel types). (free for use).	EPV	Lubricant supply system.
EPE	Treatment of other fuels (use only for combinations of different fuel types). (free for use).	EPX	Fluid supply system for control and protection equipment.
EPF	Treatment of other fuels (use only for combinations of different fuel types). (free for use).	EPY	Control and protection equipment.
EPG	Treatment of other fuels (use only for combinations of different fuel types). (free for use).	EQ	Conversion of other fuels (use only for combinations of different fuel types).
EPH	Treatment of other fuels (use only for combinations of different fuel types). (free for use).	EQA	Conversion of other fuels (use only for combinations of different fuel types). (free for use).
EPJ	Treatment of other fuels (use only for combinations of different fuel types). (free for use).	EQB	Conversion of other fuels (use only for combinations of different fuel types). (free for use).
EPK	Treatment of other fuels (use only for combinations of different fuel types). (free for use).	EQC	Conversion of other fuels (use only for combinations of different fuel types). (free for use).
EPL	Treatment of other fuels (use only for combinations of different fuel types). (free for use).	EQD	Conversion of other fuels (use only for combinations of different fuel types). (free for use).

CONVENTIONAL FUEL SUPPLY AND RESIDUES DISPOSAL

EQE	Conversion of other fuels (use only for combinations of different fuel types). (free for use).	EQU	Conversion of other fuels (use only for combinations of different fuel types). (free for use).
EQF	Conversion of other fuels (use only for combinations of different fuel types). (free for use).	EQV	Lubricant supply system.
EQG	Conversion of other fuels (use only for combinations of different fuel types). (free for use).	EQX	Fluid supply system for control and protection equipment.
EQH	Conversion of other fuels (use only for combinations of different fuel types). (free for use).	EQY	Control and protection equipment.
EQJ	Conversion of other fuels (use only for combinations of different fuel types). (free for use).	ER	Ignition fuel supply.
EQK	Conversion of other fuels (use only for combinations of different fuel types). (free for use).	ERA	Pulverized coal supply system. from excl. receiving point to excl. branch to boiler or to excl. other users.
EQL	Conversion of other fuels (use only for combinations of different fuel types). (free for use).	ERB	Oil supply system. from excl. receiving point to excl. branch to boiler or to excl. other users.
EQM	Conversion of other fuels (use only for combinations of different fuel types). (free for use).	ERC	Gas supply system. from excl. receiving point to excl. branch to boiler or to excl. other users.
EQN	Conversion of other fuels (use only for combinations of different fuel types). (free for use).	ERV	Lubricant supply system.
EQP	Conversion of other fuels (use only for combinations of different fuel types). (free for use).	ERX	Fluid supply system for control and protection equipment.
EQQ	Conversion of other fuels (use only for combinations of different fuel types). (free for use).	ERY	Control and protection equipment.
EQR	Conversion of other fuels (use only for combinations of different fuel types). (free for use).	ET	Ash and slag removal system (from excl. removal equipment).
EQS	Conversion of other fuels (use only for combinations of different fuel types). (free for use).	ETA	Wet ash conveying system. from excl. removal equipment or from excl. receiving point to excl. storage facility or to excl. discharge into other system.
EQT	Conversion of other fuels (use only for combinations of different fuel types). (free for use).	ETB	Storage or settling pond for wet ash. from excl. receiving point to incl. removal equipment.
		ETC	Wet ash dredger. from excl. receiving point to excl. discharge into other system.
		ETD	Conveying system for granulate. from excl. removal equipment to excl. storage facility or to excl. discharge into other system.
		ETE	Storage system for granulate. from excl. receiving point to incl. removal equipment
		ETG	Conveying system for dry ash. from excl. removal equipment (boiler, electrostatic precipitator) to excl. storage facilities or to excl. discharge into other system.

CONVENTIONAL FUEL SUPPLY AND RESIDUES DISPOSAL

ETH	Storage system for dry ash. from excl. receiving point to incl. removal equipment.	EUE	Treatment system for gas generation and treatment residues. from incl. inlet to incl. outlet.
ETK	Common conveying system for wet and dry ash. from excl. receiving point to excl. storage facility or to excl. discharge into other system.	EUF	Treatment system for fuel treatment residues (only if *EUA* is not sufficient for identification). from incl. inlet to incl. outlet.
ETL	Common storage system for wet and dry ash. from excl. receiving point to incl. removal equipment.	EUG	Treatment system for fuel treatment residues (only if *EUA* and *EUF* are not sufficient for identification). from incl. inlet to incl. outlet.
ETM	Settling plant for wet and dry ash. from excl. receiving point from conveying system to excl. pit or to excl. discharge into other system	EUH	Treatment system for fuel conversion residues (only if *EUB* is not sufficient for identification). from incl. inlet to incl. outlet.
ETN	Forwarding, distribution, recovery and disposal systems for flushing and ash water. from excl. inlet or from incl. branch t o excl. inlet to other system.	EUJ	Treatment system for fuel conversion residues (only if *EUB* and *EUH* are not sufficient for identification). from incl. inlet to incl. outlet.
ETP	Generation and distribution systems for carrier air. from excl. branch or from incl. compressor system to excl. inlet to conveying system..	EUK	Treatment system for fuel combustion residues (only if *EUC* is not sufficient for identification). from incl. inlet to incl. outlet.
ETV	Lubricant supply system.	EUL	Treatment system for fuel combustion residues (only if *EUC* and *EUK* are not sufficient for identification). from incl. inlet to incl. outlet.
ETX	Fluid supply system for control and protection equipment.	EUM	Treatment system for flue gas cleaning residues (only if *EUD* is not sufficient for identification). from incl. inlet to incl. outlet.
ETY	Control and protection equipment.	EUN	Treatment system for flue gas cleaning residues (only if *EUD* and *EUM* are not sufficient for identification). from incl. inlet to incl. outlet.
EU	Treatment and transport system for combustion, fuel treatment, fuel conversion, flue gas cleaning, gas generation residues.	EUP	Treatment system for gas generation and treatment residues (only if *EUE* is not sufficient for identification). from incl. inlet to incl. outlet.
EUA	Treatment system for fuel treatment residues. from incl. inlet to incl. outlet.		
EUB	Treatment system for fuel conversion residues. from incl. inlet to incl. outlet.		
EUC	Treatment system for fuel combustion residues. from incl. inlet to incl. outlet.		
EUD	Treatment system for flue gas cleaning residues. from incl. inlet to incl. outlet.		

CONVENTIONAL FUEL SUPPLY AND RESIDUES DISPOSAL

EUQ	Treatment system for gas generation and treatment residues (only if *EUE* and *EUP* are not sufficient for identification). from incl. inlet to incl. outlet.	EWB	Sealing/loosening medium supply. (free for use).
EUV	Lubricant supply system.	EWG	Sealing/loosening medium supply. (free for use).
EUX	Fluid supply system for control and protection equipment.	EWD	Sealing/loosening medium supply. (free for use).
EUY	Control and protection equipment.	EWE	Sealing/loosening medium supply. (free for use).
EV	Lubricant supply system.	EWF	Sealing/loosening medium supply. (free for use).
EVA	Lubricant supply system. (free for use).	EWG	Sealing/loosening medium supply. (free for use).
EVB	Lubricant supply system. (free for use).	EWH	Sealing/loosening medium supply. (free for use).
EVC	Lubricant supply system. (free for use).	EWJ	Sealing/loosening medium supply. (free for use).
EVD	Lubricant supply system. (free for use).	EWK	Sealing/loosening medium supply. (free for use).
EVE	Lubricant supply system. (free for use).	EWL	Sealing/loosening medium supply. (free for use).
EVF	Lubricant supply system. (free for use).	EWM	Sealing/loosening medium supply. (free for use).
EVG	Lubricant supply system. (free for use).	EWN	Sealing/loosening medium supply. (free for use).
EVH	Lubricant supply system. (free for use).	EWQ	Sealing/loosening medium supply. (free for use).
EVJ	Lubricant supply system. (free for use).	EWR	Sealing/loosening medium supply. (free for use).
EVK	Lubricant supply system. (free for use).	EWS	Sealing/loosening medium supply. (free for use).
EVL	Lubricant supply system. (free for use).	EWT	Sealing/loosening medium supply. (free for use).
EVM	Lubricant supply system. (free for use).	EWU	Sealing/loosening medium supply. (free for use).
EVN	Lubricant supply system. (free for use).	EWV	Lubricant supply system. (free for use).
EVP	Lubricant supply system. (free for use).	EWX	Fluid supply system for control and protection equipment.
EVQ	Lubricant supply system. (free for use).	EWY	Control and protection equipment.
EVR	Lubricant supply system. (free for use).	EX	Fluid supply system for control and protection equipment.
EVS	Lubricant supply system. (free for use).	EXA	Fluid supply system for control and protection equipment. (free for use).
EVT	Lubricant supply system. (free for use).	EXB	Fluid supply system for control and protection equipment. (free for use).
EVU	Lubricant supply system. (free for use).	EXC	Fluid supply system for control and protection equipment. (free for use).
EW	Sealing/loosening medium supply.		
EWA	Sealing/loosening medium supply. (free for use).		

CONVENTIONAL FUEL SUPPLY AND RESIDUES DISPOSAL

EXD Fluid supply system for control and protection equipment.
(free for use).

EXE Fluid supply system for control and protection equipment.
(free for use).

EXF Fluid supply system for control and protection equipment.
(free for use).

EXG Fluid supply system for control and protection equipment.
(free for use).

EXH Fluid supply system for control and protection equipment.
(free for use).

EXJ Fluid supply system for control and protection equipment.
(free for use).

EXK Fluid supply system for control and protection equipment.
(free for use).

EXL Fluid supply system for control and protection equipment.
(free for use).

EXM Fluid supply system for control and protection equipment.
(free for use).

EXN Fluid supply system for control and protection equipment.
(free for use).

EXP Fluid supply system for control and protection equipment.
(free for use).

EXQ Fluid supply system for control and protection equipment.
(free for use).

EXR Fluid supply system for control and protection equipment.
(free for use).

EXS Fluid supply system for control and protection equipment.
(free for use).

EXT Fluid supply system for control and protection equipment.
(free for use).

EXU Fluid supply system for control and protection equipment.
(free for use).

EYC Control and protection equipment.
(free for use).

EYD Control and protection equipment.
(free for use).

EYE Control and protection equipment.
(free for use).

EYF Control and protection equipment.
(free for use).

EYG Control and protection equipment.
(free for use).

EYH Control and protection equipment.
(free for use).

EYJ Control and protection equipment.
(free for use).

EYK Control and protection equipment.
(free for use).

EYL Control and protection equipment.
(free for use).

EYM Control and protection equipment.
(free for use).

EYN Control and protection equipment.
(free for use).

EYP Control and protection equipment.
(free for use).

EYQ Control and protection equipment.
(free for use).

EYR Control and protection equipment.
(free for use).

EYS Control and protection equipment.
(free for use).

EYT Control and protection equipment.
(free for use).

EYU Control and protection equipment.
(free for use).

EY Control and protection equipment.

EYA Control and protection equipment.
(free for use).

EYB Control and protection equipment.
(free for use).



WATER SUPPLY AND DISPOSAL

GA Raw water supply.

GB Treatment system (carbonate hardness removal) incl. cooling tower make-up water treatment system.

GC Treatment system (demineralization).

GD Treatment system (others).

GH Distribution systems (not drinking water).

GK Drinking water supply.

GM Process drainage system.

GN Process drains treatment system.

GQ Domestic waste water collection and drainage systems.

GR Domestic waste water treatment system.

GT Water recovery from waste water.

GU Rainwater collection and drainage systems incl. treatment system.

GV Lubricant supply system.

GW Sealing fluid supply system.

GX Fluid supply system for control and protection equipment.

GY Control and protection equipment.



WATER SUPPLY AND DISPOSAL

GA Raw water supply.

- GAA Extraction, mechanical cleaning.**
from incl. intake
to incl. mechanical cleaning system outlet.
- GAC Piping and channel system.**
from excl. intake or
from excl. outlet of other system
to excl. inlet to other system.
- GAD Storage system.**
from incl. storage system inlet t
o incl. storage system outlet, incl. intake
and outfall.
- GAF Pump system.**
from incl. pump system suction nozzle
to incl. pump system discharge nozzle.
- GAV Lubricant supply system.**
- GAX Fluid supply system for control
and protection equipment.**
- GAY Control and protection equipment.**

GB Treatment system (carbonate hardness removal) incl. cooling tower make-up water treatment system.

- GBB Filtering, mechanical cleaning system.**
from incl. separation equipment inlet
to incl. separation equipment outlet.
- GBC Aeration, gas injection system.**
from excl. atmosphere or
from incl. gas supply.
- GBD Precipitation system (e.g. for carbonate hardness removal).**
from incl. precipitation equipment inlet
to incl. precipitation equipment outlet.
- GBE Acid proportioning system (e.g. for carbonate hardness removal).**
from incl. acid proportioning equipment
from excl. branch off chemicals supply system
to excl. inlet to other system.
- GBF Ion exchange, reverse osmosis system (e.g. for carbonate hardness removal).**
from incl. ion exchanger inlet or from incl.
isolating valve of chemicals supply system
or auxiliary fluid supply system upstream
of ion exchanger.
- GBG Evaporation system (e.g. for carbonate hardness removal).**
from incl. water inlet
to incl. steam outlet and
from incl. heating steam inlet
to incl. condensate outlet.

GBH Deaeration system.

from incl. deaerator or tank inlet
to incl. tank outlet incl. warm-up equipment
of vapour condenser.

GBJ Preheating, cooling system.

**from incl. preheater or cooler inlet
to incl. preheater or cooler outlet.**

GBK Piping system, temporary storage system, pump system for main fluid.

Piping system:

from excl. intake or
from excl. outlet of other system
to excl. inlet to other system
to incl. fluid treatment system outlet.

Temporary storage system:

from incl. temporary storage system inlet
to incl. temporary storage system outlet.

Pump system:

from incl. pump system suction nozzle
to incl. pump system discharge nozzle.

GBL Storage system outside fluid treatment system (if not part of another system).

from incl. inlet storage system
to excl. outlet storage system incl. intake
and outfall.

GBN Chemicals supply system.

from incl. intake or
from incl. storage tank
to excl. discharge into other system.

GBP Regeneration, flushing equipment.

from incl. system inlet
to excl. inlet to other system
from excl. chemicals or auxiliary fluid
supply system and flushing air supply
system to incl. regenerating, flushing
equipment.

GBQ Injection system for main fluid (for hardness stabilisation).

from incl. injection equipment
from excl. branch off chemicals supply system
to excl. inlet to other system.

GBR Flushing water and residues removal system incl. neutralization.

from excl. outlet of respective system
to excl. discharge into disposal system.

GBS Sludge thickening system.

from excl. outlet of respective system
to excl. discharge into other system.

GBT Heating, cooling and flushing fluid distribution system.

from incl. heating, cooling, flushing fluid
generation equipment or from excl. branch
off heating, cooling, flushing fluid supply
system
to excl. user and from excl. user.

WATER SUPPLY AND DISPOSAL

- GBV** Lubricant supply system.
GBX Fluid supply system for control and protection equipment.
GBY Control and protection equipment.

GC Treatment system (demineralization).

- GCB** **Filtering, mechanical cleaning system.**
 from incl. separation equipment inlet
 to incl. separation equipment outlet.
- GCC** **Aeration, gas injection system.**
 from excl. atmosphere or
 from incl. gas supply.
- GCD** **Precipitation system (e.g. for carbonate hardness removal).**
 from incl. precipitation equipment inlet
 to incl. precipitation equipment outlet.
- GCE** **Acid proportioning system (e.g. for carbonate hardness removal).**
 from incl. acid proportioning equipment or
 from excl. branch off chemicals supply system
 to excl. inlet to other system.
- GCF** **Ion exchange, reverse osmosis system (e.g. for demineralization).**
 from incl. ion exchanger inlet
 from incl. isolating valve of chemicals supply system or auxiliary fluid supply system upstream of ion exchanger.
- GCG** **Evaporation system (e.g. for demineralization).**
 from incl. water inlet
 to incl. steam outlet and
 from incl. heating steam inlet
 to incl. condensate outlet.
- GCH** **Deaeration system.**
 from incl. deaerator or tank inlet
 to incl. tank outlet incl. warm-up equipment of vapour condenser.
- GCI** **Preheating, cooling system.**
 from incl. preheater or cooler inlet
 to incl. preheater or cooler outlet.
- GCK** **Piping system, temporary storage system, pump system for main fluid.**
 Piping system:
 from excl. intake or
 from excl. outlet of other system
 to excl. inlet to other system
 to incl. fluid treatment system outlet.
 Temporary storage system:
 from incl. temporary storage system inlet
 to incl. temporary storage system outlet.
 Pump System:
 from incl. pump system suction nozzle
 to incl. pump system discharge nozzle.

- GCL** **Storage system outside fluid treatment system (if not part of another system).**
 from incl. inlet storage system
 to incl. outlet storage system incl. intake and outfall.
- GCN** **Chemicals supply system.**
 from incl. intake or
 from incl. storage tank
 to excl. discharge into other system.
- GCP** **Regeneration, flushing equipment.**
 from incl. system inlet
 to excl. inlet to other system
 from excl. chemicals or auxiliary fluid supply system and flushing air supply system
 to incl. regenerating, flushing equipment.
- Gcq** **Injection system for main fluid.**
 from incl. injection equipment or
 from excl. branch off chemicals supply system
 to excl. inlet to other system.
- GCR** **Flushing water and residues removal system incl. neutralization.**
 from excl. outlet of respective system
 to excl. discharge into disposal system.
- GCS** **Sludge thickening system.**
 from excl. outlet of respective system
 to excl. discharge into other system.
- GCT** **Heating, cooling and flushing fluid distribution system.**
 from incl. heating, cooling and flushing fluid generation equipment or
 from excl. branch off heating, cooling, flushing fluid supply system
 to excl. user and
 from excl. user.
- GCV** Lubricant supply system.
GcX Fluid supply system for control and protection equipment.
GcY Control and protection equipment.

GD Treatment system (others).

- GDB** **Filtering, mechanical cleaning system.**
 from incl. separation equipment inlet
 to incl. separation equipment outlet.
- GDC** **Aeration, gas injection system.**
 from excl. atmosphere or
 from incl. gas supply.
- GDD** **Precipitation system (e.g. for carbonate hardness removal).**
 from incl. precipitation equipment inlet
 to incl. precipitation equipment outlet.

WATER SUPPLY AND DISPOSAL

- GDE Acid proportioning system (e.g. for carbonate hardness removal).**
from incl. acid proportioning equipment or
from excl. branch off chemicals supply
system
to excl. inlet to other system.
- GDF Ion exchange, reverse osmosis system (e.g. for demineralization).**
from incl. ion exchanger inlet or
from incl. isolating valve of chemicals supply
system or auxiliary fluid supply system
upstream of ion exchanger.
- GDG Evaporation system (e.g. for demineralization).**
from incl. water inlet
to incl. steam outlet and
from incl. heating steam inlet
to incl. condensate outlet.
- GDH Deaeration system.**
from incl. deaerator or tank inlet
to incl. tank outlet incl. warm-up equipment
of vapour condenser.
- GDJ Preheating, cooling system.**
from incl. preheater or cooler inlet
to incl. preheater or cooler outlet.
- GDK Piping system, temporary storage system, pump system for main fluid.**
Piping system:
from excl. intake or
from excl. outlet of other system
to excl. inlet to other system
to incl. fluid treatment system outlet.
Temporary storage system:
from incl. temporary storage system inlet
to incl. temporary storage system outlet.
Pump System:
from incl. pump system suction nozzle
to incl. pump system discharge nozzle.
- GDL Storage system outside fluid treatment system (if not part of another system).**
from incl. inlet storage system
to incl. outlet storage system incl. intake
and outfall.
- GDN Chemicals supply system.**
from incl. intake or
from incl. storage tank
to excl. discharge into other system.
- GDP Regeneration flushing equipment.**
from incl. system inlet
to excl. inlet to other system
from excl. chemicals or auxiliary fluid supply
system and flushing air supply system
to incl. regenerating, flushing equipment.
- GDQ Injection system for main fluid.**
from incl. injection equipment or
from excl. branch off chemicals supply system
to excl. inlet to other system.
- GDR Flushing water and residues removal system incl. neutralization.**
from excl. outlet of respective system
to excl. discharge into disposal system.
- GDS Sludge thickening system.**
from excl. outlet of respective system
to excl. discharge into other system.
- GDT Heating cooling and flushing fluid distribution system.**
from incl. heating, cooling and flushing
fluid generation equipment or
from excl. branch off heating, cooling,
flushing fluid supply system
to excl. user and
from excl. user.
- GDV Lubricant supply system.**
- GDX Fluid supply system for control and protection equipment.**
- GDY Control and protection equipment.**
- GH Distribution systems (not drinking water).**
- GHA Distribution systems (not drinking water).**
(free for use for other kinds of water).
- GHB Distribution systems after treatment (carbonate hardness removal).**
to excl. inlet to other system.
- GHC Distribution systems after treatment (demineralization).**
to excl. inlet into other system.
- GHD Distribution systems after treatment (others).**
to excl. inlet into other system.
- GHE Distribution systems (not drinking water).**
(free for use for other kinds of water).
- GHF Distribution systems (not drinking water).**
(free for use for other kinds of water).
- GHG Distribution systems (not drinking water).**
(free for use for other kinds of water).
- GHH Distribution systems (not drinking water).**
(free for use for other kinds of water).
- GHJ Distribution systems (not drinking water).**
(free for use for other kinds of water).

WATER SUPPLY AND DISPOSAL

- GHK** Distribution systems (not drinking water).
(free for use for other kinds of water).
- GHL** Distribution systems (not drinking water).
(free for use for other kinds of water).
- GHM** Distribution systems for main machine sets.
(free for use for other kinds of water).
- GHN** Distribution systems (not drinking water).
(free for use for other kinds of water).
- GHP** Distribution systems (not drinking water).
(free for use for other kinds of water).
- GHQ** Distribution systems (not drinking water).
(free for use for other kinds of water).
- GHR** Distribution systems (not drinking water).
(free for use for other kinds of water).
- GHS** Distribution systems (not drinking water).
(free for use for other kinds of water).
- GHT** Distribution systems (not drinking water).
(free for use for other kinds of water).
- GHU** Distribution systems (not drinking water).
(free for use for other kinds of water).
- GHX** Fluid supply system for control and protection equipment.
- GHY** Control and protection equipment.
- GK** Drinking water supply.
- GKA** Receiving point.
- GKB** Storage, forwarding, distribution system.
- GKC** Drinking water supply in structures for switchgear.
- GKE** Drinking water supply in structures for conventional fuel supply and residue disposal.
- GKG** Drinking water supply in structures for water supply and discharge.
- GKH** Drinking water supply in structures for conventional heat generation.
- GKL** Drinking water supply in structures for steam, water, gas cycle.
- GKM** Drinking water supply in structures for main machine sets.
- GKN** Drinking water supply in structures for process energy supply systems.

- GKP** Drinking water supply in structures for circulation (cooling) water systems.
- GKQ** Drinking water supply in structures for circulatig (cooling) water systems (e.g. circulating water pumps and outfall).
- GKR** Drinking water supply in structures for circulatig (cooling) water systems (e.g. recirculation cooling).
- GKS** Drinking water supply in structures for ancillary systems.
- GKU** Drinking water supply in general service structures: transport, traffic, fencing, gardens and other purposes.
- GKX** Fluid supply system for control and protection equipment.
- GKY** Control and protection equipment.

GM Process drainage system.

- GMA** Central process drainage system.
- GMB** Process drainage system.
(free for use e.g. building-specific).
- GMC** Process drainage system in structures for switch gear.
- GMD** Process drainage system.
(free for use e.g. building-specific).
- GME** Process drainage system in structures for conventional fuel supply and residue disposal.
- GMF** Process drainage system for chemical flue gas treatment.
(free for use e.g. building-specific).
- GMG** Process drainage system in structures for water supply and discharge.
- GMH** Process drainage system in structures for conventional heat generation.
- GMJ** Process drainage system in structures for external systems (power plant specific).
(free for use e.g. building-specific).
- GMK** Process drainage system in general service structures.
(free for use e.g. building-specific).
- GML** Process drainage system in structures for steam, water, gas cycles.
- GMM** Process drainage system in structures for main machine sets.
- GMN** Process drainage system in structures for process energie supply.

WATER SUPPLY AND DISPOSAL

GMP	Process drainage system in structures for circulating (cooling) water systems (e.g. circulating water intake).	GNH	Deaeration system. from incl. deaerator or tank inlet to incl. tank outlet incl. warm-up equipment of vapour condenser.
GMQ	Process drainage system in structures for circulating (cooling) water systems (e.g. circulating water pumps and outfall).	GNJ	Preheating, cooling system. from incl. preheater or cooler inlet to incl. preheater or cooler outlet.
GMR	Process drainage system in structures for circulating (cooling) water systems (e.g. circulation cooling).	GNK	Piping system, temporary storage system, pump system for main fluid. Piping system: from excl. intake or from excl. outlet of other system to excl. inlet to other system to incl. fluid treatment system outlet. Temporary storage system: from incl. temporary storage system inlet to incl. temporary storage system outlet. Pump System: from incl. pump system suction nozzle to incl. pump system discharge nozzle.
GMS	Process drainage system in structures for ancillary systems.	GNL	Storage system outside fluid treatment system (if not part of another system). from incl. inlet storage system to incl. outlet storage system incl. intake and outfall.
GMT	Process drainage system in structures for auxiliary systems.	GNN	Chemicals supply system. from incl. intake or from incl. storage tank to excl. discharge into other system.
GMU	Process drainage system in general service structures: transport, traffic, fencing, gardens and other purposes.	GNP	Regeneration, flushing equipment. from incl. system inlet to excl. inlet to other system from excl. chemicals or auxiliary fluid supply system and flushing air supply system to incl. regenerating, flushing equipment.
GMX	Fluid supply system for control and protection equipment.	GNQ	Injection system for main fluid. from incl. injection equipment or from excl. branch off chemicals supply system to excl. inlet to other system.
GMY	Control and protection equipment.	GNR	Flushing water and residues removal system incl. neutralization. from excl. outlet of respective system to excl. discharge into disposal system.
GN	Process drains treatment system.	GNS	Sludge thickening system. from excl. outlet of respective system to excl. discharge into other system.
GNB	Filtering mechanical cleaning system. from incl. separation equipment inlet to incl. separation equipment outlet.	GNT	Heating, cooling and flushing fluid distribution system. from incl. heating, cooling and flushing fluid generation equipment or from excl. branch off heating, cooling, flushing fluid supply system to excl. user and from excl. user.
GNC	Aeration, gas injection system. from excl. atmosphere or from incl. gas supply.		
GND	Precipitation system (e.g. for carbonate hardness removal). from incl. precipitation equipment inlet to incl. precipitation equipment outlet.		
GNE	Acid proportioning system (e.g. for carbonate hardness removal). from incl. acid proportioning equipment or from excl. branch off chemicals supply system to excl. inlet to other system.		
GNF	Ion exchange system (e.g. for demineralization). from incl. ion exchanger inlet from incl. isolating valve of chemicals supply system or auxiliary fluid supply system upstream of ion exchanger.		
GNG	Evaporation system (e.g. for demineralization). from incl. feedwater inlet to incl. steam outlet and from incl. heating steam inlet to incl. condensate outlet.		

WATER SUPPLY AND DISPOSAL

GNV	Lubricant supply system.	GQX	Fluid supply system for control and protection equipment.
GNX	Fluid supply system for control and protection equipment.	GQY	Control and protection equipment.
GNY	Control and protection equipment.		
GQ	Domestic waste water collection and drainage systems.	GR	Domestic waste water treatment system.
GQA	Central domestic waste water collection and drainage systems.	GRB	Filtering, mechanical cleaning system. from incl. separation equipment inlet to incl. separation equipment outlet.
GQB	Domestic waste water collection and drainage systems, storage system.	GRC	Aeration, gas injection system. from excl. atmosphere or from incl. gas supply.
GQC	Domestic waste water collection and drainage systems in structures for switch gear.	GRD	Precipitation system (e.g. for carbonate hardness removal). from incl. precipitation equipment inlet to incl. precipitation equipment outlet.
GQE	Domestic waste water collection and drainage systems in structures for conventional fuel supply and residue disposal.	GRE	Acid proportioning system (e.g. for carbonate hardness removal). from incl. acid proportioning equipment or from excl. branch off chemicals supply system to excl. inlet to other system.
GQG	Domestic waste water collection and drainage systems in structures for water supply and discharge.	GRF	Ion exchange system (e.g. for demineralization). from incl. ion exchanger inlet from incl. isolating valve of chemicals supply system or auxiliary fluid supply system upstream of ion exchanger.
GQH	Domestic waste water collection and drainage systems in structures for conventional heat generation.	GRG	Evaporation system (e.g. for demineralization). from incl. feedwater inlet to incl. steam outlet and from incl. heating steam inlet to incl. condensate outlet.
GQL	Domestic waste water collection and drainage systems in structures for steam, water, gas cycle.	GRH	Deaeration system. from incl. deaerator or tank inlet to incl. tank outlet incl. warm-up equipment of vapour condenser.
GQM	Domestic waste water collection and drainage systems in structures for main machine sets.	GRJ	Preheating, cooling system. from incl. preheater or cooler inlet to incl. preheater or cooler outlet.
GQN	Domestic waste water collection and drainage systems in structures for process energy supply.	GRK	Piping system. temporary storage system, pump system for main fluid. Piping system: from excl. intake or from excl. outlet of other system to excl. inlet to other system to incl. fluid treatment system outlet. Temporary storage system: from incl. temporary storage system inlet to incl. temporary storage system outlet. Pump System: from incl. pump system suction nozzle to incl. pump system discharge nozzle.
GQP	Domestic waste water collection and drainage systems in structures for circulating (cooling) water systems (e.g. circulating water intake).		
GQQ	Domestic waste water collection and drainage systems in structures for circulating (cooling) water systems (e.g. circulating water pumps and outfall).		
GQS	Domestic waste water collection and drainage systems in structures for ancillary systems.		
GQT	Domestic waste water collection and drainage systems in structures for auxiliary systems.		
GQU	Domestic waste water collection and drainage systems in general service structures: transport, traffic, fencing, gardens and other purposes.		

WATER SUPPLY AND DISPOSAL

GRL	Storage system outside fluid treatment system (if not part of another system). from incl. inlet storage system to incl. outlet storage system incl. intake and outfall.	GTH	Water recovery from waste water. (free for use).
GRN	Chemicals supply system. from incl. intake or from incl. storage tank to excl. discharge into other system.	GTJ	Water recovery from waste water. (free for use).
GRP	Regeneration, flushing equipment. from incl. system inlet to excl. inlet to other system from excl. chemicals or auxiliary fluid supply system and flushing air supply system to incl. regenerating, flushing equipment.	GTK	Water recovery from waste water. (free for use).
GRQ	Injection system for main fluid. from incl. injection equipment or from excl. branch off chemicals supply system to excl. inlet to other system.	GTL	Water recovery from waste water. (free for use).
GRR	Flushing water and residues removal system incl. neutralization. from excl. outlet of respective system to excl. discharge into disposal system.	GTM	Water recovery from waste water. (free for use).
GRS	Sludge thickening system. from excl. outlet of respective system to excl. discharge into other system.	GTN	Water recovery from waste water. (free for use).
GRT	Heating, cooling and flushing fluid distribution system. from incl. heating, cooling and flushing fluid generation equipment or from excl. branch off heating, cooling, flushing fluid supply system to excl. user and from excl. user.	GTP	Water recovery from waste water. (free for use).
GRV	Lubricant supply system.	GTQ	Water recovery from waste water. (free for use).
GRX	Fluid supply system for control and protection equipment.	GTR	Water recovery from waste water. (free for use).
GRY	Control and protection equipment.	GTS	Water recovery from waste water. (free for use).
GT	Water recovery from waste water.	GTT	Water recovery from waste water. (free for use).
GTA	Water recovery from waste water. (free for use).	GTU	Water recovery from waste water. (free for use).
GTB	Water recovery from waste water. (free for use).	GTX	Fluid supply system for control and protection equipment.
GTC	Water recovery from waste water. (free for use).	GTY	Control and protection equipment.
GTD	Water recovery from waste water. (free for use).	GU	Rainwater collection and drainage systems incl. treatment system.
GTE	Water recovery from waste water. (free for use).	GUA	Central rainwater collection and drainage system.
GTF	Water recovery from waste water. (free for use).	GUB	Central rainwater treatment system.
GTG	Water recovery from waste water. (free for use).	GUC	Rainwater collection and drainage systems incl. treatment system in structures for switchgear.
		GUE	Rainwater collection and drainage systems incl. treatment system in structures for conventional fuel supply and residue disposal.
		GUG	Rainwater collection and drainage systems incl. treatment system in structures for water supply and discharge.
		GUH	Rainwater collection and drainage systems incl. treatment system in structures for conventional heat generation.
		GUL	Rainwater collection and drainage systems incl. treatment system in structures for steam, water, gas cycle.

WATER SUPPLY AND DISPOSAL

- GUM** Rainwater collection and drainage systems incl. treatment system in structures for main machine sets.
- GUN** Rainwater collection and drainage systems incl. treatment system in structures for process energy supply.
- GUP** Rainwater collection and drainage systems incl. treatment system in structures for circulating (cooling) water systems (e.g. circulating water intake).
- GUQ** Rainwater collection and drainage systems incl. treatment system in structures for circulating (cooling) water systems (e.g. circulating water pumps and outfall).
- GUS** Rainwater collection and drainage systems incl. treatment system in structures for ancillary systems.
- GUT** Rainwater collection and drainage systems incl. treatment system in structures for auxiliary systems.
- GUU** Rainwater collection and drainage systems incl. treatment system in general service structures: transport, traffic, fencing, gardens and other purposes.
- GUX** Fluid supply system for control and protection equipment.
- GUY** Control and protection equipment.

GV Lubricant supply system.

- GVA** Lubricant supply system.
(free for use).
- GVB** Lubricant supply system.
(free for use).
- GVC** Lubricant supply system.
(free for use).
- GVD** Lubricant supply system.
(free for use).
- GVE** Lubricant supply system.
(free for use).
- GVF** Lubricant supply system.
(free for use).
- GVG** Lubricant supply system.
(free for use).
- GVH** Lubricant supply system.
(free for use).
- GVJ** Lubricant supply system.
(free for use).
- GVK** Lubricant supply system.
(free for use).
- GVL** Lubricant supply system.
(free for use).

- GVM** Lubricant supply system.
(free for use).
- GVN** Lubricant supply system.
(free for use).
- GVP** Lubricant supply system.
(free for use).
- GVQ** Lubricant supply system.
(free for use).
- GVR** Lubricant supply system.
(free for use).
- GVS** Lubricant supply system.
(free for use).
- GVT** Lubricant supply system.
(free for use).
- GVU** Lubricant supply system.
(free for use).

GW Sealing fluid supply system.

- GWA** Sealing fluid supply system.
(free for use).
- GWB** Sealing fluid supply system.
(free for use).
- GWC** Sealing fluid supply system.
(free for use).
- GWD** Sealing fluid supply system.
(free for use).
- GWE** Sealing fluid supply system.
(free for use).
- GWF** Sealing fluid supply system.
(free for use).
- GWG** Sealing fluid supply system.
(free for use).
- GWH** Sealing fluid supply system.
(free for use).
- GWJ** Sealing fluid supply system.
(free for use).
- GWK** Sealing fluid supply system.
(free for use).
- GWL** Sealing fluid supply system.
(free for use).
- GWM** Sealing fluid supply system.
(free for use).
- GWN** Sealing fluid supply system.
(free for use).
- GWP** Sealing fluid supply system.
(free for use).
- GWQ** Sealing fluid supply system.
(free for use).
- GWR** Sealing fluid supply system.
(free for use).
- GWS** Sealing fluid supply system.
(free for use).
- GWT** Sealing fluid supply system.
(free for use).
- GWU** Sealing fluid supply system.
(free for use).

WATER SUPPLY AND DISPOSAL

GX Fluid supply system for control and protection equipment.

GXA Fluid supply system for control and protection equipment.
(free for use).

GXB Fluid supply system for control and protection equipment.
(free for use).

GXC Fluid supply system for control and protection equipment.
(free for use).

GXD Fluid supply system for control and protection equipment.
(free for use).

GXE Fluid supply system for control and protection equipment.
(free for use).

GXF Fluid supply system for control and protection equipment.
(free for use).

GXG Fluid supply system for control and protection equipment.
(free for use).

GXH Fluid supply system for control and protection equipment.
(free for use).

GXJ Fluid supply system for control and protection equipment.
(free for use).

GXK Fluid supply system for control and protection equipment.
(free for use).

GXL Fluid supply system for control and protection equipment.
(free for use).

GXM Fluid supply system for control and protection equipment.
(free for use).

GXN Fluid supply system for control and protection equipment.
(free for use)

GXP Fluid supply system for control and protection equipment.
(free for use).

GXQ Fluid supply system for control and protection equipment.
(free for use)

GXR Fluid supply system for control and protection equipment.
(free for use)

GXS Fluid supply system for control and protection equipment.
(free for use).

GXT Fluid supply system for control and protection equipment.
(free for use).

GXU Fluid supply system for control and protection equipment.
(free for use).

GY Control and protection equipment.

GYA Control and protection equipment.
(free for use).

GYB Control and protection equipment.
(free for use).

GYC Control and protection equipment.
(free for use).

GYD Control and protection equipment.
(free for use).

GYE Control and protection equipment.
(free for use).

GYF Control and protection equipment.
(free for use).

GYG Control and protection equipment.
(free for use).

GYH Control and protection equipment.
(free for use).

GYJ Control and protection equipment.
(free for use).

GYK Control and protection equipment.
(free for use).

GYL Control and protection equipment.
(free for use).

GYM Control and protection equipment.
(free for use).

GYN Control and protection equipment.
(free for use).

GYP Control and protection equipment.
(free for use).

GYQ Control and protection equipment.
(free for use).

GYR Control and protection equipment.
(free for use).

GYS Control and protection equipment.
(free for use).

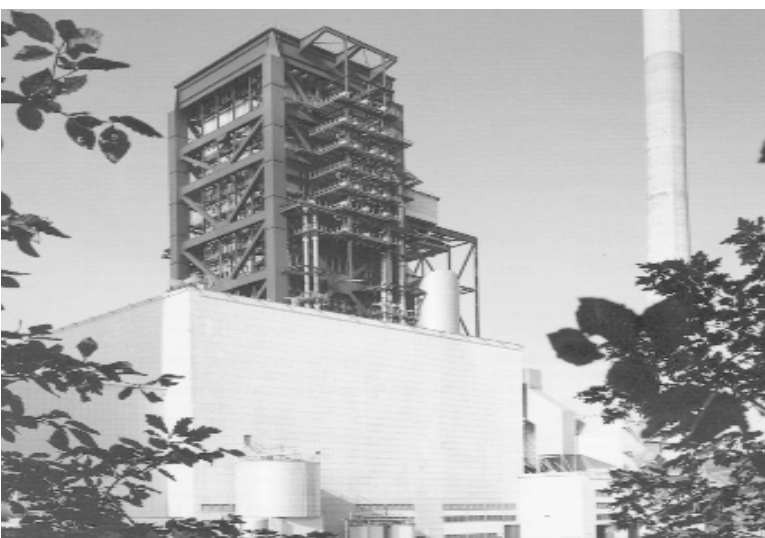
GYT Control and protection equipment.
(free for use).

GYU Control and protection equipment.
(free for use).



CONVENTIONAL HEAT GENERATION (HRSG)

- HA Pressure system.**
- HB Support structure, enclosure, steam generator interior.**
- HC Fireside heat transfer surface cleaning equipment.**
- HD Ash and slag removal.**
- HF Bunker, feeder and pulverizing system.**
- HH Main firing system (electric-powered as well).**
- HJ Ignition firing equipment (if separate), supplementary firing as well (e.g. for HRSG).**
- HL Combustion air system (primary air, secondary air).**
- HM Gas heating system (for closed cycle).**
- HN Flue gas exhaust (without flue gas treatment).**
- HR Chemical flue gas treatment system incl. residues removal, adsorptive process.**
- HS Chemical flue gas treatment system incl. residues removal, catalytic process.**
- HT Chemical flue gas treatment system incl. residues removal, absorptive process.**
- HU Flue gas reheating system.**
- HV Lubricant supply system.**
- HW Sealing fluid supply system.**
- HX Fluid supply system for control and protection equipment.**
- HY Control and protection equipment.**



CONVENTIONAL HEAT GENERATION (HRSG)

HA Pressure system.

- HAA LP part-flow feed heating system (flue-gas-heated).**
from incl. LP part-flow feed heating system inlet
to incl. LP part-flow feed heating system outlet.
- HAB HP part-flow feed heating system (flue-gas-heated).**
from incl. HP part-flow feed heating system inlet
to incl. HP part-flow feed heating system outlet.
- HAC Economizer system.**
from incl. boiler inlet header
to excl. evaporator inlet incl. control and auxiliary heat transfer surfaces.
Low pressure: HAC01 - 19
Intermediate pressure: HAC20 - 49
High pressure: HAC50 - 99.
- HAD Evaporator system.**
from incl. evaporator system inlet
to incl. evaporator system outlet and
to excl. water/steam separator in once-through boiler
to excl. water/steam separator and water collecting vessel in drum boiler.
Low pressure: HAD01 - 19
Intermediate pressure: HAD20 - 49
High pressure: HAD50 - 99.
- HAG Circulation system.**
from excl. water/steam separator and
from incl. water/steam separator (start-up vessel) in once-through boiler or
from excl. water/steam separator and water collecting vessel in drum boiler.
to excl. inlet to heat transfer surface system (removal systems are classified under *L*)
or
to excl. feedwater system.
- HAH Superheater system.**
from excl. evaporator system outlet
to incl. boiler outlet header.
Low pressure: HAH01 - 19
Intermediate pressure: HAH20 - 49
High pressure: HAH50 - 99.
- HAJ Reheat system.**
from incl. reheater inlet header
to incl. reheater outlet header.
- HAK Secondary reheat system.**
from incl. second reheater inlet header
to incl. second reheater outlet header.
- HAM Triflux system.**
from incl. triflux system inlet
to incl. triflux system outlet

HAN Pressure system drainage and venting systems.

from incl. collecting point or
from excl. final drain/vent valve
to excl. discharge into other system.
Low pressure: HAN01 - 19
Intermediate pressure: HAN20 - 49
High pressure: HAN50 - 99.

HAV Lubricant supply system.

HAW Sealing fluid supply system.

HAX Fluid supply system for control and protection equipment.

HAY Control and protection equipment.

HB Support structure, enclosure, steam generator interior.

HBA Frame incl. foundations.

HBB Enclosures, insulations.

HBC Brick linings incl. insulating brickwork.

HBD Platforms, stairways.

HBK Steam generator interior.

from excl. inlet duct
to excl. flue gas exhaust.

HC Fireside heat transfer surface cleaning equipment.

HCA Air sootblowing system.

from excl. branch off supply system.

HCB Steam sootblowing system.

from excl. branch off supply system.

HCC Water sootblowing system.

from excl. branch off supply system.

HCD Flushing equipment.

from excl. branch off supply system.

HCE Rapping gear.

HCF Shot cleaning system.

HCH Sonic cleaning system.

HCV Lubricant supply system.

HCW Sealing fluid supply system.

HCX Fluid supply system for control and protection equipment.

HCY Control and protection equipment.

HD Ash and slag removal.

HDA Furnace ash and slag removal.

from incl. removal equipment
to excl. discharge into disposal system.

HDB Bed ash return.

from excl. furnace outlet or
from excl. bed ash removal outlet
to excl. furnace inlet.

CONVENTIONAL HEAT GENERATION (HRSG)

HDC	Flue dust return system. from excl. removal equipment for heat recovery surfaces or from excl. boiler flue gas dust separation equipment or from excl. branch for flue dust return system to excl. flue dust burner or to excl. inlet to other system..	HFG	Pulverized coal temporary storage bunker after central pulverizing system (indirect firing). from excl. pulverizing system outlet to incl. temporary storage bunker outlet, incl. rotary vane feeder.
HDD	Mechanical dust handling system. from excl. flue gas duct inlet to excl. flue gas duct outlet or to excl. discharge to other system.	HFV	Lubricant supply system.
HDE	Electrstatic precipitator. from excl. flue gas duct inlet to excl. flue gas duct outlet or to excl. discharge to other system.	HFW	Sealing fluid supply system.
HDF	Cyclone dust removal and return system. from excl. flue gas duct inlet to excl. flue gas duct outlet to excl. discharge to other system.	HFX	Fluid supply system for control and protection equipment.
HDV	Lubricant supply system.	HFY	Control and protection equipment.
HDW	Sealing fluid supply system.	HH	Main firing system (electric-powered as well).
HDX	Fluid supply system for control and protection equipment.	HHA	Main burners. from incl. respective fuel-side and air-side burner inlet.
HDY	Control and protection equipment.	HHB	Retarded combustion grate. from incl. retarded combustion grate inlet to excl. inlet to other system.
HF	Bunker, feeder and pulverizing system.	HHC	Grate combustion system. from incl. fuel receiving point or from incl. grate combustion inlet to excl. inlet to other system.
HFA	Bunker for pulverizing system. from excl. receiving point to incl. outlet	HHD	Other burner equipment (e.g. vapour burner, flue dust burner). from incl. inlet to incl. outlet.
HFB	Feeder system. from excl. bunker outlet to excl. oversize reject shaft or pulverizing system.	HHE	Pulverized coal bin, forwarding and distribution system. from excl. pulverizing system outlet or from excl. pulverized coal bin outlet after central pulverizing system (indirect feed) *HFG* or from excl. outlet of other system to excl. main burner equipment.
HFC	Pulverizing system (incl. classifier). from incl. pulverizing system inlet to excl. pulverizing coal line.	HHF	Oil temporary storage, pump and distribution system. from excl. branch off main supply line or from incl. temporary storage tank to excl. main burner equipment.
HFD	Flue gas return system. from excl. outlet of other system to excl. pulverizing system	HHG	Gas pressure reduction, distribution system. from excl. branch off main supply line to excl. main burner equipment.
HFE	Mill air system, carrier air system. from incl. air inlet or from excl. branch off *HLA* to excl. downcomer shaft or to excl. pulverizing system or to excl. flue gas recirculation system.	HHH	Temporary storage, forwarding and distribution system for other fuels, fluid 1.
HFF	Vapour/exhaust gas system. from excl. separation equipment to excl. other system.	HHJ	Temporary storage, forwarding and distribution system for other fuels, fluid 2.
		HHK	Temporary storage, forwarding and distribution system for other fuels, fluid 3.

CONVENTIONAL HEAT GENERATION (HRSG)

HHL	Combustion air supply system. from incl. branch off ducting system (*HLA*) to excl. user.	HJE	Pulverized coal bin, forwarding and distribution system. from excl. pulverizing system outlet or from excl. pulverized coal temporary storage bunker outlet (if central pulverizing system, * HFG*) or from excl. outlet of other system to excl. ignition burner equipment.
HHM	Atomizer medium supply system (steam). from excl. branch off supply system to excl. user.	HJF	Oil temporary storage, pump and distribution system. from excl. branch off main supply line or from incl. temporary storage tank to excl. ignition burner equipment.
HHN	Atomizer medium supply system (air). from excl. branch off supply system to excl. user.	HJG	Gas pressure reduction, distribution system. from excl. branch off main supply line to excl. supplementary burner equipment.
HHP	Coolant supply system (steam). from excl. branch off supply system to excl. user.	HJL	Combustion air supply system. from incl. branch off ducting system (*HLA*) or from incl. air inlet, incl. fan to excl. user.
HHQ	Coolant supply system (air). from excl. branch off supply system to excl. user.	HJM	Atomizer medium supply system (steam). from excl. branch off supply system to excl. user.
HHR	Purging medium supply system (steam). from excl. branch off supply system to excl. user.	HJN	Atomizer medium supply system (air). from excl. branch off supply system to excl. user.
HHS	Purging medium supply system (air). from excl. branch off supply system to excl. user.	HJP	Cooling supply system (steam). from excl. branch off supply system to excl. user.
HHT	Heating medium supply system (steam). from excl. branch off supply system to excl. user and from excl. user to excl. inlet to other system.	HJQ	Cooling supply system (air). from excl. branch off supply system to excl. user.
HHU	Heating medium supply system (hot water). from excl. branch off supply system to excl. user and from excl. user to excl. inlet to other system.	HJR	Purging medium supply system (steam). from excl. branch off supply system to excl. user.
HHV	Lubricant supply system.	HJS	Purging medium supply system (air). from excl. branch off supply system to excl. user.
HHW	Sealing fluid supply system.	HJT	Heating medium supply system (steam). from excl. branch off supply system to excl. user and from excl. user to excl. inlet to other system.
HHX	Fluid supply system for control and protection equipment.	HJU	Heating medium supply system (hot water). from excl. branch off supply system to excl. user and from excl. user to excl. inlet to other system.
HHY	Control and protection equipment.		
HHZ	Electric heating system.		
HJ	Ignition firing equipment (if separate), supplementary firing as well (e.g. for HRSG).		
HJA	Ignition burners, supplementary burners for HRSG. from incl. resp. fuel-side and air-side burner inlet.		

CONVENTIONAL HEAT GENERATION (HRSG)

HJV	Lubricant supply system.	HMD	Reheat system. from incl. reheater inlet header to incl. reheater outlet header.
HJW	Sealing fluid supply system.	HMV	Lubricant supply system.
HJX	Fluid supply system for control and protection equipment.	HMW	Sealing fluid supply system.
HJY	Control and protection equipment.	HMX	Fluid supply system for control and protection equipment.
HL	Combustion air system (primary air, secondary air).	HMY	Control and protection equipment.
HLA	Ducting system. from incl. air inlet to excl. branch off firing system or mill air system, carrier air system to excl. branch to *HFE*, excl. fan system, air heater system and gas turbine exhaust gas system.	HN	Flue gas exhaust (without flue gas treatment).
HLB	Fan system, forced-draught fan system. from incl. fan system inlet to excl. fan system outlet.	HNA	Ducting system. from excl. boiler outlet or from excl. outlet of other system to excl. smoke stack, excl. air heater, flue gas dust handling system, induced-draught fan system, gas scrubber system, chemical flue gas treatment system.
HLC	External air heating system (not flue-gas-heated). from incl. heater inlet to excl. heater outlet.	HNC	Induced-draught fan system. from incl. induced-draught fan system inlet to incl. induced-draught fan system outlet.
HLD	Air heating system (flue-gas-heated). from incl. heater inlet to incl. heater outlet.	HNE	Smoke stack system (chimney). from incl. inlet.
HLU	Air pressure relief system. Task: controlled relief of pressure in combustion air system.	HNF	Flue gas recirculation system. from excl. branch off main flue gas exhaust system to excl. inlet to other system, incl. fan system.
HLV	Lubricant supply system.	HNU	Flue gas pressure relief system. Task: controlled relief of pressure in flue gas exhaust system.
HLW	Sealing fluid supply system.	HNV	Lubricant supply system.
HLX	Fluid supply system for control and protection equipment.	HNW	Sealing fluid supply system.
HLY	Control and protection equipment.	HNX	Fluid supply system for control and protection equipment.
HM	Gas heating system (for closed cycle).	HNH	Control and protection equipment.
HMA	Primary heater (primary convection section). from incl. cold gas inlet header or from excl. heat exchanger cold gas outlet to incl. primary heater outlet or to excl. mixing header inlet.	HR	Chemical flue gas treatment system incl.residues removal, adsorptive process.
HMB	Radiation section. from incl. mixing header inlet to incl. radiation section outlet or to excl. secondary heater inlet header.	HRA	Flue gas ducting system within *HR*. from excl. *HNA*, to excl. inlet to *HNA*.
HMC	Secondary heater (second convection section). from incl. inlet header to incl. hot gas outlet header.	HRB	Flue gas-side heat exchange. from incl. inlet to incl. outlet.
		HRC	Flue gas fan system. from incl. inlet to incl. outlet.
		HRD	Adsorber (reactor) plant. from incl. inlet to incl. outlet.

CONVENTIONAL HEAT GENERATION (HRSG)

HRE	Flue gas side cleaning equipment. from excl. branch-off supply system.	HSD	Reactor (reduction). from incl. inlet to incl. outlet.
HRH	Residues separator. from incl. inlet to incl. outlet.	HSE	Converter (oxidation). from incl. inlet to incl. outlet.
HRJ	Fresh coke supply system incl. storage. from excl. receiving point to excl. fresh coke distribution or to excl. fresh coke treatment.	HSF	Flue gas-side cleaning equipment for reactor. from excl. branch off supply system.
HRK	Fresh coke treatment and distribution. from excl. fresh coke supply system to excl. reactor.	HSG	Reduction agent dilution system. from excl. outlet of other system or from incl. supply system to excl. reduction agent treatment system.
HRL	Water supply and disposal system.	HSJ	(Residues) separator. from incl. inlet to incl. outlet.
HRM	Coke removal system. from excl. reactor to excl. used coke transport or to excl. used coke treatment.	HSK	Reduction agent supply system incl. storage. from excl. reduction agent supply to incl. reduction agent injection incl. coolant inlet.
HRN	Used coke transport incl. storage. from excl. used coke removal or from excl. used coke treatment to excl. inlet to disposal system.	HSK	Reduction agent treatment and distribution system. from excl. reduction agent supply to incl. reduction agent injection incl. coolant inlet.
HRP	Used coke treatment and disposal. from excl. used coke transport to excl. used coke transport to excl. delivery.	HSL	Water supply and disposal system.
HRQ	Dust exhaustion and disposal. from incl. inlet to incl. delivery.	HSN	Chemicals and additives supply system.
HRR	Inerting system. from incl. storage. to excl. inlet to another system.	HSN	Drainage systems. Task: water collecting, storage, return.
HRV	Lubricant supply system.	HSP	Flyash collecting (incl. filtering) and removal system. from incl. separator/filter or from excl. flue gas ducting system to excl. inlet to disposal system.
HRW	Sealing fluid supply system.	HSQ	Sprinkler system incl. drainage. from incl. inlet to excl. inlet to other system.
HRX	Fluid supply system for control and protection equipment.	HSR	Oxidizing agent treatment and distribution system. from excl. outlet converter to excl. inlet converter.
HRY	Control and protection equipment.	HSS	(Residues) forwarding, storage, loading system. from excl. outlet residues separator.
HS	Chemical flue gas treatment system incl. residues removal, catalytic process.	HST	Flushing fluid system incl. supply. Task: flushing of reducing agent systems.
HSA	Flue gas ducting system within *HS*. from excl. *HNA* to incl. evaporator system outlet and to excl. inlet to *HNA*.	HSU	Heating fluid system. from excl. heating fluid supply to excl. evaporator inlet from excl. evaporator outlet to excl. inlet to other system.
HSB	Flue gas-side heat exchanger, gas heater (not *HU*). from incl. inlet to incl. outlet.	HSV	Lubricant supply system.
HSC	Flue gas fan system. from incl. inlet to incl. outlet.	HSW	Sealing fluid supply system.
		HSX	Fluid supply system for control and protection equipment.

CONVENTIONAL HEAT GENERATION (HRSG)

HSY	Control and protection equipment.	HTW	Sealing fluid supply system.
HT	Chemical flue gas treatment system incl. residues removal, absorptive process.	HTX	Fluid supply system for control and protection equipment.
HTA	Flue gas ducting system within *HT*. from excl. *HNA* to excl. inlet to *HNA*.	HTY	Control and protection equipment.
HTB	Flue gas-side heat exchanger, gas heater (not *HU*). from incl. inlet to incl. outlet.	HU	Flue gas reheating system.
HTC	Flue gas fan system. from incl. inlet to incl. outlet.	HUA	Flue gas reheating system. (free for use).
HTD	Flue gas scrubbing system. from incl. flue gas inlet to incl. moisture separator outlet.	HUB	Flue gas reheating system. (free for use).
HTE	Flue gas cleaning and filtering system. Task: additional cleaning within *HT*; not belonging to *HP*, *HQ* and *HR*.	HUC	Flue gas reheating system. (free for use).
HTF	Absorption cycle. from incl. inlet to incl. outlet.	HUD	Flue gas reheating system. (free for use).
HTG	Oxidation system incl. supply system. to excl. user or scrubber.	HUE	Flue gas reheating system. (free for use).
HTJ	Absorbent supply system incl. storage system. to excl. mashing (*HTK*).	HUF	Flue gas reheating system. (free for use).
HTK	Absorbent preparation and distribution system. from incl. mashing, slaking to excl. user or scrubber.	HUG	Flue gas reheating system. (free for use).
HTL	Piping system for discharge of solids incl. water removal and return excl. thickening and solids dewatering systems. incl. water removal and return excl. thickening and solids dewatering systems.	HUH	Flue gas reheating system. (free for use).
HTM	Thickening and solids dewatering system. from incl. inlet to incl. outlet.	HUJ	Flue gas reheating system. (free for use).
HTN	Solids drying, compacting system.	HUK	Flue gas reheating system. (free for use).
HTP	(Solids/ product) forwarding, storage, loading system.	HUL	Flue gas reheating system. (free for use).
HTQ	Water supply and disposal system.	HUM	Flue gas reheating system. (free for use).
HTS	Chemicals and additives supply system.	HUN	Flue gas reheating system. (free for use).
HTT	Drainage systems. Task: water collecting, storage, return.	HUP	Flue gas reheating system. (free for use).
HTV	Lubricant supply system.	HUQ	Flue gas reheating system. (free for use).
		HUR	Flue gas reheating system. (free for use).
		HUS	Flue gas reheating system. (free for use).
		HUT	Flue gas reheating system. (free for use).
		HUU	Flue gas reheating system. (free for use).
		HUV	Lubricant supply system.
		HUW	Sealing fluid supply system.
		HUX	Fluid supply system for control and protection equipment.
		HUY	Control and protection equipment.

CONVENTIONAL HEAT GENERATION (HRSG)

HV Lubricant supply system.

- HVA Lubricant supply system.**
(free for use).
- HVB Lubricant supply system.**
(free for use).
- HVC Lubricant supply system.**
(free for use).
- HVD Lubricant supply system.**
(free for use).
- HVE Lubricant supply system.**
(free for use).
- HVF Lubricant supply system.**
(free for use).
- HVG Lubricant supply system.**
(free for use).
- HVH Lubricant supply system.**
(free for use).
- HVJ Lubricant supply system.**
(free for use).
- HVK Lubricant supply system.**
(free for use).
- HVL Lubricant supply system.**
(free for use).
- HVM Lubricant supply system.**
(free for use).
- HVN Lubricant supply system.**
(free for use).
- HVP Lubricant supply system.**
(free for use).
- HVQ Lubricant supply system.**
(free for use).
- HVR Lubricant supply system.**
(free for use).
- HVS Lubricant supply system.**
(free for use).
- HVT Lubricant supply system.**
(free for use).
- HVU Lubricant supply system.**
(free for use).

HW Sealing fluid supply system.

- HWA Sealing fluid supply system.**
(free for use).
- HWB Sealing fluid supply system.**
(free for use).
- HWC Sealing fluid supply system.**
(free for use).
- HWD Sealing fluid supply system.**
(free for use).
- HWE Sealing fluid supply system.**
(free for use).
- HWF Sealing fluid supply system.**
(free for use).

HWG Sealing fluid supply system. (free for use).

HWH Sealing fluid supply system. (free for use).

HWJ Sealing fluid supply system. (free for use).

HWK Sealing fluid supply system. (free for use).

HWL Sealing fluid supply system. (free for use).

HWM Sealing fluid supply system. (free for use).

HWN Sealing fluid supply system. (free for use).

HWP Sealing fluid supply system. (free for use).

HWQ Sealing fluid supply system. (free for use).

HWR Sealing fluid supply system. (free for use).

HWS Sealing fluid supply system. (free for use).

HWT Sealing fluid supply system. (free for use).

HWU Sealing fluid supply system. (free for use).

HX Fluid supply system for control and protection equipment.

HXA Fluid supply system for control and protection equipment. (free for use).

HXB Fluid supply system for control and protection equipment. (free for use).

HXC Fluid supply system for control and protection equipment. (free for use).

HXD Fluid supply system for control and protection equipment. (free for use).

HY Control and protection equipment.

HYA Control and protection equipment. (free for use).

HYB Control and protection equipment. (free for use).

HYC Control and protection equipment. (free for use).

HYD Control and protection equipment. (free for use).

HYE Control and protection equipment. (free for use).

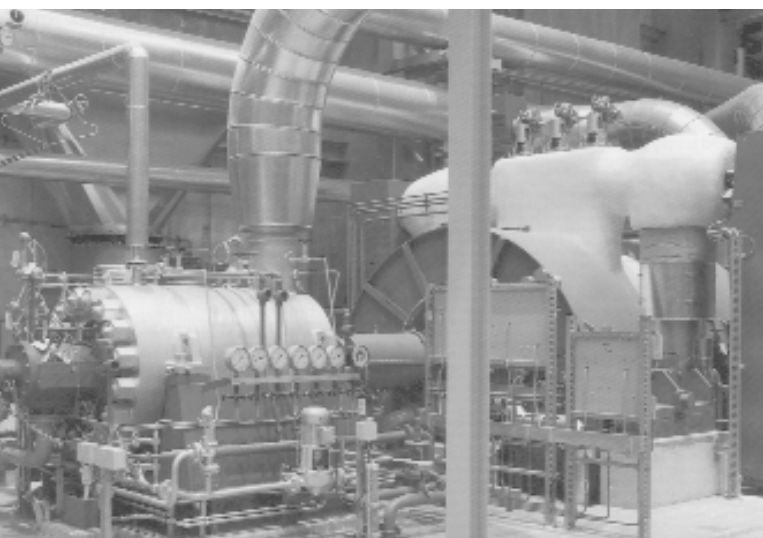
CONVENTIONAL HEAT GENERATION (HRSG)

- HYF Control and protection equipment.**
(free for use).
- HYG Control and protection equipment.**
(free for use).
- HYH Control and protection equipment.**
(free for use).
- HYJ Control and protection equipment.**
(free for use).
- HYK Control and protection equipment.**
(free for use).
- HYL Control and protection equipment.**
(free for use).
- HYM Control and protection equipment.**
(free for use).
- HYN Control and protection equipment.**
(free for use).
- HYP Control and protection equipment.**
(free for use).
- HYQ Control and protection equipment.**
(free for use).
- HYR Control and protection equipment.**
(free for use).
- HYS Control and protection equipment.**
Dummies for El. consumer No.91-99
eg. HYS91 GH001.
- HYT Control and protection equipment.**
(free for use).
- HYU Control and protection equipment.**
(free for use).



STEAM-WATER CYCLE

- LA Feedwater system.**
- LB Steam system.**
- LC Condensate system.**
- LD Condensate polishing plant.**
- LF Common installations for steam-water cycle.**
- LK Gas system (closed cycle).**
- LL Gas cleaning system (only for closed cycle).**
- LN Water impounding works for hydroelectric power plant.**
- LP Intake system, upperwater system for hydroelectric power plant.**
- LQ Tail-race system, underwater system for hydroelectric power plant.**
- LS Common installations for hydroelectric power plant.**
- LV Lubricant supply system.**
- LW Sealing fluid supply system for steam, water, gas cycles.**
- LX Fluid supply system for control and protection equipment.**
- LY Control and protection equipment.**



STEAM-WATER CYCLE

LA Feedwater system.

LAA Storage, deaeration (incl. feedwater tank).

from incl. deaerator or tank inlet
to incl. tank outlet incl. warm-up equipment
and vapour condenser.

LAB Feedwater piping system (excl. feedwater pump and feedwater heating system).

from excl. feedwater tank outlet
to excl. boiler inlet header or heat
exchanger.

Low pressure: LAB01 - 19

Intermediate pressure: LAB20 - 49

High pressure: LAB50 - 99.

LAC Feedwater pump system.

from incl. pump system suction nozzles
to incl. pump system discharge nozzles.

Low pressure: LAC01 - 19

Intermediate pressure: LAC20 - 49

High pressure: LAC50 - 99.

LAD HP feed water heating system.

from incl. respective feed heater inlet
to incl. respective feed heater outlet
incl. de-superheater and cooler.

LAE HP desuperheating spray system.

from excl. branch off feedwater piping
system
to excl. user.

LAF IP desuperheating spray system.

from excl. pump system discharge nozzle
or
from excl. branch off other system
to excl. user.

LAH Start-up and shutdown piping system.

from excl. outlet of or branch off feedwater
system
to excl. feedwater piping system or other
system.

LAJ Start-up and shutdown pump system.

from incl. pump system suction nozzle
to incl. pump system discharge nozzle.

LAR Emergency feed water piping system incl. storage (excl. emergency feed water pump system).

from excl. branch off another system
to excl. feed water piping system inlet.

LAS Emergency feed water pump system.

from incl. pump system suction nozzle
to incl. pump system discharge nozzle.

LAT Back-up emergency feed water system.

from excl. branch off another system
to excl. feed water piping system inlet.

LAV Lubricant supply system.

LAW Sealing fluid supply system.

LAX Fluid supply system for control and protection equipment.

LAY Control and protection equipment.

LB Steam system.

LBA Main steam piping system.

from excl. boiler outlet or
from excl. heat exchanger
to excl. turbine main stop valve or HP
reducing station or turbine bypass or other
user (system).

Low pressure: LBA01 - 19

Intermediate pressure: LBA20 - 49

High pressure: LBA50 - 99.

LBB Hot reheat piping system.

from excl. reheater or moisture
separator/reheater outlet
to excl. intercept valve or turbine inlet or
turbine bypass or other user (system).

LBC Cold reheat piping system.

from excl. turbine outlet or high pressure
reducing station
to excl. reheater inlet excl. moisture
separator or user (system).

LBD Extraction piping system.

from excl. branch off crossover line
to excl. user (system).

LBE Back-pressure piping system.

from excl. turbine outlet
to excl. user (system).

LBF Overpressure limitation and safety equipment incl. water injection and pressure measuring device.

from excl. branch off *LAE*/*LAF*, *LBA*
to excl. inlet to *LBC*.

LBG Auxiliary steam piping system.

from excl. receiving point from other system
to excl. user (system).

LBH Start-up steam system, shutdown steam system.

from excl. boiler outlet or
from excl. branch off main steam line, incl.
start-up condenser or
from excl. outlet of other system
to excl. inlet to other systems.

Low pressure: LBH01 - 19

Intermediate pressure: LBH20 - 49

High pressure: LBH50 - 99.

STEAM-WATER CYCLE

LBQ	Extraction steam piping system for feed water heating. from excl. turbine outlet or branch off other system to excl. feed water heating system or user (system).	LCE	Condensate desuperheating spray system. from excl. branch off main condensate piping system or from excl. branch off auxiliary steam turbine condensate piping system to excl. user.
LBR	Piping system for auxiliary (feed water pump) steam turbine. from excl. branch off main steam turbine or from excl. branch off other system to excl. auxiliary (feed water pump) steam turbine isolating valve from excl. auxiliary (feed water pump) steam turbine outlet to excl. inlet to other system.	LCF	Auxiliary (feed water pump) steam turbine condensate piping system. from excl. (auxiliary) condenser outlet to excl. inlet to other system excl. auxiliary steam turbine condensate pump system.
LBS	Extraction steam piping system for main condensate heating. from excl. turbine outlet or branch off other system to excl. LP feedwater heating system or deaerator or user (system).	LCG	Auxiliary (feed water pump) steam turbine condensate pump system. from incl. pump system suction nozzle to incl. pump system discharge nozzle.
LBT	Emergency condensing system. from excl. steam generator outlet or from excl. branch off main steam piping system incl. condenser to excl. inlet to other system.	LCH	HP heater drains system. from excl. heater outlet to excl. inlet to other systems.
LBU	Common dump line. eg. Blow-out.	LCJ	LP heater drains system. from excl. heater outlet to excl. inlet to other systems.
LBV	Lubricant supply system.	LCL	Steam generator drain system. from excl. branch off pressure system or from incl. start-up flash tank to excl. inlet to other systems.
LBW	Sealing fluid supply system.	LCM	Clean drains system (collecting and return system). from incl. collecting tank or from excl. final drain valve or from excl. inlet from other collecting system to excl. inlet to other systems.
LBX	Fluid supply system for control and protection equipment.	LCN	Auxiliary steam condensate system (collecting and return system). from excl. (steam) user to excl. inlet to other systems.
LBY	Control and protection equipment.	LCP	Standby condensate system, incl. storage and pump system. from excl. branch off other system to excl. inlet to other systems.
LC	Condensate system.	LCQ	Steam generator blowdown system. from excl. steam generator outlet from incl. blowdown flash tank (caustic flash tank) to excl. inlet to other systems
LCA	Main condensate piping system (excl. main condensate pump system, LP feedwater heating system, condensate polishing plant). from excl. condenser outlet to excl. deaerator inlet or to excl. feedwater pump system (in plants without feedwater tank).	LCR	Standby condensate distribution system. from excl. branch off other system to excl. inlet to other systems.
LCB	Main condensate pump system. from incl. pump system suction nozzle to incl. pump system discharge nozzle.	LCV	Lubricant supply system.
LCC	Main condensate heating system. from incl. heater inlet to incl. heater outlet incl. desuperheater and cooler.		

LCW Sealing and cooling condensate system.

from excl. branch off other system

to excl. user, incl. recirculation.

LCX Fluid supply system for control and protection equipment.

LCY Control and protection equipment.

Dummies for El. consumer No.91-99
eg.LCY91 GH001.

LD Condensate polishing plant.

LDA Fluid treatment extraction system (if not part of another system).

from excl. fluid treatment system outlet
to excl. inlet into other system.

LDB Filtering, mechanical cleaning.

from incl. separation equipment inlet
to incl. separation equipment outlet.

LDC Aeration, gas injection system.

from excl. atmosphere or
from incl. gas supply.

LDD Electromagnetic polishing system.

from incl. electromagnetic polishing system inlet
to incl. electromagnetic polishing system outlet.

LDE Acid proportioning system (e.g. for carbonate hardness removal).

from incl. acid proportioning equipment or
from excl. branch off chemicals supply system
to excl. inlet to other system.

LDI Ion exchange, reverse osmosis system (e.g. for demineralization).

from incl. ion exchanger inlet
to incl. ion exchanger outlet.

LDG Evaporation system (e.g. for demineralization).

from incl. feedwater inlet
to incl. steam outlet and
from incl. heating steam inlet
to incl. condensate outlet.

LDH Deaeration system.

from incl. deaerator or tank inlet
to incl. tank outlet, incl. warm-up equipment
of vapour condenser.

LDJ Preheating, cooling system.

from incl. preheater or cooler inlet
to incl. preheater or cooler outlet.

LDK Piping system, temporary storage system, pump system for main fluid.

Piping system:

from excl. intake or
from excl. outlet of other system
to excl.inlet to other system
to incl. fluid treatment system outlet.

Temporary storage system:

from incl. temporary storage system inlet
to incl. temporary storage system outlet.

Pump System:

from incl. pump system suction nozzle
to incl. pump system discharge nozzle.

LDL Storage system outside fluid treatment system (if not part of another system).

from incl. storage system inlet
to incl. storage system outlet, incl. intake
and outfall.

LDN Chemicals supply system.

from incl. intake or
from incl. storage tank
to excl. discharge into other system.

LDP Regeneration, flushing equipment.

from incl. system inlet
to excl. inlet to other system
from excl. chemicals or auxiliary fluid supply
system and flushing air supply system
to incl. regeneration, flushing equipment.

LDQ Injection system for main fluid.

from incl. injection equipment or
from excl. branch off chemicals supply
system
to excl. inlet to other system.

LDR Flushing water and residues removal system incl. neutralization.

from excl. outlet of respective system
to excl. discharge into disposal system.

LDS Sludge thickening system.

from excl. outlet of respective system
to excl. discharge into disposal system.

LDT Heating, cooling and flushing fluid distribution system.

from incl. heating, cooling, flushing fluid
generation equipment or
from excl. branch off heating, cooling,
flushing fluid supply system
to excl. user and
from excl. user.

LDV Lubricant supply system.

STEAM-WATER CYCLE

LDX	Fluid supply system for control and protection equipment.	LLC	Gas cleaning system (only for closed cycle). (free for use).
LDY	Control and protection equipment.	LLD	Gas cleaning system (only for closed cycle). (free for use).
LF	Common installations for steam-water cycle.	LLE	Gas cleaning system (only for closed cycle). (free for use).
LFC	Common drain and vent systems.	LLF	Gas cleaning system (only for closed cycle). (free for use).
LFJ	Steam generator layup system.	LLG	Gas cleaning system (only for closed cycle). (free for use).
LFN	Proportioning system for feedwater, condensate system, incl. proportioning in boiler and turbine area.	LLH	Gas cleaning system (only for closed cycle). (free for use).
LK	Gas system (closed cycle).	LLJ	Gas cleaning system (only for closed cycle). (free for use).
LKA	Storage system. from excl. receiving point from gas supply system to excl. piping system inlet.	LLK	Gas cleaning system (only for closed cycle). (free for use).
LKB	Piping system. from excl. gas heater outlet to excl. gas heater inlet, excl. turbine, compressor, preheater, cooler.	LLL	Gas cleaning system (only for closed cycle). (free for use).
LKC	Compressor system (if separate from gas turbine). from incl. compressor inlet to incl. compressor outlet.	LLM	Gas cleaning system (only for closed cycle). (free for use).
LKD	Preheating system. from incl. preheater inlet to incl. preheater outlet.	LLN	Gas cleaning system (only for closed cycle). (free for use).
LKE	Precooling system. from incl. precooler inlet to incl. precooler outlet.	LLP	Gas cleaning system (only for closed cycle). (free for use).
LKF	Intercooling system. from incl. intercooler inlet to incl. intercooler outlet.	LLQ	Gas cleaning system (only for closed cycle). (free for use).
LKG	Pressurizing system. from incl. pressurizing system inlet to excl. discharge into piping systems.	LLR	Gas cleaning system (only for closed cycle). (free for use).
LKV	Lubricant supply system.	LLS	Gas cleaning system (only for closed cycle). (free for use).
LKW	Sealing fluid supply system.	LLT	Gas cleaning system (only for closed cycle). (free for use).
LKX	Fluid supply system for control and protection equipment.	LLU	Gas cleaning system (only for closed cycle). (free for use).
LKY	Control and protection equipment.	LLW	Sealing fluid supply system.
LL	Gas cleaning system (only for closed cycle).	LLX	Fluid supply system for control and protection equipment.
LLA	Gas cleaning system (only for closed cycle). (free for use).	LLY	Control and protection equipment.
LLB	Gas cleaning system (only for closed cycle). (free for use).		

STEAM-WATER CYCLE

LN Water impounding works for hydroelectric power plant.

LNA Head and tail race system, storage system.

from incl. water intake incl. transfer conduit and pump system
to excl. dam (incl. measuring equipment) or
to excl. headwater piping system or
to excl. headrace system.

LNB Trash rack, trash/fish barrier.

to excl. inlet
to other system

LNC Dam, weir system.

Dam:
from incl. foundation
to incl. crest, incl. inspection and instrumentation & equipment
Weir system:
from incl. inlet
to incl. outlet to other system.

LND Spillway.

LNE Drainage system.

LNV Lubricant supply system.

LNW Sealing fluid supply system.

LNX Fluid supply system for control and protection equipment.

LNZ Control and protection equipment.

LP Intake system, upperwater system for hydroelectric power plant.

LPA Rake and rake cleaning system.

Task: Separate and collect floating matter from headrace.

LPB Isolating equipment.

from incl. isolating equipment inlet
to incl. outlet to other system.

LPC Piping and penstock system.

from incl. receiving point
to incl. discharge into turbine system, incl. connecting piping for pump system.

LPE Surge tank.

LPV Lubricant supply system.

LPW Sealing fluid supply system.

LPX Fluid supply system for control and protection equipment.

LPY Control and protection equipment.

LQ Tail-race system, underwater system for hydroelectric power plant.

LQA Underwater piping and culvert system.

from excl. main machine set *ME*, *MF*, *MG* inlet
to incl. outlet to other system.

LQB Surge tank.

LQC Isolating equipment.

from incl. isolating equipment inlet
to incl. outlet to other system.

LQE Rake and rake cleaning system for pumped-storage operation.

Task: Separate out and collect floating matter from tailrace

LQV Lubricant supply system.

LQW Sealing fluid supply system.

LQX Fluid supply system for control and protection equipment.

LQY Control and protection equipment.

LS Common installations for hydroelectric power plant.

LSA Common installations for hydroelectric power plant. (free for use).

LSB Common installations for hydroelectric power plant. (free for use).

LSC Common installations for hydroelectric power plant. (free for use).

LSD Common installations for hydroelectric power plant. (free for use).

LSE Common installations for hydroelectric power plant. (free for use).

LSF Common installations for hydroelectric power plant. (free for use).

LSG Common installations for hydroelectric power plant. (free for use).

LSH Common installations for hydroelectric power plant. (free for use).

LSJ Common installations for hydroelectric power plant. (free for use).

LSK Common installations for hydroelectric power plant. (free for use).

LSL Drainage system.

LSM Common installations for hydroelectric power plant. (free for use).

STEAM-WATER CYCLE

LSN	Common installations for hydroelectric power plant. (free for use).	LVS	Lubricant supply system. (free for use).
LSP	Common installations for hydroelectric power plant. (free for use).	LVT	Lubricant supply system. (free for use).
LSQ	Common installations for hydroelectric power plant. (free for use).	LVU	Lubricant supply system. (free for use).
LSR	Common installations for hydroelectric power plant. (free for use).	LW	Sealing fluid supply system for steam, water, gas cycles.
LSS	Common installations for hydroelectric power plant. (free for use).	LWA	Sealing fluid supply system for steam, water, gas cycles.
LST	Common installations for hydroelectric power plant. (free for use).	LWB	Sealing fluid supply system for steam, water, gas cycles.
LSU	Common installations for hydroelectric power plant. (free for use).	LWC	Sealing fluid supply system for steam, water, gas cycles.
LV	Lubricant supply system.	LWD	Sealing fluid supply system for steam, water, gas cycles.
LVA	Lubricant supply system. (free for use).	LWE	Sealing fluid supply system for steam, water, gas cycles.
LVB	Lubricant supply system. (free for use).	LWF	Sealing fluid supply system for steam, water, gas cycles.
LVC	Lubricant supply system. (free for use).	LWG	Sealing fluid supply system for steam, water, gas cycles.
LVD	Lubricant supply system. (free for use).	LWH	Sealing fluid supply system for steam, water, gas cycles.
LVE	Lubricant supply system. (free for use).	LWJ	Sealing fluid supply system for steam, water, gas cycles.
LVF	Lubricant supply system. (free for use).	LWK	Sealing fluid supply system for steam, water, gas cycles.
LVG	Lubricant supply system. (free for use).	LWL	Sealing fluid supply system for steam, water, gas cycles.
LVH	Lubricant supply system. (free for use).	LWM	Sealing fluid supply system for steam, water, gas cycles.
LVJ	Lubricant supply system. (free for use).	LWN	Sealing fluid supply system for steam, water, gas cycles.
LVK	Lubricant supply system. (free for use).	LWP	Sealing fluid supply system for steam, water, gas cycles.
LVL	Lubricant supply system. (free for use).	LWQ	Sealing fluid supply system for steam, water, gas cycles.
LVM	Lubricant supply system. (free for use).	LWR	Sealing fluid supply system for steam, water, gas cycles.
LVN	Lubricant supply system. (free for use).	LWS	Sealing fluid supply system for steam, water, gas cycles.
LVP	Lubricant supply system. (free for use).	LWT	Sealing fluid supply system for steam, water, gas cycles.
LVQ	Lubricant supply system. (free for use).	LWU	Sealing fluid supply system for steam, water, gas cycles.
LVR	Lubricant supply system. (free for use).	LX	Fluid supply system for control and protection equipment.
		LXA	Fluid supply system for control and protection equipment. (free for use).

M

MAIN MACHINE SETS

MA Steam turbine plant.

MB Gas turbine plant.

ME Hydraulic turbine plant.

MF Pumping turbine plant in pumped-storage power plants.

MG Pumped-storage plant.

MJ Diesel engine plant.

MK Generator plant.

ML Electro-motive plant (incl. motor generator).

MM Compressor plant.

MP Common installations for main machine sets.

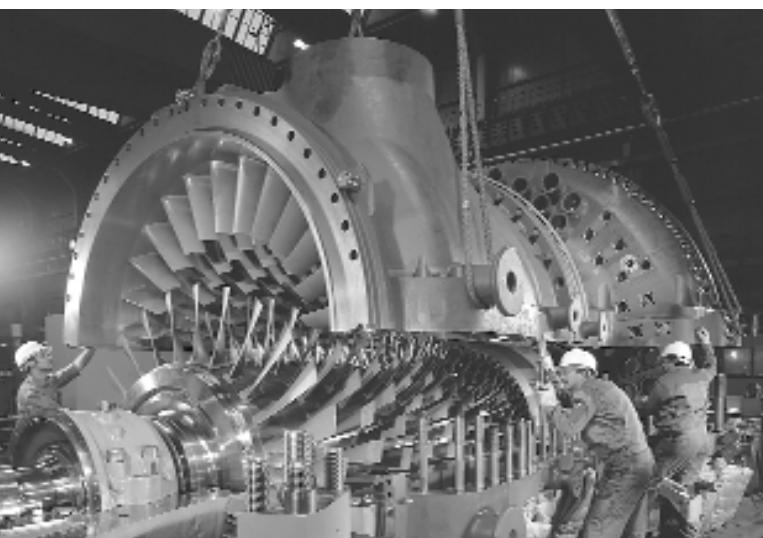
MR Gas engine plant.

MV Lubricant supply system.

MW Sealing fluid supply system.

MX Fluid supply system for control and protection equipment.

MY Control and protection equipment.



MAIN MACHINE SETS

MA Steam turbine plant.

MAA HP turbine.

from incl. steam admission (main stop valve) or combined main stop and control valve
to incl. automatic/non-automatic extraction and exhaust nozzles and
to incl. the interfaces with other turbine-internal systems.

MAB IP turbine.

from incl. crossover line, incl. control element or
from incl. intercept valve
to incl. automatic/non-automatic extraction and exhaust nozzles and
to incl. the interfaces with other turbine-internal systems.

MAC LP turbine.

from incl. crossover line, incl. control element or
from incl. intercept valve or steam inlet nozzle (in reheat system without intercept valves).
to incl. automatic/non-automatic extraction and exhaust nozzles and
to incl. the interfaces with other turbine-internal systems.

MAD Bearings.

MAG Condensing system.

from incl. condenser neck or inlet nozzle
to incl. condenser outlet nozzle, incl. connected flash tanks, incl. instrumentation equipment associated with condenser.

MAH Motive water system (if separate from *MAJ*).

from excl. outlet of other system
to excl. water-operated air ejector inlet.

MAJ Air removal system.

from excl. condenser outlet
to excl. inlet atmosphere.

MAK Transmission gear between prime mover and driven machine, incl. turning gear.

MAL Drain and vent systems.

from incl. collector point or
from incl. final drain
to excl. discharge into other system.

MAN Turbine bypass station, incl. desuperheating spray system.

from incl. bypass valve and
from incl. desuperheating spray valve
to incl. steam inlet to condenser.
Low pressure: MAN01 - 19
Intermediate pressure: MAN20 - 49
High pressure: MAN50 - 99.

MAV Lubricant supply system.

from incl. dedicated lubricant tank or common lubricant and control fluid tank or
from excl. branch off lubricant supply system
to excl. user and
from excl. user.

MAW Sealing, heating and cooling steam system.

from excl. branch
to excl. casing nozzle of seal steam user and leak-off
to excl. condenser or
to incl. gland steam condenser or
to excl. heating/cooling steam user.

MAX Non-electric control and protection equipment, incl. fluid supply system.

MAY Electrical control and protection equipment.

MB Gas turbine plant.

MBA Turbine, compressor rotor with common casing.

from incl. compressor inlet
to incl. compressor outlet
from incl. turbine inlet
to incl. turbine outlet, incl. exhaust gas diffuser.

MBB Turbine casing and rotor.

from incl. turbine inlet
to incl. turbine outlet, incl. exhaust gas diffuser.

MBC Compressor casing and rotor.

from incl. compressor inlet
to incl. compressor outlet.

MBD Bearings.

MBE Cooling medium system.

from excl. branch-off cooling medium supply
to excl. user and incl. pump system, control equipment, recirculation.

MBH Cooling and sealing gas system.

from incl. extraction point
to excl. user and
from excl. user, incl. leak-off
to incl. inlet to other system.

MBJ Start-up unit.

MBK Transmission gear between prime mover and driven machine, incl. turning gear, barring gear.

MAIN MACHINE SETS

MBL Intake air, cold gas system (open cycle).

from excl. atmosphere,
to excl. combustion chamber or
to excl. compressor inlet or
to incl. exhaust gas heat exchanger, excl.
compressor.

MBM Combustion chamber (gas heating, combustion).

from incl. cold gas, fuel inlet
to incl. hot gas outlet.

MBN Fuel supply system (liquid).

from excl. branch off main supply line or
from incl. temporary (day) tank
to excl. combustion chamber or
to excl. motive gas generating unit, incl.
fuel return system.

MBP Fuel supply system (gaseous).

from excl. branch off main supply line
to excl. combustion chamber or
to excl. motive gas generating unit.

MBQ Ignition fuel supply system (if separate).

from excl. branch off main supply line or
from incl. storage tank to excl. combustion
chamber or
to excl. motive gas generating unit.

MBR Exhaust gas system (open cycle).

from excl. combustion chamber or
from excl. exhaust gas diffuser
to excl. discharge into atmosphere, excl.
turbine or
to excl. inlet to other system
(e.g. combustion air system).

MBS Storage system.

to excl. connection to main system and
from excl. connection to main system.

MBT Motive gas generator unit, incl. combustion chamber.

from incl. air/fuel inlet
to incl. motive gas outlet of motive gas
generating unit.

MBU Additive system.

from incl. supply
to incl. injection.

MBV Lubricant supply system.

from incl. dedicated lubricant tank or
common lubricant and control fluid tank or
from excl. branch off control fluid supply
system
to excl. user and
from excl. user.

MBW Sealing oil supply system.

from incl. dedicated seal oil tank or
from excl. seal oil pump suction line
to excl. user and
from excl. user.

MBX Non-electric control and protection equipment, incl. fluid supply system.

MBY Electrical control and protection equipment.

MBZ Lubricant and control fluid treatment system.

ME Hydraulic turbine plant.

MEA Turbine (casing, shaft, runner etc.).

from excl. turbine inlet or
from excl. isolating valve
to incl. turbine outlet or
to excl. isolating valve, incl. draft tube.

MEB Isolating valve.

from incl. isolating valve inlet
to incl. isolating valve outlet.

MED Bearings.

MEG Stabilizing air system.

from incl. air compressor
to incl. outlet to turbine

MEK Transmission gear between prime mover and driven machine.

MEL Water depression air supply system.

from incl. air compressor
to incl. outlet to turbine

MES Shaft gland cooling water system.

MEV Lubricant supply system.

from incl. lubricant tank
to excl. user and
from excl. user.

MEW Sealing water supply system.

from incl. sealing water supply main isolat-
ing valve
to excl. casing nozzle of sealing water
user.

MEX Non-electric control and protection equipment, incl. fluid supply system.

MEY Electrical control and protection equipment.

MAIN MACHINE SETS

MF Pumping turbine plant in pumped-storage power plants.

MFA Pumping turbine, pump and turbine as a constructive unit.

Task: A constructive unit which on different times serves as a prime mover (turbine) or as a driven machine (pump).
from excl. pump, pumping turbine inlet or from excl. isolating valve
to incl. pump, pumping turbine outlet or to excl. isolating valve.

MFB Isolating valve.

from incl. isolating valve inlet
to incl. isolating valve outlet.

MFD Bearings.

MFG Stabilizing air system.

from incl. air compressor
to incl. inlet to turbine.

MFK Transmission gear between motor generator and pumping turbine.

MFL Compressed air supply system.

from incl. air compressor
to incl. inlet to pumping turbine, singleshaft pump and turbine unit.

MFM Start-up unit.

MFS Shaft gland cooling water system.

MFV Lubricant supply system.

from incl. lubricant tank
to excl. user and
from excl. user.

MFV Sealing water supply system.

from incl. sealing water supply main isolating valve
to excl. casing nozzle of sealing water user.

MFV Non-electric control and protection equipment, incl. fluid supply system.

MFY Electrical control and protection equipment.

MG Pumped-storage plant.

MGA Storage pump (casing, shaft, runner etc.).

from excl. storage pump inlet or from excl. isolating valve
to incl. storage pump outlet or to excl. isolating valve.

MGB Isolating valve.

from incl. isolating valve inlet
to incl. isolating valve outlet.

MGD Bearings.

MGK Transmission gear between motor generator and pumping turbine.

MGL Water depression air supply system.

from incl. air compressor
to incl. inlet to storage pump.

MGM Start-up unit.

MGR locked.

MGS Shaft gland cooling water system.

MGV Lubricant supply system.

from incl. lubricant tank
to excl. user and
from excl. user.

MGW Sealing water supply system.

from incl. sealing water supply main isolating valve
to excl. casing nozzle of sealing water user.

MGX Non-electric control and protection equipment, incl. fluid supply system.

MGY Electrical control and protection equipment.

MJ Diesel engine plant.

MJA Engine.

from incl. fuel injection nozzle inlet or from incl. air intake nozzles or from incl. cooling water inlet nozzles
to incl. exhaust nozzle outlet or to incl. cooling water nozzle outlet, incl. engine- internal systems.

MJB Turbocharger, blower.

from incl. turbocharger or blower inlet
to incl. turbocharger or blower outlet.

MJG Liquid cooling system.

from excl. engine cooling water nozzle outlet or from incl. turbocharger air cooling system inlet
to excl. engine cooling water nozzle inlet or to incl. turbocharger air cooler outlet.

MJH Air intercooling system.

from incl. intercooler inlet
to incl. intercooler outlet
to excl. outlet to other cooling systems.

MJK Transmission gear between prime mover and driven machine.

MJN Fuel systems.

from incl. temporary (day) tank or from excl. branch off piping system
to excl. fuel injection nozzle inlet.

MJP Start-up unit (incl. flywheel).

MAIN MACHINE SETS

MJQ Air intake system.

from excl. atmosphere
to excl. turbocharger or
to excl. engine air intake nozzles.

MJR Exhaust gas system.

from excl. engine exhaust nozzle outlet
to excl. discharge into atmosphere.

MJV Lubricant supply system.

from incl. dedicated lubricant tank or common lubricant and control fluid tank or
from excl. branch off control fluid supply system
to excl. user from excl. user.

MJW Sealing fluid supply system.

MJX Fluid supply system for control and protection equipment.

MJY Control and protection equipment.

MK Generator plant.

MKA Generator, complete, incl. stator, rotor and all integral cooling equipment.

to incl. generator bushing.

MKB Generator exciter set, including set with electrical braking system.

(use only if *MKC* is not sufficient for identification).

MKC Exciter set.

MKD Bearings.

MKF Stator/rotor liquid cooling system, incl. coolant supply system (for cooling oil see *MKU*).

Task: dissipate heat produced by
stator/rotor to coolant
from excl. stator/rotor outlet
to excl. stator/rotor inlet.

MKG Stator/rotor gas cooling system, incl. coolant supply system (Note: for nitrogen cooling see *MKH*).

Task: dissipate heat produced by
stator/rotor to coolant
from excl. stator/rotor outlet
to excl. stator/rotor inlet.

MKH Stator/rotor nitrogen cooling system, incl. coolant supply system (Note: for other gas cooling see *MKG*).

Task: dissipate heat produced by
stator/rotor to coolant
from excl. stator/rotor outlet
to excl. stator/rotor inlet.

MKP Secondary cooling water system.

from excl. cooling water system (*PC*, *PG*)
to excl. users incl. blowdown, discharge, venting
from excl. users
to excl. inlet into other system.

MKQ Exhaust gas system (if separate from *MKG* and *MKH*).

MKU Stator/rotor cooling oil cooling system, incl. coolant supply system (Note: for other liquid cooling see *MKF*).

Task: dissipate heat produced by
stator/rotor to coolant
from excl. stator/rotor outlet
to excl. stator/rotor inlet.

MKV Lubricant supply system (if separate system for generator).

MKW Sealing fluid supply system (Sealing oil system incl. supply and treatment).

from excl. branch off sealing oil supply system
to excl. stator inlet and
from excl. stator outlet
to excl. inlet to other system or in closed systems
from excl. stator outlet
to excl. stator inlet.

MKX Fluid supply system for control and protection equipment.

MKY Control and protection equipment.

ML Electro-motive plant (incl. motor generator).

MLA Motor frame, motor generator frame, incl. stator, rotor and all integral cooling equipment.

to incl. motor or generator bushing.

MLC Exciter set.

MLD Bearings.

MLF Stator/rotor liquid cooling system, incl. coolant supply system (Note: for cooling oil see *MLU*).

Task: dissipate heat produced by
stator/rotor to coolant
from excl. stator/rotor outlet
to excl. stator/rotor inlet.

MAIN MACHINE SETS

MLG Stator/rotor gas cooling system, incl. coolant supply system(Note: for nitrogen cooling see *MLH*).

Task: dissipate heat produced by stator/rotor to coolant from excl. stator/rotor outlet to excl. stator/rotor inlet.

MLH Stator/rotor nitrogen cooling system, incl. coolant supply system (Note: for other gas cooling see *MLG*).

Task: dissipate heat produced by stator/rotor to coolant from excl. stator/rotor outlet to excl. stator/rotor inlet.

MLQ Exhaust gas system (if separate from *MLG* and *MLH*).

MLU Stator/rotor cooling oil cooling system, incl. coolant supply system (Note: for other liquid cooling see *MLF*).

Task: dissipate heat produced by stator/rotor to coolant from excl. stator/rotor outlet to excl. stator/rotor inlet.

MLV Lubricant supply system (if separate system for electro-motive units).

MLW Sealing fluid supply system (Sealing oil system, incl. supply and treatment).

from excl. branch off sealing oil supply system to excl. stator inlet or from excl. stator outlet to excl. inlet to other system or in closed systems from excl. stator outlet to excl. stator inlet.

MLX Fluid supply system for control and protection equipment.

MLY Control and protection equipment.

MM Compressor plant.

MMA Compressor incl. internal systems.

from incl. air, cooling fluid inlet to incl. air, cooling fluid outlet.

MMC Air intake piping system.

from excl. atmosphere to excl. compressor inlet.

MMD Bearings.

MME Intercooling system.

MMF Aftercooling system.

MMG Final cooling system.

MMH Discharge piping system.

from excl. compressor outlet to excl. inlet to other system.

MMV Lubricant supply system.

MMW Sealing fluid supply system.

MMX Fluid supply system for control and protection equipment.

MMY Control and protection equipment.

MP Common installations for main machine sets.

MPA Foundation.

MPB Sheathing.

MPG Frame, support structure.

MPR Forced cooling system.

MPS Drying and layup system.

MR Gas engine plant.

MRA Gas engine plant.

(free for use).

MRB Gas engine plant.

(free for use).

MRC Gas engine plant.

(free for use).

MRD Gas engine plant.

(free for use).

MRE Gas engine plant.

(free for use).

MRF Gas engine plant.

(free for use).

MRG Gas engine plant.

(free for use).

MRH Gas engine plant.

(free for use).

MRJ Gas engine plant.

(free for use).

MRK Gas engine plant.

(free for use).

MRL Gas engine plant.

(free for use).

MRM Gas engine plant.

(free for use).

MRN Gas engine plant.

(free for use).

MRP Gas engine plant.

(free for use).

MRQ Gas engine plant.

(free for use).

MRR Gas engine plant.

(free for use).

MRS Gas engine plant.

(free for use).

MAIN MACHINE SETS

- MRT** Gas engine plant.
(free for use).
- MRU** Gas engine plant.
(free for use).
- MRV** Lubricant supply system.
- MRW** Sealing fluid supply system.
- MRX** Fluid supply system for control and protection equipment.
- MRY** Control and protection equipment.

MV Lubricant supply system.

- MVC** Lubricant supply system.
(free for use).

MW Sealing fluid supply system.

- MWA** Sealing fluid supply system.
(free for use).
- MWB** Sealing fluid supply system.
(free for use).
- MWC** Sealing fluid supply system.
(free for use).
- MWD** Sealing fluid supply system.
(free for use).
- MWE** Sealing fluid supply system.
(free for use).
- MWF** Sealing fluid supply system.
(free for use).
- MWG** Sealing fluid supply system.
(free for use).
- MWH** Sealing fluid supply system.
(free for use).
- MWJ** Sealing fluid supply system.
(free for use).
- MWK** Sealing fluid supply system.
(free for use).
- MWL** Sealing fluid supply system.
(free for use).
- MWM** Sealing fluid supply system.
(free for use).
- MWN** Sealing fluid supply system.
(free for use).
- MWP** Sealing fluid supply system.
(free for use).
- MWQ** Sealing fluid supply system.
(free for use).
- MWR** Sealing fluid supply system.
(free for use).
- MWS** Sealing fluid supply system.
(free for use).
- MWT** Sealing fluid supply system.
(free for use).
- MWU** Sealing fluid supply system.
(free for use).

MX Fluid supply system for control and protection equipment.

- MXA** Fluid supply system for control and protection equipment.
(free for use).
- MXB** Fluid supply system for control and protection equipment.
(free for use).
- MXC** Fluid supply system for control and protection equipment.
(free for use).
- MXD** Fluid supply system for control and protection equipment.
(free for use).
- MXE** Fluid supply system for control and protection equipment.
(free for use).
- MXF** Fluid supply system for control and protection equipment.
(free for use).
- MXG** Fluid supply system for control and protection equipment.
(free for use).
- MXH** Fluid supply system for control and protection equipment.
(free for use).
- MXJ** Fluid supply system for control and protection equipment.
(free for use).
- MXK** Fluid supply system for control and protection equipment.
(free for use).
- MXL** Fluid supply system for control and protection equipment.
(free for use).
- MXM** Fluid supply system for control and protection equipment.
(free for use).
- MXN** Fluid supply system for control and protection equipment.
(free for use).
- MXP** Fluid supply system for control and protection equipment.
(free for use).
- MXQ** Fluid supply system for control and protection equipment.
(free for use).
- MXR** Fluid supply system for control and protection equipment.
(free for use).
- MXS** Fluid supply system for control and protection equipment.
(free for use).

MXT Fluid supply system for control and protection equipment.

(free for use).

MXU Fluid supply system for control and protection equipment.

(free for use).

MY Control and protection equipment.

MYA Control and protection equipment.

(free for use).

MYB Control and protection equipment.

(free for use).

MYC Control and protection equipment.

(free for use).

MYD Control and protection equipment.

(free for use).

MYE Control and protection equipment.

(free for use).

MYF Control and protection equipment.

(free for use).

MYG Control and protection equipment.

(free for use).

MYH Control and protection equipment.

(free for use).

MYJ Control and protection equipment.

(free for use).

MYK Control and protection equipment.

(free for use).

MYL Control and protection equipment.

(free for use).

MYM Control and protection equipment.

(free for use).

MYN Control and protection equipment.

(free for use).

MYP Control and protection equipment.

(free for use).

MYQ Control and protection equipment.

(free for use).

MYR Control and protection equipment.

(free for use).

MYS Control and protection equipment.

(free for use).

MYT Control and protection equipment.

(free for use).

MYU Control and protection equipment.

(free for use).

MYV Control and protection equipment.

(free for use).

MYW Control and protection equipment.

(free for use).

MYX Control and protection equipment.

(free for use).

MYY Control and protection equipment.

(free for use).



PROCESS ENERGY / MEDIA SUPPLY FOR EXTERNAL USERS (e.g. District Heating)

NA Process steam system (incl. condensate return).

ND Process hot water system.

NE Process chilled water.

NG Process air system.

NK Process gas system.



PROCESS ENERGY / MEDIA SUPPLY FOR EXTERNAL USERS

NA Process steam system (incl. condensate return).

- NAA** Piping system (steam).
- NAB** Piping system (condensate).
incl. pumps
- NAD** Process heat transmission system.
from incl. heat exchanger inlet
to incl. heat exchanger outlet.
- NAE** Piping system (steam) *NAE* to
NAJ only to be used if *NAA* is
not sufficient for identification.
(free for use e.g. to pressure level).
- NAF** Piping system (steam) *NAE* to
NAJ only to be used if *NAA* is
not sufficient for identification.
(free for use e.g. to pressure level).
- NAG** Piping system (steam) *NAE* to
NAJ only to be used if *NAA* is
not sufficient for identification.
(free for use e.g. to pressure level).
- NAH** Piping system (steam) *NAE* to
NAJ only to be used if *NAA* is
not sufficient for identification.
(free for use e.g. to pressure level).
- NAJ** Piping system (steam) *NAE* to
NAJ only to be used if *NAA* is
not sufficient for identification.
(free for use e.g. to pressure level).
- NAK** Piping system (condensate) *NAK*
to *NAM* only to be used if *NAB*
is not sufficient for identification.
- NAL** Piping system (condensate) *NAK*
to *NAM* only to be used if *NAB*
is not sufficient for identification.
(free for use e.g. according to process).
- NAM** Piping system (condensate) *NAK*
to *NAM* only to be used if *NAB*
is not sufficient for identification.
(free for use e.g. according to process).
- NAX** Fluid supply system for control and
protection equipment.
- NAY** Control and protection equipment.

ND Process hot water system.

- NDA** Piping system (forward).
- NDB** Piping system (return).
- NDC** Process hot water pump system.
- NDD** Process heat transfer.
from incl. heat exchanger inlet
to incl. heat exchanger outlet.
- NDE** Hot water storage system.
from incl. tank inlet
to incl. tank outlet.

- NDF** Distribution systems.
(free for use e.g. for temperature levels).
- NDG** Distribution systems.
(free for use e.g. for temperature levels).
- NDH** Distribution systems.
(free for use e.g. for temperature levels).
- NDJ** Distribution systems.
(free for use e.g. for temperature levels).
- NDK** Pressurizing system.
- NDV** Lubricant supply system.
- NDX** Fluid supply system for control
and protection equipment.
- NDY** Control and protection equipment.

NE Process chilled water.

NG Process air system.

- NGB** Piping system.
- NGC** Forwarding system.
- NGW** Sealing fluid supply system.
- NGX** Fluid supply system for control
and protection equipment.
- NGY** Control and protection equipment.

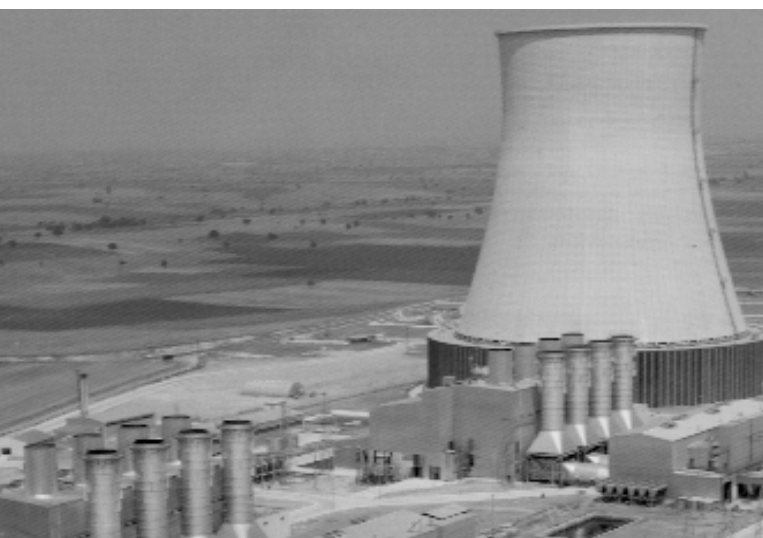
NK Process gas system.

- NKB** Piping system.
- NKW** Sealing fluid supply system.
- NKX** Fluid supply system for control
and protection equipment.
- NKY** Control and protection equipment.

P

COOLING WATER SYSTEMS

- PA** Circulating (main cooling) water system.
- PB** Circulating (main cooling) water treatment system.
- PC** Service (secondary cooling) water system, conventional area.
- PD** Service (secondary cooling) water treatment system, conventional area.
- PG** Closed cooling water system.
- PH** Closed cooling water treatment system for conventional area.
- PM** Closed cooling water system for transformers (if separate from closed cooling water system).
- PS** Cooling tower blowdown system (if separate from *PAB*).
- PU** Common equipment for cooling water systems.
- PV** Lubricant supply system.
- PW** Sealing fluid supply system.
- PX** Fluid supply system for control and protection equipment.
- PY** Control and protection equipment.



COOLING WATER SYSTEMS

PA Circulating (main cooling) water system.

PAA Extraction, mechanical cleaning for direct cooling.

from incl. intake system
to incl. mechanical cleaning system outlet.

PAB Circulating (main cooling) water piping and culvert system.

from excl. outlet of circulating (main cooling) water extraction system (excl. any user)
to incl. outfall in direct cooling system or from excl. cooling tower outlet
to excl. cooling tower inlet in recirculation cooling system.

PAC Circulating (main cooling) water pump system.

from incl. pump system suction nozzle
to incl. pump system discharge nozzle.

PAD Recirculating cooling system, outfall cooling system.

from incl. hot water riser
to incl. basin outlet.

PAE Cooling tower pump system (if separate).

PAH Condenser tubes cleaning system, incl. all appurtenances.

PAR Make-up water piping system.

from excl. intake
to excl. inlet to other system.

PAS Make-up water pump system.

from incl. pump system suction nozzle
to incl. pump system discharge nozzle.

PAV Lubricant supply system.

PAX Fluid supply system for control and protection equipment.

PAY Control and protection equipment.

PB Circulating (main cooling) water treatment system.

PBA Discharge from fluid treatment system (if not part of another system).

from excl. fluid treatment system outlet
to excl. inlet to other system.

PBB Filtering, mechanical cleaning system.

from incl. separation equipment inlet
to incl. separation equipment outlet

PBC Aeration, gas injection system.

from excl. atmosphere or
from incl. gas supply

PBD Precipitation system (e.g. for carbonate hardness removal).

from incl. precipitation equipment inlet
to incl. precipitation equipment outlet

PBE Acid proportioning system (e.g. for carbonate hardness removal).

from incl. acid dosing equipment or from excl. branch off chemicals supply system

to excl. inlet to other system

PBF Ion exchange system (e.g. for demineralization).

from incl. ion exchanger inlet and from incl. isolating valve of chemicals supply system or auxiliary fluid supply system upstream of ion exchanger.

PBG Evaporation system (e.g. for demineralisation).

from incl. feedwater inlet
to incl. steam outlet and from incl. heating steam inlet
to incl. condensate outlet.

PBH Deaeration system.

from incl. deaerator or tank inlet
to incl. tank outlet incl. warm-up equipment of vapour condenser.

PBJ Preheating, cooling system.

from incl. preheater or cooler inlet
to incl. preheater or cooler outlet.

PBK Piping system, temporary storage system, pump system for main fluid.

Piping system:

from excl. intake or from excl. outlet of other system
to excl. inlet to other system
to incl. fluid treatment system outlet.

Temporary storage system:

from incl. temporary storage system inlet
to incl. temporary storage system outlet.

Pump system:

from incl. pump system suction nozzle
to incl. pump system discharge nozzle.

PBL Storage system outside fluid treatment system (if not part of another system).

from incl. storage system inlet
to incl. storage system outlet incl. intake and outfall.

PBN Chemicals supply system (eg. Electrochlorination system).

from incl. intake or from incl. storage tank
to excl. discharge into other system.

COOLING WATER SYSTEMS

PBP	Regeneration, flushing equipment. from incl. system inlet to excl. inlet to other system from excl. chemicals or auxiliary fluid supply system and flushing air supply system to incl. regenerating, flushing equipment.	PCH	Heat exchanger cleaning system.
PBQ	Injection system for main fluid (eg. Hypochlorite solution injection system). from incl. injection equipment or from excl. branch off chemicals supply system to excl. inlet to other system.	PCM	Service (secondary cooling) water system for generator, motor generator cooling. from excl. branch off *PCB* to excl. generator cooler and from excl. generator cooler to excl. inlet to *PCB* or other systems.
PBR	Flushing water and residues removal system, incl. neutralization. from excl. outlet of respective system to excl. discharge into other system.	PCV	Lubricant supply system.
PBS	Sludge thickening system. from excl. outlet of respective system to excl. discharge into other system.	PCX	Fluid supply system for control and protection equipment.
PBT	Heating, cooling and flushing fluid distribution system. from incl. heating, cooling, flushing fluid generation equipment or from excl. branch off heating, cooling, flushing fluid supply system to excl. user and from excl. user.	PCY	Control and protection equipment.
PBV	Lubricant supply system.	PD	Service (secondary cooling) water treatment system, conventional area.
PBX	Fluid supply system for control and protection equipment.	PDA	Discharge from fluid treatment system (if not part of another system). from excl. fluid treatment system outlet to excl. inlet to other system.
PBY	Control and protection equipment.	PDB	Filtering, mechanical cleaning system. from incl. separation equipment inlet to incl. separation equipment outlet.
PC	Service (secondary cooling) water system, conventional area.	PDC	Aeration, gas injection system. from excl. atmosphere or from incl. gas supply.
PCA	Extraction, mechanical cleaning for direct cooling. from incl. intake system to incl. mechanical cleaning system outlet.	PDD	Precipitation system (e.g. for carbonate hardness removal). from incl. precipitation equipment inlet to incl. precipitation equipment outlet.
PCB	Piping and culvert system. from excl. outlet of service (secondary cooling) extraction system or from excl. branch off circulating water system to excl. inlet to other system, excl. respective user and from excl. make-up water treatment and distribution system.	PDE	Acid proportioning system (e.g. for carbonate hardness removal). from incl. acid proportioning equipment or from excl. branch off chemicals supply system to excl. inlet to other system.
PCC	Pump system. from incl. pump system suction nozzle to incl. pump system discharge nozzle.	PDF	Ion exchange system (e.g. for demineralization). from incl. ion exchanger inlet and to incl. isolating valve of chemicals supply system or auxiliary fluid supply system upstream of ion exchanger.
PCD	Recirculation cooling system, outfall cooling system. from incl. hot water riser to incl. basin outlet.	PDG	Evaporation system (e.g. for demineralization). from incl. feedwater inlet to incl. steam outlet and from incl. heating steam inlet to incl. condensate outlet.
		PDH	Deaeration system. from incl. deaerator or tank inlet to incl. tank outlet incl. warm-up equipment of vapour condenser.
		PDJ	Preheating, cooling system. from incl. preheater or cooler inlet to incl. preheater or cooler outlet.

COOLING WATER SYSTEMS

PDK Piping system, temporary storage system, pump system for main fluid.

Piping system:
from excl. intake or
from excl. outlet of other system
to excl. inlet to other system
to incl. fluid treatment system outlet.
Temporary storage system:
from incl. temporary storage system inlet
to incl. temporary storage system outlet.
Pump system:
from incl. pump system suction nozzle
to incl. pump system discharge nozzle.

PDL Storage system outside fluid treatment system (if not part of another system).

from incl. storage system inlet
to incl. storage system outlet incl. intake
and outfall.

PDN Chemicals supply system.

from incl. intake or
from incl. storage tank
to excl. discharge into other system.

PDP Regeneration, flushing equipment.

from incl. system inlet
to excl. inlet to other system
from excl. chemicals or auxiliary fluid supply
system and flushing air supply system
to incl. regenerating, flushing equipment.

PDQ Injection system for main fluid.

from incl. injection equipment or
from excl. branch off chemicals supply
system
to excl. inlet to other system.

PDR Flushing water and residues removal system, incl. neutralization.

from excl. outlet of respective system
to excl. discharge into disposal system.

PDS Sludge thickening system.

from excl. outlet of respective system
to excl. discharge into other system.

PDT Heating, cooling and flushing fluid distribution system.

from incl. heating, cooling, flushing fluid
generation equipment or
from excl. branch off heating, cooling,
flushing fluid supply system
to excl. user and
from excl. user.

PDV Lubricant supply system.

PDX Fluid supply system for control and protection equipment.

PDY Control and protection equipment.

PG Closed cooling water system.

PGA Closed cooling water piping system (forward).

from excl. discharge nozzle of closed
cooling water pump system
to excl. user, incl. blowdown, discharge,
venting
to excl. inlet into other system excl.
intercooler (see *PGD*).

PGB Closed cooling water piping system (return).

from excl. outlet user
to excl. closed cooling water pump system
suction nozzles incl. blowdown, discharge,
venting
to excl. inlet to other system.

PGC Closed cooling water pump system.

from incl. pump system suction nozzles
to incl. pump system discharge nozzles
incl. discharge, flushing, venting
to excl. inlet into other system.

PGD Intercooler.

from incl. intercooler inlet nozzle
to incl. intercooler discharge nozzle, incl.
discharge, venting
to excl. inlet to other system incl. safety
valves with their blow-off lines
to excl. inlet to other system (secondary
cooling water side safety valves only when
fitted directly to the cooler, otherwise on
system *PCB*).

PGF Pressure system.

from excl. branch of piping system forward
(*PGA*)
from incl. first shut-off valve of control
station in make-up water supply
to excl. inlet to piping system return
(*PGB*) incl. head tank with associated
equipment incl. control station in make-up
watersupply incl. discharge, venting
to excl. inlet to other system.

PGH Closed cooling water system for main groups *E* and *H*.

PGL Closed cooling water system for main groups *G*, *L* and *P*.

PGM Closed cooling water system for main groups *B*, *M*, and *X*.

PGN Closed cooling water system for main group *N*.

PGQ Closed cooling water system for main groups "Q" and "S".

PGR Closed cooling water system for main group *R*.

COOLING WATER SYSTEMS

- PGX** Fluid supply system for control and protection equipment.
PGY Control and protection equipment.

PH Closed cooling water treatment system for conventional area.

- PHA** Discharge from fluid treatment system (if not part of another system).
 from excl. fluid treatment system outlet
 to excl. inlet to other system.
- PHB** Filtering, mechanical cleaning system.
 from incl. separation equipment inlet
 to incl. separation equipment outlet.
- PHC** Aeration, gas injection system.
 from excl. atmosphere or
 from incl. gas supply.
- PHD** Precipitation system (e.g. for carbonate hardness removal).
 from incl. precipitation equipment inlet
 to incl. precipitation equipment outlet.
- PHE** Acid proportioning system (e.g. for carbonate hardness removal).
 from incl. acid proportioning equipment or
 from excl. branch off chemicals supply system
 to excl. inlet to other system.
- PHF** Ion exchangesystem (e.g. for demineralization).
 from incl. ion exchanger inlet and
 to incl. isolating valve of chemicals supply system or auxiliary fluid supply system upstream of ion exchanger.
- PHG** Evaporation system (e.g. for demineralization).
 from incl. feedwater inlet
 to incl. steam outlet and
 from incl. heating steam inlet
 to incl. condensate outlet.
- PHH** Deaeration system.
 from incl. deaerator or tank inlet
 to incl. tank outlet, incl. warm-up equipment of vapour condenser.
- PHJ** Preheating, cooling system.
 from incl. preheater or cooler inlet
 to incl. preheater or cooler outlet.

PHK Piping system, temporary storage system, pump system for main fluid.

Piping system:

from excl. intake or
 from excl. outlet of other system
 to excl. inlet to other system
 to incl. fluid treatment system outlet.

Temporary storage system:

from incl. temporary storage system inlet
 to incl. temporary storage system outlet.

Pump system:

from incl. pump system suction nozzle
 to incl. pump system discharge nozzle.

PHL Storage system outside fluid treatment system (if not part of another system).

from incl. storage system inlet
 to incl. storage system outlet, incl. intake and outfall.

PHN Chemicals supply system.

from incl. intake or
 from incl. storage tank
 to excl. discharge into other system.

PHP Regeneration, flushing equipment.

from incl. system inlet
 to excl. inlet to other system
 from excl. chemicals or auxiliary fluid supply system and flushing air supply system
 to incl. regenerating, flushing equipment.

PHQ Injection system for main fluid.

from incl. injection equipment or
 from excl. branch off chemicals supply system
 to excl. inlet to other system.

PHR Flushing water and residues removal system, incl. neutralization.

from excl. outlet of respective system
 to excl. discharge into disposal system.

PHS Sludge thickening system.

from excl. outlet of respective system
 to excl. discharge into other system.

PHT Heating, cooling and flushing fluid distribution system.

from incl. heating, cooling, flushing fluid generation equipment or
 from excl. branch off heating, cooling, flushing fluid supply system
 to excl. user and
 from excl. user.

PHV Lubricant supply system.

PHX Fluid supply system for control and protection equipment.

PHY Control and protection equipment.

COOLING WATER SYSTEMS

PM Closed cooling water system for transformers (if separate from closed cooling water system).

PMA Closed cooling water system for transformers (if separate from closed cooling water system).
(free for use).

PMB Closed cooling water system for transformers (if separate from closed cooling water system).
(free for use).

PMC Closed cooling water system for transformers (if separate from closed cooling water system).
(free for use).

PMD Closed cooling water system for transformers (if separate from closed cooling water system).
(free for use).

PME Closed cooling water system for transformers (if separate from closed cooling water system).
(free for use).

PMF Closed cooling water system for transformers (if separate from closed cooling water system).
(free for use).

PMG Closed cooling water system for transformers (if separate from closed cooling water system).
(free for use).

PMH Closed cooling water system for transformers (if separate from closed cooling water system).
(free for use).

PMJ Closed cooling water system for transformers (if separate from closed cooling water system).
(free for use).

PMK Closed cooling water system for transformers (if separate from closed cooling water system).
(free for use).

PML Closed cooling water system for transformers (if separate from closed cooling water system).
(free for use).

PMM Closed cooling water system for transformers (if separate from closed cooling water system).
(free for use).

PMN Closed cooling water system for transformers (if separate from closed cooling water system).
(free for use).

PMP Closed cooling water system for transformers (if separate from closed cooling water system).
(free for use).

PMQ Closed cooling water system for transformers (if separate from closed cooling water system).
(free for use).

PMR Closed cooling water system for transformers (if separate from closed cooling water system).
(free for use).

PMS Closed cooling water system for transformers (if separate from closed cooling water system).
(free for use).

PMT Closed cooling water system for transformers (if separate from closed cooling water system).
(free for use).

PMU Closed cooling water system for transformers (if separate from closed cooling water system).
(free for use).

PMV Lubricant supply system.

PMX Fluid supply system for control and protection equipment.

PMY Control and protection equipment.

PS Cooling tower blowdown system (if separate from *PAB*).

PSA Cooling tower blowdown system.
(free for use).

PSB Cooling tower blowdown system.
(free for use).

PSC Cooling tower blowdown system (if separate from *PAB*).*
(free for use).

PSD Cooling tower blowdown system (if separate from *PAB*).*
(free for use).

PSE Cooling tower blowdown system (if separate from *PAB*).*
(free for use).

PU Common equipment for cooling water systems.

PUC Defrosting equipment.

PUD Fish barrier.

PUE Boat barrier.

COOLING WATER SYSTEMS

PUG Debris filter equipment.
PUH Corrosion protection for cooling water purification.
PUK Cooling water biocide treatment.
PUN Proportioning equipment.

PV Lubricant supply system.

PVA Lubricant supply system.
 (free for use).
PVB Lubricant supply system.
 (free for use).
PVC Lubricant supply system.
 (free for use).
PVD Lubricant supply system.
 (free for use).

PW Sealing fluid supply system.

PWA Sealing fluid supply system.
 (free for use).
PWB Sealing fluid supply system.
 (free for use).
PWC Sealing fluid supply system.
 (free for use).
PWD Sealing fluid supply system.
 (free for use).
PWE Sealing fluid supply system.
 (free for use).
PWF Sealing fluid supply system.
 (free for use).
PWG Sealing fluid supply system.
 (free for use).
PWH Sealing fluid supply system.
 (free for use).
PWJ Sealing fluid supply system.
 (free for use).
PWK Sealing fluid supply system.
 (free for use).
PWL Sealing fluid supply system.
 (free for use).
PWM Sealing fluid supply system.
 (free for use).
PWN Sealing fluid supply system.
 (free for use).
PWP Sealing fluid supply system.
 (free for use).
PWQ Sealing fluid supply system.
 (free for use).
PWR Sealing fluid supply system.
 (free for use).
PWS Sealing fluid supply system.
 (free for use).
PWT Sealing fluid supply system.
 (free for use).

PWU Sealing fluid supply system.
 (free for use).

PX Fluid supply system for control and protection equipment.

PXA Fluid supply system for control and protection equipment.
 (free for use).
PXB Fluid supply system for control and protection equipment.
 (free for use).
PXC Fluid supply system for control and protection equipment.
 (free for use).
PXD Fluid supply system for control and protection equipment.
 (free for use).

PY Control and protection equipment.

PYA Control and protection equipment.
 (free for use).
PYB Control and protection equipment.
 (free for use).
PYC Control and protection equipment.
 (free for use).
PYD Control and protection equipment.
 (free for use).



AUXILIARY SYSTEMS

- QC** Central chemicals supply.
- QE** General compressed air and carrier air supply.
- QF** General control air supply.
- QG** Central gas supply for closed gas cycles (as working fluid).
- QH** Auxiliary steam generating system.
- QJ** Central gas supply, also inert gas.
- QK** Chilled water systems for conventional area.
- QL** Feedwater, steam, condensate cycle of auxiliary steam generating and distribution system.
- QM** Air humidifying system.
- QS** Central oil supply and disposal system (for systems assignable to more than one F1-function).
- QU** Sampling systems for conventional area.



AUXILIARY SYSTEMS

QC Central chemicals supply.

- QCA** Common filling equipment for liquid chemical products.
- QCB** Common filling equipment for solid chemical products.
- QCC** Hydrazine supply and distribution.
- QCD** Phosphate supply and distribution.
- QCE** Ammonia supply and distribution.
- QCF** Iron-sulphate supply and distribution.
- QCG** Sodium-nitrite supply and distribution.
- QCH** Sulphuric acid supply and distribution.
- QCJ** Ammonia storage and feed for HRSG catalytic converter *HSJ*.
- QCK** Hypochlorite generation, supply and distribution.
- QCV** Lubricant supply system.
- QCX** Fluid supply system for control and protection equipment.
- QCY** Control and protection equipment.

QE General compressed air and carrier air supply.

- QEA** Central compressed air and carrier air generation system.
- QEB** Central compressed air and carrier air distribution system.
- QEE** Compressed air and carrier air supply system for main group *E*.
- QEH** Compressed air and carrier air supply system for main group *H*.
- QEL** Compressed air and carrier air supply system for main groups *G*, *L* and *P*.
- QEM** Compressed air and carrier air supply system for main groups *B* and *M*.
- QEN** Central compressed air and carrier air supply for main group *N*.
- QEQ** Compressed air and carrier air supply system for main groups *Q*.
- QER** Central compressed air and carrier air supply for main group *R*.
- QES** Compressed air and carrier air supply system for main group *S*.
- QEV** Lubricant supply system.
- QEW** Sealing fluid supply system.
- QEX** Fluid supply system for control and protection equipment.
- QEY** Control and protection equipment.

QF General control air supply.

- QFA** Central control air generation system.
- QFB** Central control air distribution system.
- QFE** Control air supply system for main group *E*.
- QFF** Control air treatment system.
- QFH** Control air supply system for main group *H*.
- QFL** Control air supply system for main groups *G*, *L* and *P*.
- QFM** Control air supply system for main groups *B* and *M*.
- QFN** General control air supply for main group *N*.
- QFQ** Control air supply system for main group *Q*.
- QFR** General control air supply for main group *R*.
- QFS** Control air supply system for main group *S*.
- QFV** Lubricant supply system.
- QFW** Sealing fluid supply system.
- QFX** Fluid supply system for control and protection equipment.
- QFY** Control and protection equipment.

QG Central gas supply for closed gas cycles (as working fluid).

- QGA** Central gas supply for closed gas cycles (as working fluid).
(free for use).
- QGB** Central gas supply for closed gas cycles (as working fluid).
(free for use).
- QGC** Central gas supply for closed gas cycles (as working fluid).
(free for use).
- QGD** Central gas supply for closed gas cycles (as working fluid).
(free for use).
- QGE** Central gas supply for closed gas cycles (as working fluid).
(free for use).
- QGF** Central gas supply for closed gas cycles (as working fluid).
(free for use).
- QGG** Central gas supply for closed gas cycles (as working fluid).
(free for use).

AUXILIARY SYSTEMS

QGH	Central gas supply for closed gas cycles (as working fluid). (free for use).	QHG	Boiler water circulation system (also for electrode steam boiler).
QGJ	Central gas supply for closed gas cycles (as working fluid). (free for use).	QHH	Main firing system (also for electric heating).
QGK	Central gas supply for closed gas cycles (as working fluid). (free for use).	QHJ	Ignition firing equipment (if separate).
QGL	Central gas supply for closed gas cycles (as working fluid). (free for use).	QHL	Combustion air system (primary air, secondary air).
QGM	Central gas supply for closed gas cycles (as working fluid). (free for use).	QHM	Gas heating system (for closed cycle).
QGN	Central gas supply for closed gas cycles (as working fluid). (free for use).	QHN	Flue gas exhaust (without flue gas treatment).
QGP	Central gas supply for closed gas cycles (as working fluid). (free for use).	QHP	Mechanical dust handling system.
QGQ	Central gas supply for closed gas cycles (as working fluid). (free for use).	QHQ	Electrostatic precipitator.
QGR	Central gas supply for closed gas cycles (as working fluid). (free for use).	QHR	Chemical flue gas treatment system incl. residues removal adsorptive process.
QGS	Central gas supply for closed gas cycles (as working fluid). (free for use).	QHS	Chemical flue gas treatment system incl. residues removal catalytic process.
QGT	Central gas supply for closed gas cycles (as working fluid). (free for use).	QHT	Chemical flue gas treatment system incl. residues removal absorptive process.
QGU	Central gas supply for closed gas cycles (as working fluid). (free for use).	QHU	Flue gas reheating system.
QGV	Lubricant supply system.	QHV	Lubricant supply system.
QGW	Sealing fluid supply system.	QHX	Fluid supply system for control and protection equipment.
QGX	Fluid supply system for control and protection equipment.	QHY	Control and protection equipment.
QGY	Control and protection equipment.	QJ	Central gas supply, also inert gas.
QH	Auxiliary steam generating system.	QJB	Gas supply and distribution system O₂ (oxygen).
QHA	Pressure system.	QJD	Gas supply and distribution system N₂ (nitrogen).
QHB	Support structure, enclosure, steam generator interior.	QJE	Gas generation system N₂ (nitrogen).
QHC	Fireside heat transfer surface cleaning equipment.	QJF	Gas supply and distribution system H₂ (hydrogen).
QHD	Ash and slag removal.	QJG	Central gas supply, gas generation system H₂ (hydrogen).
QHE	Blowdown system, flash drain system.	QJH	Gas supply and distribution system CH₄ (methane).
QHF	Bunker, feeder and pulverizing system.	QJK	Gas supply and distribution system CO₂ (carbon dioxide).
		QJM	Gas supply and distribution system C₂H₂ (acetylene).
		QJN	Gas supply and distribution system He (helium).
		QJP	Gas supply and distribution system C₃H₈ (propane gas).
		QJQ	Gas supply and distribution system AR (argon).
		QJR	Gas supply and distribution system N₂O (laughing gas).

AUXILIARY SYSTEMS

QJV	Lubricant supply system.	QKR	Chilled water systems for conventional area. (free for use e.g. building specific).
QJW	Sealing fluid supply system.	QKS	Chilled water systems for conventional area. (free for use e.g. building specific).
QJX	Fluid supply system for control and protection equipment.	QKT	Chilled water systems for conventional area. (free for use e.g. building specific).
QJY	Control and protection equipment.	QKU	Chilled water systems for conventional area. (free for use e.g. building specific).
QK	Chilled water systems for conventional area.	QKV	Lubricant supply system.
QKA	Chilled water systems for conventional area. (free for use e.g. building specific).	QKX	Fluid supply system for control and protection equipment.
QKB	Chilled water systems for conventional area. (free for use e.g. building specific).	QKY	Control and protection equipment.
QKC	Chilled water systems for conventional area. (free for use e.g. building specific).	QL	Feedwater, steam, condensate cycle of auxiliary steam generating and distribution system.
QKD	Chilled water systems for conventional area. (free for use e.g. building specific).	QLA	Feedwater system.
QKE	Chilled water systems for conventional area. (free for use e.g. building specific).	QLB	Steam system.
QKF	Chilled water systems for conventional area. (free for use e.g. building specific).	QLC	Condensate system.
QKG	Chilled water systems for conventional area. (free for use e.g. building specific).	QLD	Condensate polishing plant.
QKH	Chilled water systems for conventional area. (free for use e.g. building specific).	QLF	Common equipment for auxiliary steam generation and distribution systems.
QKJ	Chilled water systems for conventional area. (free for use e.g. building specific).	QLV	Lubricant supply system.
QKK	Chilled water systems for conventional area. (free for use e.g. building specific).	QLX	Fluid supply system for control and protection equipment.
QKL	Chilled water systems for conventional area. (free for use e.g. building specific).	QLY	Control and protection equipment.
QKM	Chilled water systems for conventional area. (free for use e.g. building specific).	QM	Air humidifying system.
QKN	Chilled water systems for conventional area. (free for use e.g. building specific).	QMA	Air humidifying system. (free for use e.g. building specific).
QKP	Chilled water systems for conventional area. (free for use e.g. building specific).	QMB	Air humidifying system. (free for use e.g. building specific).
QKQ	Chilled water systems for conventional area. (free for use e.g. building specific).	QMC	Air humidifying system. (free for use e.g. building specific).
		QMD	Air humidifying system. (free for use e.g. building specific).
		QME	Air humidifying system. (free for use e.g. building specific).
		QMF	Air humidifying system. (free for use e.g. building specific).
		QMG	Air humidifying system. (free for use e.g. building specific).
		QMH	Air humidifying system. (free for use e.g. building specific).
		QMJ	Air humidifying system. (free for use e.g. building specific).
		QMK	Air humidifying system. (free for use e.g. building specific).

AUXILIARY SYSTEMS

- QML Air humidifying system.**
(free for use e.g. building specific).
- QMM Air humidifying system.**
(free for use e.g. building specific).
- QMN Air humidifying system.**
(free for use e.g. building specific).
- QMP Air humidifying system.**
(free for use e.g. building specific).
- QMQ Air humidifying system.**
(free for use e.g. building specific).
- QMR Air humidifying system.**
(free for use e.g. building specific).
- QMS Air humidifying system.**
(free for use e.g. building specific).
- QMT Air humidifying system.**
(free for use e.g. building specific).
- QMU Air humidifying system.**
(free for use e.g. building specific).
- QMV Lubricant supply system.**
- QMX Fluid supply system for control and protection equipment.**
- QMY Control and protection equipment.**
- QS Central oil supply and disposal system (for systems assignable to more than one F1-function).**
- QSA Central oil supply and disposal system (for systems assignable to more than one F1-function).**
(free for use).
- QSB Central oil supply and disposal system (for systems assignable to more than one F1-function).**
(free for use).
- QSC Central oil supply and disposal system (for systems assignable to more than one F1-function).**
(free for use).
- QSD Central oil supply and disposal system (for systems assignable to more than one F1-function).**
(free for use).
- QSE Central oil supply and disposal system (for systems assignable to more than one F1-function).**
(free for use).
- QSF Central oil supply and disposal system (for systems assignable to more than one F1-function).**
(free for use).
- QSG Central oil supply and disposal system (for systems assignable to more than one F1-function).**
(free for use).
- QSH Central oil supply and disposal system (for systems assignable to more than one F1-function).**
(free for use).
- QSJ Central oil supply and disposal system (for systems assignable to more than one F1-function).**
(free for use).
- QSK Central oil supply and disposal system (for systems assignable to more than one F1-function).**
(free for use).
- QSL Central oil supply and disposal system (for systems assignable to more than one F1-function).**
(free for use).
- QSM Central oil supply and disposal system (for systems assignable to more than one F1-function).**
(free for use).
- QSN Central oil supply and disposal system (for systems assignable to more than one F1-function).**
(free for use).
- QSP Central oil supply and disposal system (for systems assignable to more than one F1-function).**
(free for use).
- QSQ Central oil supply and disposal system (for systems assignable to more than one F1-function).**
(free for use).
- QSR Central oil supply and disposal system (for systems assignable to more than one F1-function).**
(free for use).
- QSS Central oil supply and disposal system (for systems assignable to more than one F1-function).**
(free for use).
- QST Central oil supply and disposal system (for systems assignable to more than one F1-function).**
(free for use).
- QSU Central oil supply and disposal system (for systems assignable to more than one F1-function).**
(free for use).
- QSV Lubricant supply system.**
- QSX Fluid supply system for control and protection equipment.**
- QSY Control and protection equipment.**

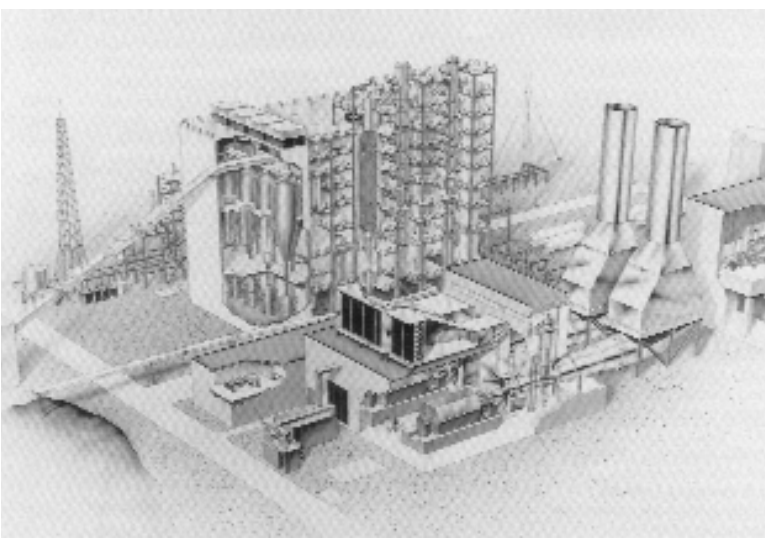
QU Sampling systems for conventional area.

- QUE** Sampling systems for main group *E*.
- QUH** Sampling systems for main group *H* (gas-side).
- QUL** Sampling systems for main group *H* (steam-water-side), *L* and *M*.
- QUN** Sampling systems for main group *N*.
- QUP** Sampling systems for main groups *G* and *P*.
- QUQ** Sampling systems for main group *Q*.
- QUR** Sampling systems for main group *R*.
- QUS** Sampling systems for main group *S*.
- QUX** Fluid supply system for control and protection equipment.
- QUY** Control and protection equipment.

R

GAS GENERATION AND TREATMENT

- RA Gas generation.**
- RB Support structure.**
- RC Feedstock systems.**
- RD Discharge systems for gasification residues.**
- RE Gasifying agent generation and distribution.**
- RG Main gas cooling systems (if not *RA*).**
- RH Main gas piping systems, storage, compression, expansion.**
- RJ Main gas precipitator.**
- RK Main gas clean-up (not *RJ*) including regeneration.**
- RL Acid gas, including treatment systems.**
- RM Gas recycle, storage and compression systems.**
- RN Collection, storage and recycle systems for gas condensate.**
- RP Inert gas, including recovery systems.**
- RS Supply and removal systems for water, steam and condensate.**
- RT Waste water collection and drainage systems.**
- RU Waste water treatment systems.**
- RV Lubricant supply systems.**
- RW Sealing fluid supply systems.**
- RX Fluid supply systems for control and protection equipment.**
- RY Control and protection equipment.**
- RZ Injection and proportioning systems.**



GAS GENERATION AND TREATMENT

RA Gas generation.

RAA Gas generator.

from incl. fuel, ignition fuel and fluxing agent inlet
or from excl. burner and nozzle system
to excl. removal system or
to excl. solids separator and recycle system
or
to excl. gas generator quench gas section
or
to incl. main gas outlet.

RAB Gas generator interior.

RAC Gas generator internals.

RAD Lining and refractory brickwork.

RAE Cooling system.

from incl. gas generator cooling system inlet
to incl. gas generator cooling system outlet.

RAF Solids separator and recycle system.

from excl. gas generator outlet
to excl. gas generator inlet
to incl. main gas outlet.

RAG Gas generator quench gas section.

from incl. quench gas section inlet
to incl. quench gas section outlet.

RAH Main burner system.

from excl. feedstock system or
from excl. gasifying agent generation and distribution system
to excl. gas generator interior.

RAJ Pilot burner system.

from excl. pilot burner fuel storage and distribution system or
from excl. gasifying agent generation and distribution system
to excl. gas generator interior.

RAK Gas generator ignition and standby burner system incl. combustion air system.

from excl. ignition fuel storage and distribution system or
from incl. combustion air inlet
to excl. gas generator interior.

RAL Ignition and standby hot gas generation and feed system incl. combustion air system.

from excl. ignition fuel storage and distribution system or
from incl. combustion air inlet
to excl. gas generator interior.

RAM Separate gasifying agent injection system.

from excl. gasifying agent generation and distribution system
to excl. gas generator interior.

RAN Injection from gas recycle system.

from excl. gas recycle system
to excl. gas generator interior.

RAP Gas condensate injection system.

from excl. gas recycle system for gas condensate
to excl. gas generator interior.

RAQ Injection system for other feedstock.

from excl. other feedstock system
to excl. gas generator interior.

RAR Fluxing agent injection system.

from excl. feed system for fluxing agents
to excl. gas generator interior.

RAS Filter cleaning system.

from excl. branch off supply system.

RAT Fluidizing air injection system.

from excl. fluidizing gas piping system
to excl. gas generator interior.

RAV Lubricant supply system.

RAW Sealing fluid supply system.

RAX Fluid supply system for control and protection equipment.

RAY Control and protection equipment.

RB Support structure.

RBA Frame including foundations.

RBB Enclosures, insulations.

RBD Platforms and stairways.

RBE Support structure.

(free for use as process demands).

RBF Support structure.

(free for use as process demands).

RC Feedstock systems.

RCA Fuel storage bunker, hopper (atmospheric pressure).

from excl. receiving point
to excl. distribution and transport system (atmospheric pressure).

RCB Distribution and transport system (atmospheric pressure).

from excl. fuel storage bunker, hopper outlet
to excl. pulverizing and mixing system (slurry) or
to excl. air lock system, slurry pump system.

RCC Pulverizing and mixing system (slurry).

from excl. distribution and transport system (atmospheric pressure)
to excl. airlock system, slurry pump system.

GAS GENERATION AND TREATMENT

RCE	Airlock system, slurry pump system. from excl. distribution and transport system (atmospheric pressure) or from excl. pulverizing and mixing system (slurry) to excl. fuel hopper (pressurized).	RDB	Pressure relief system (system 1). from excl. collection, transport and treatment system (pressurized) to excl. collection, transport and treatment system (atmospheric pressure).
RCF	Fuel storage bunker, container (pressurized). from excl. airlock system, slurry pump system to excl. distribution and transport system (pressurized).	RDC	Collection, transport and treatment system (atmospheric pressure) (system 1). from excl. pressure relief system to excl. transport and treatment system for fuel conversion residues or to excl. feedstock system.
RCG	Distribution and transport system (pressurized). from excl. fuel hopper (pressurized) to excl. gas generator or to excl. main burner system.	RDE	Collection, transport and treatment system (pressurized) (system 2). from excl. receiving point to excl. gas generator or to excl. pressure relief system.
RCJ	Ignition fuel storage and distribution system. from excl. ignition fuel supply system to excl. gas generator or to excl. gas generator ignition and standby burner system or to excl. ignition and standby hot gas generation and feed system or to excl. feedstock system.	RDF	Pressure relief system (system 2). from excl. collection, transport and treatment system (pressurized) to excl. collection, transport and treatment system (atmospheric pressure).
RCK	Pilot burner fuel storage and distribution system. from excl. fuel supply system to excl. pilot burner system.	RDG	Collection, transport and treatment system (atmospheric pressure) (system 2). from excl. pressure relief system to excl. transport and treatment system for fuel conversion residues or to excl. feedstock system.
RCL	Proportioning system for other feedstock. from excl. supply system to excl. gas generator or to excl. feedstock system.	RDJ	Collection, transport and treatment system (pressurized) (system 3). from excl. receiving point to excl. gas generator or to excl. pressure relief system.
RCM	Proportioning system for fluxing agents. from excl. supply system to excl. gas generator or to excl. feed system for fluxing agents.	RDK	Pressure relief system (system 3). from excl. collection, transport and treatment system (pressurized) to excl. collection, transport and treatment system (atmospheric pressure).
RCS	Filter cleaning system. from excl. branch off supply system.	RDL	Collection, transport and treatment system (atmospheric pressure) (system 3). from excl. pressure relief system to excl. transport and treatment system for fuel conversion residues or to excl. feedstock system.
RCV	Lubricant supply system.	RDN	Collection, transport and treatment system (pressurized) (system 4). from excl. receiving point to excl. gas generator or to excl. pressure relief system.
RCW	Sealing fluid supply system.	RDP	Pressure relief system (system 4). from excl. collection, transport and treatment system (pressurized) to excl. collection, transport and treatment system (atmospheric pressure).
RCX	Fluid supply system for control and protection equipment.		
RCY	Control and protection equipment.		
RD	Discharge systems for gasification residues.		
RDA	Collection, transport and treatment system (pressurized) (system 1). from excl. receiving point to excl. gas generator or to excl. pressure relief system.		

GAS GENERATION AND TREATMENT

RDQ	Collection, transport and treatment system (atmospheric pressure) (system 4). from excl. pressure relief system to excl. transport and treatment system for fuel conversion residues or to excl. feedstock system.	REF	Air separation plant. from excl. air (gasifying agent) compression and supercompression system outlet upstream of air separation plant to excl. oxygen piping system (pressurized) or to excl. oxygen compression and supercompression system or to excl. liquid oxygen discharge sys. or to excl. discharge sys. for other gases (inert gases) from air separation plant or to excl. liquid nitrogen discharge sys. or to excl. inert gas, including recovery sys. or to excl. nitrogen piping sys. (*RPG*).
RDS	Filter cleaning system. from excl. branch off supply system.	REG	Oxygen compression and supercompression system. from incl. oxygen compression and supercompression system inlet to incl. oxygen compression and supercompression system outlet.
RDV	Lubricant supply system.	REH	Oxygen piping system (pressurized). from excl. air separation plant or from excl. oxygen compression and supercompression system or from excl. other system to excl. other system or to excl. main burner system or to excl. pilot burner system or to excl. separate gasifying agent injection system.
RDW	Sealing fluid supply system.	REJ	Liquid oxygen outlet system. from excl. air separation plant to excl. oxygen piping system (pressurized) or to excl. other system.
RDX	Fluid supply system for control and protection equipment.	REK	Discharge system for other gases from air separation plant (inert gases). from excl. air separation plant to incl. dispatch facility or to excl. other system.
RDY	Control and protection equipment.	REL	Liquid nitrogen discharge system. from excl. air separation plant to excl. inert gas, including recovery system or to excl. other system or to incl. dispatch facility.
RE	Gasifying agent generation and distribution.	REP	External oxygen supply system. from excl. receiving point to excl. oxygen compression and supercompression system or to excl. oxygen piping system (pressurized).
REA	Air (gasifying agent) compression and supercompression system. from excl. atmosphere or from excl. compressor casing outlet of gas turbine to excl. compressed air distribution system or to excl. gasifying agent heater.		
REB	Gasifying agent heater (other than *RGC*). from incl. preheater inlet to incl. preheater outlet or to incl. preheater outlet incl. desuperheater.		
REC	Compressed air distribution system (incl. oxygen-enriched air). from excl. air (gasifying agent) compression and supercompression system outlet or from excl. oxygen piping system (pressurized) or from excl. other system to excl. other system or to excl. main burner system or to excl. pilot burner system or to excl. separate gasifying agent injection system.		
REE	Air compression and supercompression system upstream of air separation plant. from excl. atmosphere or from excl. compressor casing outlet of gas turbine to incl. air compression and supercompression system outlet upstream of air separation plant.		

GAS GENERATION AND TREATMENT

- RER Gasifying steam system.**
from excl. water/steam system or
from excl. other system
to excl. main burner system or
to excl. pilot burner system or
to excl. separate gasifying agent injection
system.
- REV Lubricant supply system.**
- REW Sealing fluid supply system.**
- REX Fluid supply system for control.
and protection equipment.**
- REY Control and protection equipment.**

RG Main gas cooling systems (if not *RA*).

- RGA Indirect heat transfer to
water/steam cycle.**
from incl. main gas inlet or
from incl. water/steam inlet
to incl. main gas outlet or
to incl. water/steam outlet or
to incl. gas condensate outlet.
- RGB Indirect heat transfer to gas.**
from incl. main gas inlet or
from incl. gas recycle system inlet or
from incl. inert gas inlet
to incl. main gas outlet or
to incl. gas recycle system outlet or
to incl. inert gas outlet or
to incl. gas condensate outlet.
- RGC Indirect heat transfer to gasifying
agent.**
from incl. main gas inlet or
from incl. gasifying agent inlet
to incl. main gas outlet or
to incl. gasifying agent outlet or
to incl. gas condensate outlet.
- RGD Indirect heat transfer to combustion
air.**
from incl. main gas inlet or
from incl. combustion air inlet
to incl. main gas outlet or
to incl. combustion air outlet or
to incl. gas condensate outlet.
- RGE Indirect heat transfer to cooling
water.**
from incl. main gas inlet or
from incl. cooling water inlet
to incl. main gas outlet or
to incl. cooling water outlet or
to incl. gas condensate outlet.
- RGJ Direct heat transfer (quench) to gas.**
from incl. main gas inlet or
from incl. coolant inlet
from excl. coolant supply line
to incl. main gas outlet.

- RGK Direct heat transfer (quench,
saturation) to steam.**
- RGL Direct heat transfer (quench,
saturation) to water.**
- RGM Direct heat transfer (quench) to
other fluids.**
- RGS Filter cleaning system from excl.
branch off supply system.**
- RGV Lubricant supply system.**
- RGW Sealing fluid supply system.**
- RGX Fluid supply system for control
and protection equipment.**
- RGY Control and protection equipment.**
- RGZ Injection and proportioning system.**

RH Main gas piping systems, storage, compression, expansion.

- RHA Main gas piping system.**
from excl. other system
to excl. other system.
- RHE Humidification system.**
from excl. main gas piping system or
from excl. water/steam cycle or
from excl. collection, storage and recycle
system for gas condensate
to excl. main gas piping system or
to excl. water/steam cycle or
to excl. collection, storage and recycle sys-
tem for gas condensate.
- RHF Conditioning system 1.**
from excl. branch off supply system or
from incl. unloading facility.
- RHG Conditioning system 2.**
from excl. branch off supply system or
from incl. unloading facility.
- RHK Gas storage system.**
from excl. main gas piping system or
from excl. gas compression system
to excl. main gas piping system or
to excl. gas pressure reducing system.
- RHM Gas compression system.**
from excl. main gas piping system
to excl. main gas piping system or
to excl. gas storage system or
to excl. gas recycle system.
- RHN Gas pressure reducing system.**
from excl. main gas piping system or
from excl. gas storage system
to excl. main gas piping system or
to excl. other system.
- RHP Flare system 1.**
from excl. main gas piping system
to incl. flare system outlet.

GAS GENERATION AND TREATMENT

RHQ	Flare system 2. from excl. main gas piping system to incl. flare system outlet.	RK	Main gas clean-up (not *RJ*) including regeneration.
RHS	Filter cleaning system. from excl. branch off supply system.	RKA	Scrubber system. from incl. inlet to incl. outlet.
RHV	Lubricant supply system.	RKB	Conversion system including hydrolysis. from incl. inlet to incl. outlet.
RHW	Sealing fluid supply system.	RKC	Gas purification system. from incl. inlet to incl. outlet.
RHX	Fluid supply system for control and protection equipment.	RKG	Scrubber regeneration system 1. from incl. inlet to incl. outlet.
RHY	Control and protection equipment.	RKH	Scrubber regeneration system 2. from incl. inlet to incl. outlet.
RHZ	Injection and proportioning system.	RKJ	Scrubber regeneration system 3. from incl. inlet to incl. outlet.
RJ	Main gas precipitator.	RKL	Regeneration system 1 for purification system. from incl. inlet to incl. outlet.
RJA	Cartridge filter system. from incl. cartridge filter inlet from excl. filter cleaning system to incl. cartridge filter outlet to excl. gasification residues removal system.	RKM	Regeneration system 2 for purification system. from incl. inlet to incl. outlet.
RJB	Cyclone filter system. from incl. cyclone filter inlet from excl. filter cleaning system to incl. cyclone filter outlet to excl. gasification residues removal system.	RKN	Regeneration system 3 for purification system. from incl. inlet to incl. outlet.
RJC	Bag filter system. from incl. bag filter inlet from excl. filter cleaning system to incl. bag filter outlet to excl. gasification residues removal system.	RKQ	Refrigeration system. from incl. inlet to incl. outlet.
RJD	Packed-bed filter system. from incl. packed-bed filter inlet from excl. filter cleaning system to incl. packed-bed filter outlet to excl. gasification residues removal system.	RKV	Lubricant supply system.
RJE	Electrostatic precipitator system. from incl. electrostatic precipitator inlet from excl. filter cleaning system to incl. electrostatic precipitator outlet to excl. gasification residues removal system.	RKW	Sealing fluid supply system.
RJS	Filter cleaning system. from excl. branch off supply system.	RKX	Fluid supply system for control and protection equipment.
RJV	Lubricant supply system.	RKY	Control and protection equipment.
RJW	Sealing fluid supply system.	RKZ	Injection and proportioning system.
RJX	Fluid supply system for control and protection equipment.	RL	Acid gas, including treatment systems.
RJY	Control and protection equipment.	RLA	Thermal treatment (Claus process). from incl. inlet to incl. outlet.
		RLB	Catalytic treatment (Claus process). from incl. inlet to incl. outlet.
		RLE	Absorber, wet oxidation system. from incl. inlet to incl. outlet.

GAS GENERATION AND TREATMENT

RLH	Thermal treatment (tail gas). from incl. inlet to incl. outlet.	RMF	Other gas piping system. from excl. branch off other system to excl. other system.
RLJ	Catalytic treatment (tail gas). from incl. inlet to incl. outlet.	RMH	Storage system 1. from excl. store inlet to excl. other gas piping system.
RLK	Absorption treatment (tail gas). from incl. inlet to incl. outlet.	RMJ	Storage system 2. from excl. store inlet to excl. other gas piping system.
RLN	Sulphur treatment system. from incl. inlet to incl. outlet.	RMM	Quench gas compressor system. from incl. compressor system inlet to incl. compressor system outlet.
RLP	Sulphur forwarding and storage system. from incl. inlet to incl. outlet.	RMN	Expanded gas compressor system. from incl. compressor system inlet to incl. compressor system outlet.
RLR	Sulphuric acid system. from incl. inlet to incl. outlet.	RMP	Acid gas compressor system. from incl. compressor system inlet to incl. compressor system outlet.
RLS	Sulphuric acid storage system. from incl. inlet to incl. outlet.	RMQ	Fluidizing gas compressor system. from incl. compressor system inlet to incl. compressor system outlet.
RLW	Sealing fluid supply system.	RMR	Airlock gas compressor system (not inert gas). from incl. compressor system inlet to incl. compressor system outlet.
RLX	Fluid supply system for control and protection equipment.	RMS	Compressor system for other gases. from incl. compressor system inlet to incl. compressor system outlet.
RLY	Control and protection equipment.	RMV	Lubricant supply system.
RLZ	Injection and proportioning system.	RMW	Sealing fluid supply system.
RM	Gas recycle, storage and compression systems.	RMX	Fluid supply system for control and protection equipment.
RMA	Quench gas piping system. from excl. main gas piping system or from excl. branch off other system to excl. direct heat transfer (quench) to gas or to excl. gas generator quench gas section.	RMY	Control and protection equipment.
RMB	Piping system for expanded gas (not from airlock system). from excl. pressure reducing valve to excl. other system (excl. quench gas compressor).	RMZ	Injection and proportioning system.
RMC	Acid gas piping system. from excl. regeneration system to excl. acid gas, incl. treatment system.	RN	Collection, storage and recycle systems for gas condensate.
RMD	Fluidizing gas piping system. from excl. main gas piping system to excl. fluidizing air injection system or to excl. load system (excl. fluidizing gas compressor).	RNA	Collection, storage and recycle systems for gas condensate. (free for use, process specific).
RME	Airlock gas piping system (not inert gas). from excl. branch off other system to excl. airlocks (excl. airlock gas compressor).	RNB	Collection, storage and recycle systems for gas condensate. (free for use, process specific).
		RNC	Collection, storage and recycle systems for gas condensate. (free for use, process specific).
		RND	Collection, storage and recycle systems for gas condensate. (free for use, process specific).
		RNE	Collection, storage and recycle systems for gas condensate. (free for use, process specific).
		RNF	Collection, storage and recycle systems for gas condensate. (free for use, process specific).

GAS GENERATION AND TREATMENT

RNG	Collection, storage and recycle systems for gas condensate. (free for use, process specific).	RPC	Inert gas generation by combustion. from excl. atmosphere to incl. inert gas distribution system inlet.
RNH	Collection, storage and recycle systems for gas condensate. (free for use, process specific).	RPG	Nitrogen piping system for addition ahead of burner gas turbine. from excl. air separation plant outlet or from excl. compressor outlet to excl. compressor inlet or to excl. combustion chamber inlet.
RNJ	Collection, storage and recycle systems for gas condensate. (free for use, process specific).	RPH	Supercompression of nitrogen from *RPG*. from incl. compressor inlet to incl. compressor outlet
RNK	Collection, storage and recycle systems for gas condensate. (free for use, process specific).	RPJ	Inert gas distribution system including storage system. from excl. inert gas generator outlet or from excl. air separation plant *REF* outlet or from excl. compressor outlet to excl. compressor inlet or to excl. inert gas load system inlet. (free for use, process specific).
RNL	Collection, storage and recycle systems for gas condensate. (free for use, process specific).	RPK	Inert gas distribution system including storage system. from excl. inert gas generator outlet or from excl. air separation plant *REF* outlet or from excl. compressor outlet to excl. compressor inlet or to excl. inert gas load system inlet. (free for use, process specific).
RNM	Collection, storage and recycle systems for gas condensate. (free for use, process specific).	RPL	Inert gas distribution system including storage system. from excl. inert gas generator outlet or from excl. air separation plant *REF* outlet or from excl. compressor outlet to excl. compressor inlet or to excl. inert gas load system inlet. (free for use, process specific).
RNN	Collection, storage and recycle systems for gas condensate. (free for use, process specific).	RPM	Inert gas distribution system including storage system. from excl. inert gas generator outlet or from excl. air separation plant *REF* outlet or from excl. compressor outlet to excl. compressor inlet or to excl. inert gas load system inlet. (free for use, process specific).
RNP	Collection, storage and recycle systems for gas condensate. (free for use, process specific).		
RNQ	Collection, storage and recycle systems for gas condensate. (free for use, process specific).		
RNR	Collection, storage and recycle systems for gas condensate. (free for use, process specific).		
RNS	Collection, storage and recycle systems for gas condensate. (free for use, process specific).		
RNT	Collection, storage and recycle systems for gas condensate. (free for use, process specific).		
RNU	Collection, storage and recycle systems for gas condensate. (free for use, process specific).		
RNV	Lubricant supply system.		
RNW	Sealing fluid supply system.		
RNX	Fluid supply system for control and protection equipment.		
RNY	Control and protection equipment.		
RP	Inert gas, including recovery systems.		
RPA	Inert gas generation by air fractionation. from excl. atmosphere to incl. inert gas distribution system inlet.		
RPB	Inert gas generation by molecular sieving. from excl. atmosphere to incl. inert gas distribution system inlet.		

GAS GENERATION AND TREATMENT

RPN Inert gas distribution system including storage system.

from excl. inert gas generator outlet or
from excl. air separation plant *REF* outlet
or
from excl. compressor outlet
to excl. compressor inlet or
to excl. inert gas load system inlet.
(free for use, process specific).

RPP Inert gas compressor system.

from incl. compressor inlet
to incl. compressor outlet.
(free for use, process specific).

RPQ Inert gas compressor system.

from incl. compressor inlet
to incl. compressor outlet.
(free for use, process specific).

RPR Inert gas compressor system.

from incl. compressor inlet
to incl. compressor outlet.
(free for use, process specific).

RPS Inert gas compressor system.

from incl. compressor inlet
to incl. compressor outlet.
(free for use, process specific).

RPW Sealing fluid supply system.

RPX Fluid supply system for control and protection equipment.

RPY Control and protection equipment.

RS Supply and removal systems for water, steam and condensate.

RSA Supply and removal systems for water, steam and condensate.

The following applies to supply systems:
from excl. branch off other system
to excl. user
The following applies to removal systems:
from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RSB Supply and removal systems for water, steam and condensate.

The following applies to supply systems:
from excl. branch off other system
to excl. user
The following applies to removal systems:
from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RSC Supply and removal systems for water, steam and condensate.

The following applies to supply systems:
from excl. branch off other system
to excl. user
The following applies to removal systems:
from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RSD Supply and removal systems for water, steam and condensate.

The following applies to supply systems:
from excl. branch off other system
to excl. user
The following applies to removal systems:
from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RSE Supply and removal systems for water, steam and condensate.

The following applies to supply systems:
from excl. branch off other system
to excl. user
The following applies to removal systems:
from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RSF Supply and removal systems for water, steam and condensate.

The following applies to supply systems:
from excl. branch off other system
to excl. user
The following applies to removal systems:
from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RSG Supply and removal systems for water, steam and condensate.

The following applies to supply systems:
from excl. branch off other system
to excl. user
The following applies to removal systems:
from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RSH Supply and removal systems for water, steam and condensate.

The following applies to supply systems:
from excl. branch off other system
to excl. user
The following applies to removal systems:
from excl. producer
to excl. discharge into other system.
(free for use, process specific).

GAS GENERATION AND TREATMENT

RSJ Supply and removal systems for water, steam and condensate.

The following applies to supply systems:
from excl. branch off other system
to excl. user

The following applies to removal systems:
from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RSK Supply and removal systems for water, steam and condensate.

The following applies to supply systems:
from excl. branch off other system
to excl. user

The following applies to removal systems:
from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RSL Supply and removal systems for water, steam and condensate.

The following applies to supply systems:
from excl. branch off other system
to excl. user

The following applies to removal systems:
from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RSM Supply and removal systems for water, steam and condensate.

The following applies to supply systems:
from excl. branch off other system
to excl. user

The following applies to removal systems:
from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RSN Supply and removal systems for water, steam and condensate.

The following applies to supply systems:
from excl. branch off other system
to excl. user

The following applies to removal systems:
from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RSP Supply and removal systems for water, steam and condensate.

The following applies to supply systems:
from excl. branch off other system
to excl. user

The following applies to removal systems:
from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RSQ Supply and removal systems for water, steam and condensate.

The following applies to supply systems:
from excl. branch off other system
to excl. user

The following applies to removal systems:
from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RSR Supply and removal systems for water, steam and condensate.

The following applies to supply systems:
from excl. branch off other system
to excl. user

The following applies to removal systems:
from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RSS Supply and removal systems for water, steam and condensate.

The following applies to supply systems:
from excl. branch off other system
to excl. user

The following applies to removal systems:
from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RST Supply and removal systems for water, steam and condensate.

The following applies to supply systems:
from excl. branch off other system
to excl. user

The following applies to removal systems:
from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RSU Supply and removal systems for water, steam and condensate.

The following applies to supply systems:
from excl. branch off other system
to excl. user

The following applies to removal systems:
from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RSV Lubricant supply system.

RSW Sealing fluid supply system.

RSX Fluid supply system for control and protection equipment.

RSY Control and protection equipment.

RSZ Injection and proportioning system.

GAS GENERATION AND TREATMENT

RT Waste water collection and drainage systems.

RTA Waste water collection and drainage systems.

from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RTB Waste water collection and drainage systems.

from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RTC Waste water collection and drainage systems.

from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RTD Waste water collection and drainage systems.

from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RTE Waste water collection and drainage systems.

from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RTF Waste water collection and drainage systems.

from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RTG Waste water collection and drainage systems.

from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RTH Waste water collection and drainage systems.

from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RTJ Waste water collection and drainage systems.

from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RTK Waste water collection and drainage systems.

from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RTL Waste water collection and drainage systems.

from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RTM Waste water collection and drainage systems.

from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RTN Waste water collection and drainage systems.

from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RTP Waste water collection and drainage systems.

from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RTQ Waste water collection and drainage systems.

from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RTR Waste water collection and drainage systems.

from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RTS Waste water collection and drainage systems.

from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RTT Waste water collection and drainage systems.

from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RTU Waste water collection and drainage systems.

from excl. producer
to excl. discharge into other system.
(free for use, process specific).

RTV Lubricant supply system.

RTW Sealing fluid supply system.

RTX Fluid supply system for control and protection equipment.

RTY Control and protection equipment.

RTZ Injection and proportioning system.

GAS GENERATION AND TREATMENT

RU Waste water treatment systems.

RUA Waste water treatment systems.

from excl. inlet
to incl. outlet.
(free for use, process specific).

RUB Waste water treatment systems.

from excl. inlet
to incl. outlet.
(free for use, process specific).

RUC Waste water treatment systems.

from excl. inlet
to incl. outlet.
(free for use, process specific).

RUD Waste water treatment systems.

from excl. inlet
to incl. outlet.
(free for use, process specific).

RUE Waste water treatment systems.

from excl. inlet
to incl. outlet.
(free for use, process specific).

RUF Waste water treatment systems.

from excl. inlet
to incl. outlet.
(free for use, process specific).

RUG Waste water treatment systems.

from excl. inlet
to incl. outlet.
(free for use, process specific).

RUJ Inert gas compressor system.

from incl. receiving point
to excl. discharge.
(free for use, process specific).

RUK Inert gas compressor system.

from incl. receiving point
to excl. discharge.
(free for use, process specific).

RUL Inert gas compressor system.

from incl. receiving point
to excl. discharge.
(free for use, process specific).

RUM Waste water treatment systems.

from excl. inlet
to incl. outlet.
(free for use, process specific).

RUN Waste water treatment systems.

from excl. inlet
to incl. outlet.
(free for use, process specific).

RUP Waste water treatment systems.

from excl. inlet
to incl. outlet.
(free for use, process specific).

RUQ Waste water treatment systems.

from excl. inlet
to incl. outlet.
(free for use, process specific).

RUR Waste water treatment systems.

from excl. inlet
to incl. outlet.
(free for use, process specific).

RUS Residues treatment and removal systems.

from excl. receiving point
to excl. discharge.
(free for use, process specific).

RUT Residues treatment and removal systems.

from excl. receiving point
to excl. discharge.
(free for use, process specific).

RUU Residues treatment and removal systems.

from excl. receiving point
to excl. discharge.
(free for use, process specific).

RUV Lubricant supply system.

RUW Sealing fluid supply system.

RUX Fluid supply system for control and protection equipment.

RUY Control and protection equipment.

RUZ Injection and proportioning system.

RV Lubricant supply systems.

RVA Lubricant supply systems.

(free for use).

RVB Lubricant supply systems.

(free for use).

RVC Lubricant supply systems.

(free for use).

RVD Lubricant supply systems.

(free for use).

RVE Lubricant supply systems.

(free for use).

RVF Lubricant supply systems.

(free for use).

RVG Lubricant supply systems.

(free for use).

RVH Lubricant supply systems.

(free for use).

RVJ Lubricant supply systems.

(free for use).

RVK Lubricant supply systems.

(free for use).

RVL Lubricant supply systems.

(free for use).

GAS GENERATION AND TREATMENT

RVM Lubricant supply systems.
(free for use).

RVN Lubricant supply systems.
(free for use).

RVP Lubricant supply systems.
(free for use).

RVQ Lubricant supply systems.
(free for use).

RVR Lubricant supply systems.
(free for use).

RVS Lubricant supply systems.
(free for use).

RVT Lubricant supply systems.
(free for use).

RVU Lubricant supply systems.
(free for use).

RW Sealing fluid supply systems.

RWA Sealing fluid supply systems.
(free for use).

RWB Sealing fluid supply systems.
(free for use).

RWC Sealing fluid supply systems.
(free for use).

RWD Sealing fluid supply systems.
(free for use).

RWE Sealing fluid supply systems.
(free for use).

RWF Sealing fluid supply systems.
(free for use).

RWG Sealing fluid supply systems.
(free for use).

RWH Sealing fluid supply systems.
(free for use).

RWJ Sealing fluid supply systems.
(free for use).

RWK Sealing fluid supply systems.
(free for use).

RWL Sealing fluid supply systems.
(free for use).

RWM Sealing fluid supply systems.
(free for use).

RWN Sealing fluid supply systems.
(free for use).

RWP Sealing fluid supply systems.
(free for use).

RWQ Sealing fluid supply systems.
(free for use).

RWR Sealing fluid supply systems.
(free for use).

RWS Sealing fluid supply systems.
(free for use).

RWT Sealing fluid supply systems.
(free for use).

RWU Sealing fluid supply systems.
(free for use).

RX Fluid supply systems for control and protection equipment.

RXA Fluid supply systems for control and protection equipment.
(free for use).

RXB Fluid supply systems for control and protection equipment.
(free for use).

RXC Fluid supply systems for control and protection equipment.
(free for use).

RXD Fluid supply systems for control and protection equipment.
(free for use).

RXE Fluid supply systems for control and protection equipment.
(free for use).

RXF Fluid supply systems for control and protection equipment.
(free for use).

RXG Fluid supply systems for control and protection equipment.
(free for use).

RXH Fluid supply systems for control and protection equipment.
(free for use).

RXJ Fluid supply systems for control and protection equipment.
(free for use).

RXK Fluid supply systems for control and protection equipment.
(free for use).

RXL Fluid supply systems for control and protection equipment.
(free for use).

RXM Fluid supply systems for control and protection equipment.
(free for use).

RXN Fluid supply systems for control and protection equipment.
(free for use).

RXP Fluid supply systems for control and protection equipment.
(free for use).

RXQ Fluid supply systems for control and protection equipment.
(free for use).

RXR Fluid supply systems for control and protection equipment.
(free for use).

GAS GENERATION AND TREATMENT

RXS Fluid supply systems for control and protection equipment.

(free for use).

RXT Fluid supply systems for control and protection equipment.

(free for use).

RXU Fluid supply systems for control and protection equipment.

(free for use).

RY Control and protection equipment.

RYA Control and protection equipment.

(free for use).

RYB Control and protection equipment.

(free for use).

RYC Control and protection equipment.

(free for use).

RYD Control and protection equipment.

(free for use).

RYE Control and protection equipment.

(free for use).

RYF Control and protection equipment.

(free for use).

RYG Control and protection equipment.

(free for use).

RYH Control and protection equipment.

(free for use).

RYJ Control and protection equipment.

(free for use).

RYK Control and protection equipment.

(free for use).

RYL Control and protection equipment.

(free for use).

RYM Control and protection equipment.

(free for use).

RYN Control and protection equipment.

(free for use).

RYP Control and protection equipment.

(free for use).

RYQ Control and protection equipment.

(free for use).

RYR Control and protection equipment.

(free for use).

RYS Control and protection equipment.

(free for use).

RYT Control and protection equipment.

(free for use).

RYU Control and protection equipment.

(free for use).

RZ Injection and proportioning systems.

RZA Injection and proportioning systems.

(free for use).

RZB Injection and proportioning systems.

(free for use).

RZC Injection and proportioning systems.

(free for use).

RZD Injection and proportioning systems.

(free for use).

RZE Injection and proportioning systems.

(free for use).

RZF Injection and proportioning systems.

(free for use).

RZG Injection and proportioning systems.

(free for use).

RZH Injection and proportioning systems.

(free for use).

RZJ Injection and proportioning systems.

(free for use).

RZK Injection and proportioning systems.

(free for use).

RZL Injection and proportioning systems.

(free for use).

RZM Injection and proportioning systems.

(free for use).

RZN Injection and proportioning systems.

(free for use).

RZP Injection and proportioning systems.

(free for use).

RZQ Injection and proportioning systems.

(free for use).

RZR Injection and proportioning systems.

(free for use).

RZS Injection and proportioning systems.

(free for use).

RZT Injection and proportioning systems.

(free for use).

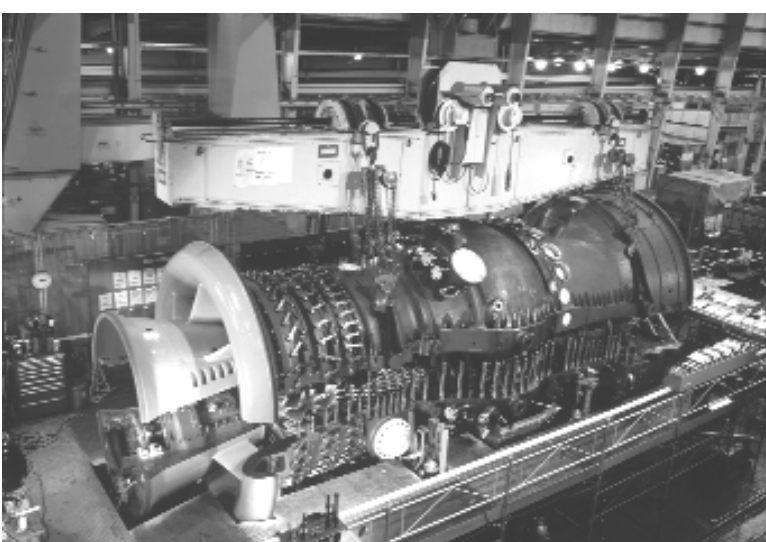
RZU Injection and proportioning systems.

(free for use).



ANCILLARY SYSTEMS

- SA Heating, ventilation, air-conditioning (HVAC) systems for conventional area.**
- SB Space heating systems.**
- SC Stationary compressed air supplies.**
- SD Cleaning system (see *FK* for decontamination systems).**
- SE Stationary welding gas systems.**
- SG Stationary fire protection systems.**
- SH Waterway facilities.**
- SM Cranes, stationary hoists and conveying appliances.**
- SN Elevators.**
- SP Railway installations.**
- ST Workshop, stores, laboratory equipment and staff amenities.**



ANCILLARY SYSTEMS

SA	Heating, ventilation, air-conditioning (HVAC) systems for conventional area.	SAT	Heating, ventilation, air-conditioning (HVAC) systems in structures for auxiliary systems.
SAA	Refrigeration plant (if not part of *QK*).	SAU	Heating, ventilation, air-conditioning (HVAC) systems in general service structures: transport, traffic, fencing, gardens and other purposes.
SAB	Air humidifying systems (if not part of *QM*).	SAV	Lubricant supply system.
SAC	Heating, ventilation, air-conditioning (HVAC) systems in structures for switch-gear.	SAX	Fluid supply system for control and protection equipment.
SAD	Heating, ventilation, air-conditioning (HVAC) systems in structures for electro modules.	SAY	Control and protection equipment.
SAE	Heating, ventilation, air-conditioning (HVAC) systems in structures for conventional fuel supply and residue disposal.	SB	Space heating systems.
SAF	Heating, ventilation, air-conditioning (HVAC) systems for chemical flue gas treatment.	SBA	Central heating station.
SAG	Heating, ventilation, air-conditioning (HVAC) systems in structures for water supply and waste water disposal.	SBC	Space heating systems in structures for switch-gear.
SAH	Heating, ventilation, air-conditioning (HVAC) systems in structures for conventional heat generation.	SBE	Space heating systems in structures for conventional fuel supply and residue disposal.
SAK	Heating, ventilation, air-conditioning (HVAC) systems for general service structures.	SBG	Space heating systems in structures for water supply and waste water disposal.
SAL	Heating, ventilation, air-conditioning (HVAC) systems in structures for steam- water cycle.	SBH	Space heating systems in structures for conventional heat generation.
SAM	Heating, ventilation, air-conditioning (HVAC) systems in structures for main machine sets.	SBL	Space heating systems in structures for steam-water cycle.
SAN	Heating, ventilation, air-conditioning (HVAC) systems in structures for process energy supply.	SBM	Space heating systems in structures for main machine sets.
SAP	Heating, ventilation, air-conditioning (HVAC) systems in structures for circulating (cooling) water systems (e.g. circulating water intake).	SBN	Space heating system in structures for process energy supply systems.
SAQ	Heating, ventilation, air-conditioning (HVAC) systems in structures for circulating (cooling) water systems (e.g. circulating water pumps and outfall).	SBP	Space heating systems in structures for circulating (cooling) water systems (e.g. circulating water intake).
SAR	Heating, ventilation, air-conditioning (HVAC) systems in structures for circulating (cooling) water systems (e.g. recirculation cooling).	SBQ	Space heating systems in structures for circulating (cooling) water systems (e.g. circulating water pumps and outfall).
SAS	Heating, ventilation, air-conditioning (HVAC) systems in structures for ancillary systems.	SBS	Space heating systems in structures for ancillary systems.
		SBT	Space heating systems for auxiliary systems. (free for use e.g. building specific).
		SBU	Space heating systems in general service structures: transport, traffic, fencing, gardens and other purposes.
		SBX	Fluid supply system for control and protection equipment.
		SBY	Control and protection equipment.

SC Stationary compressed air supplies.

- SCA** Compressed air generation system.
- SCB** Compressed air distribution system.
- SCC** Stationary compressed air supplies.
(free for use).
- SCD** Stationary compressed air supplies.
(free for use).
- SCE** Stationary compressed air supplies.
(free for use).
- SCF** Stationary compressed air supplies.
(free for use).
- SCG** Stationary compressed air supplies.
(free for use).
- SCH** Stationary compressed air supplies.
(free for use).
- SCJ** Stationary compressed air supplies.
(free for use).
- SCK** Stationary compressed air supplies.
(free for use).
- SCL** Stationary compressed air supplies.
(free for use).
- SCM** Stationary compressed air supplies.
(free for use).
- SCN** Stationary compressed air supplies.
(free for use).
- SCP** Stationary compressed air supplies in structures for circulating (cooling) water systems (e.g. circulating water intake).
- SCQ** Stationary compressed air supplies in structures for circulating (cooling) water systems (e.g. circulating water pumps and outfall).
- SCR** Stationary compressed air supplies.
(free for use).
- SCS** Stationary compressed air supplies in structures for ancillary systems.
- SCT** Stationary compressed air supplies for auxiliary systems.
(free for use).
- SCU** Stationary compressed air supplies: transport, traffic, fencing, gardens and other purposes.
- SCV** Lubricant supply system.
- SCX** Fluid supply system for control and protection equipment.
- SCY** Control and protection equipment.

SD Cleaning system (see *FK* for decontamination systems).

- SDA** Cleaning system (see *FK* for decontamination systems).
(free for use).

SDB Cleaning system (see *FK* for decontamination systems). (free for use).

SDC Cleaning system (see *FK* for decontamination systems). (free for use).

SDD Cleaning system (see *FK* for decontamination systems). (free for use).

SDE Cleaning system (see *FK* for decontamination systems). (free for use).

SDF Cleaning system (see *FK* for decontamination systems). (free for use).

SDG Cleaning system (see *FK* for decontamination systems). (free for use).

SDH Cleaning system (see *FK* for decontamination systems). (free for use).

SDJ Cleaning system (see *FK* for decontamination systems). (free for use).

SDK Cleaning system (see *FK* for decontamination systems). (free for use).

SDL Cleaning system (see *FK* for decontamination systems). (free for use).

SDM Cleaning system (see *FK* for decontamination systems). (free for use).

SDN Cleaning system (see *FK* for decontamination systems). (free for use).

SDP Cleaning system (see *FK* for decontamination systems). (free for use).

SDQ Cleaning system (see *FK* for decontamination systems). (free for use).

SDR Cleaning system (see *FK* for decontamination systems). (free for use).

SDS Cleaning system (see *FK* for decontamination systems). (free for use).

SDT Cleaning system (see *FK* for decontamination systems). (free for use).

SDU Cleaning system (see *FK* for decontamination systems). (free for use).

ANCILLARY SYSTEMS

SDV	Lubricant supply system. (free for use).	SG	Stationary fire protection systems.
SDX	Fluid supply system for control and protection equipment. (free for use).	SGA	Fire water system, conventional area.
SDY	Control and protection equipment. (free for use).	SGC	Spray deluge systems, conventional area.
SE	Stationary welding gas systems.	SGE	Sprinkler systems.
SEA	Stationary welding gas systems. (free for use).	SGF	Foam fire-fighting systems.
SEB	Stationary welding gas systems. (free for use).	SGG	Tank roof, tank shell cooling systems.
SEC	Stationary welding gas systems. (free for use).	SGJ	CO2 fire-fighting systems.
SED	Stationary welding gas systems. (free for use).	SGK	Halon fire-fighting systems.
SEE	Stationary welding gas systems. (free for use).	SGL	Powder fire-fighting systems.
SEF	Stationary welding gas systems. (free for use).	SGV	Lubricant supply system.
SEG	Stationary welding gas systems. (free for use).	SGX	Fluid supply system for control and protection equipment.
SEH	Stationary welding gas systems. (free for use).	SGY	Control and protection equipment.
SEJ	Stationary welding gas systems. (free for use).	SH	Waterway facilities.
SEK	Stationary welding gas systems. (free for use).	SHA	Intake system (upper pound). to excl. lock chamber gate (upstream).
SEL	Stationary welding gas systems. (free for use).	SHB	Lock chamber incl. chamber filling and emptying system.
SEM	Stationary welding gas systems. (free for use).	SHC	Lock chamber gate (upstream).
SEN	Stationary welding gas systems. (free for use).	SHD	Lock chamber gate (intermediate).
SEP	Stationary welding gas systems. (free for use).	SHE	Lock chamber gate (downstream).
SEQ	Stationary welding gas systems. (free for use).	SHF	Discharge system (lower pound). from excl. lock chamber gate (downstream).
SER	Stationary welding gas systems. (free for use).	SHJ	Anti-icing system.
SES	Stationary welding gas systems. (free for use).	SHK	Ship impact system.
SET	Stationary welding gas systems. (free for use).	SHL	Leakage drain system.
SEU	Stationary welding gas systems. (free for use).	SHM	Draining system.
SFT	Heating and fuel gas systems. (free for use).	SHT	Fish ladder.
		SHU	Raft canal.
		SHV	Lubricant supply system.
		SHW	Sealing fluid supply system.
		SHX	Fluid supply system for control and protection equipment.
		SHY	Control and protection equipment.
		SM	Cranes, stationary hoists and conveying appliances.
		SMA	Central equipment.
		SMC	Cranes, stationary hoists and conveying appliances in structures for switch-gear.
		SME	Cranes, stationary hoists and conveying appliances in structures for conventional fuel supply and residues discharge.

ANCILLARY SYSTEMS

SMF Cranes, stationary hoists and conveying appliances in structures for flue gas chemical treatment.

SMG Cranes, stationary hoists and conveying appliances in structures for water supply and waste water disposal.

SMH Cranes, stationary hoists and conveying appliances in structures for conventional heat generation.

SMK Cranes, stationary hoists and conveying appliances in general service structures.
(free for use e.g. building specific).

SML Cranes, stationary hoists and conveying appliances in structures for steam-water cycle.

SMM Cranes, stationary hoists and conveying appliances in structures for main machine sets.

SMN Cranes, stationary hoists and conveying appliances in structures for process energy supply for external users.

SMP Cranes, stationary hoists and conveying appliances in structures for circulating (cooling) water systems (e.g. circulating water intake).

SMQ Cranes, stationary hoists and conveying appliances in structures for auxiliary systems.

SMS Cranes, stationary hoists and conveying appliances in structures for ancillary systems.

SMT Cranes, stationary hoists and conveying appliances.
(free for use e.g. building specific).

SMU Cranes, stationary hoists and conveying appliances in general service structures.

SMX Fluid supply system for control and protection equipment.

SMY Control and protection equipment.

SN Elevators.

SNC Elevators in structures for switch-gear.

SNE Elevators in structures for conventional fuel supply and residue disposal.

SNG Elevators in structures for water supply and waste water disposal.

SNH Elevators.
(free for use e.g. building specific).

SNL Elevators in structures for steam-water cycle.

SNM Elevators in structures for main machine sets.

SNN Elevators in structures for process energy supply for external users.

SNP Elevators in structures for circulating (cooling) water systems (e.g. circulating water intake).

SNQ Elevators in structures for auxiliary systems.

SNS Elevators in structures for ancillary systems.

SNT Elevators.(free for use e.g. building specific).

SNU Elevators in general service structures.

SNY Control and protection equipment.

SP Railway installations.

SPA Railway installations.
(free for use).

SPB Railway installations.
(free for use).

SPC Railway installations.
(free for use).

SPD Railway installations.
(free for use).

SPT Railway installations.
(free for use).

SQT Road installations.
(free for use).

ST Workshop, stores, laboratory equipment and staff amenities.

STA Workshop equipment.

STC Maintenance areas.

STE Stores and filling station equipment.

STG Laboratory equipment.

STP Staff amenities.

STY Control and protection equipment.

U

STRUCTURES AND BUILDINGS FOR COMBINED CYCLE POWER PLANTS

- UA Structures for grid and distribution systems.**
- UB Structures for power transmission and auxiliary power supply.**
- UC Structures for instrumentation and control (I&C).**
- UD General items.**
- UE Structures for conventional fuel supply and residues disposal.**
- UG Structures for water supply and disposal.**
- UH Structures for conventional heat generation.**
- UL Structures for steam-, water-, gas cycles.**
- UM Structures for main machine sets.**
- UN Structures for process energy / media supply for external users.**
- UP Structures for cooling water systems (water intake and cleaning).**
- UQ Structures for cooling water systems (forwarding and outfall).**
- UR Structures for circulating (cooling) water systems (recirculation).**
- US Structures for ancillary systems.**
- UT Structures for auxiliary systems.**
- UU Shaft structures.**
- UV Structures for chemical flue gas treatment incl. residues removal (for *HR*. *HS*. *HT*).**
- UX Structures for external systems (power plant-specific).**
- UY General service structures.**
- UZ Structures for transport, traffic, fencing, gardens and other purposes.**



STRUCTURES AND BUILDINGS FOR CCPP

UA Structures for grid and distribution systems.

- UAA Structure for switchyard area.**
Steel cables, GIS or rigid bus - support structures.
- UAB Power systems switching station building.**
GIS or rigid bus - support structures.
- UAC Power system switching station control building.**
- UAE Auxiliary equipment building.**
Auxiliary equipment, compressed air station etc.
- UAF Portal frame (step-up transformer to switchyard).**
Steel cables, GIS or rigid bus - support structures.
- UAG Civil works for switchyard transformer.**
Buildings, bays.
- UAH Foundations for supports and equipment outside UAA.**
- UAJ Structure for transformer rail system in *UA* area.**
Foundation, excavation work.
- UAM High voltage outgoing portal frame / tower.**
- UAT Roads and areas in the switchyard.**
Various types, constructions.
- UAU Ground grid in *UA* area.**
Various mesh sizes.
- UAV Switchyard fencing and gates.**
- UAW Structure for transformer rail system.**
Various types, constructions.
- UAX Special structure (plant-specific).**
- UAY Pipe & cable bridge structure for main and auxiliary distribution *UA* area.**
Bridge for HV feeder (step-up transformer to switchyard).
Bridge for unit supply and control cables.
- UAZ Pipe & cable ducting structure for main and auxiliary distribution *UA* area.**
Ducting for HV feeder (step-up transformer to switchyard).
Ducting for unit supply and control cables.

UB Structures for power transmission and auxiliary power supply.

- UBA Central electrical and control building.**
module foundation also used for combined electrical / control / administration building.
- UBB GT, ST, GT/ST electrical building / module.**
module foundation
Container / building GT or ST type.
- UBC Structure for back-up transformers.**
Foundations / pit / building.
- UBD Structure for station service transformer.**
Foundations / pit / building.
- UBE Structure for unit service transformer.**
GT, ST, GT/ST unit; foundations / pit / building.
- UBF Structure for transformer area.**
Foundation / pit / building.
- UBG Structure for start-up transformer.**
Foundations / pit / building.
- UBH Structure for common oil collecting pit.**
Foundation / pit.
- UBJ Structure for transformer rail system.**
Foundation / excavation work.
- UBK Transformer assembly building.**
Shelter / building.
- UBL Structure for busbars.**
GT, ST, GT/ST unit; foundations, steel construction.
- UBM Structure for transformer cooling system.**
GT, ST, GT/ST unit; foundations / pit / building.
- UBN Structure for standby / emergency / black start generator set.**
(Diesel / turbine).
- UBP Structure for blackstart generating set.**
Diesel / Turbine.
- UBQ Structure for emergency / black start generating set fuel supply.**
Diesel / gas / kerosene etc.
- UBS Structure for MV/LV auxiliary transformer.**
Foundation / shelter / building.
- UBT Structure for LV/LV auxiliary transformer.**
Foundation / shelter / building.

STRUCTURES AND BUILDINGS FOR CCPP

UBU	Ground grid outside *UA* area.	UD	General items.
UBW	Ground grid outside UA area. Various mesh sizes.	UDA	General items (mobilisation and demobilisation).
UBX	Special structure (plant specific).	UDD	Provisional sum (PI/ROP).
UBY	Power cable bridge structure for main and auxiliary distribution ROP area.	UDE	Site dewatering (PI/ROP).
UBZ	Power cable ducting structure for main and auxiliary distribution ROP area.	UDF	Site survey (PI/ROP).
UC	Structures for instrumentation and control (I&C).	UDG	Soil improvement (PI/ROP).
UCA	Unit control building.	UDH	Piling (PI/ROP).
UCB	Central control building.	UDK	Soil investigation (PI/ROP).
UCC	Local control modules. GT local control module - container / room / annex. ST local control module - container.	UDL	Site clearing and tree cutting (PI/ROP).
UCE	Local (decentralized) control buildings. B Building / container / shelter.	UDM	Demolition work (PI/ROP).
UCG	Structure for water supply / waste water disposal control system. Building / container / shelter.	UDN	Structure for modification, repair and renovation of existing structures (PI/ROP).
UCH	Structure for HRSG control system. Building / container / shelter.	UE	Structures for conventional fuel supply and residues disposal.
UCL	Structure for local metering equipment - steam water cycle. Building / container / shelter.	UEG	Structures for coal gasification / liquefaction.
UCM	Structure for control system of air cooled condenser. Building / container / shelter.	UEH	Structures for unloading and reception of liquid fuels. Oil discharge station - lorry / train / ship discharge tank pit, Building, shelter.
UCN	Structure for metering & control system of process steam / process hot water. Building / container / shelter.	UEJ	Structures for storage of liquid fuels. Tank foundation / containment - diesel oil / dirty oil / heavy oil / naphta / kerosene etc.
UCP	Structure for cooling water measuring equipment. Building / container / shelter.	UEK	Structure for oil separator at oil unloading station and storage.
UCR	Structure for cooling tower control system. Building / container / shelter.	UEL	Structure for oil forwarding pump station. Building / shelter.
UCV	Structure for sea water desalination plant control system. Building / container / shelter.	UEM	Structure for treatment and handling of liquid fuels. Oil conditioning plant, oil additive supply and store.
UCX	Special structure (plant specific).	UEN	NG/LG/BFG - gas terminal and treatment station. Gas terminal (metering, control etc.), discharge station, pressure reduction station gas conditioning plant (filters, condensers, separators etc.), booster plant incl. cooling.
UCY	Control cable bridge structure for main and auxiliary distribution ROP area.	UEP	LG/BFG - condensate collection tank (residue disposal). Foundations / shelter / building for gas condensate collection tank.
UCZ	Control cable ducting structure for main and auxiliary distribution ROP area.	UEQ	LG (liquid gas) treatment plant. incl. heating / cooling medium system foundation / shelter / building.

STRUCTURES AND BUILDINGS FOR CCPP

- UER Structures for forwarding of gaseous fuels.**
Foundations / supports for gas pipeline.
- UES Structure for oil sludge disposal.**
Building / shelter / pit.
- UEW Structure for combustion residue-handling.**
Foundation / shelter / building.
- UEX Special structure (plant specific).**
- UEY Pipe & cable bridge structure for fuel distribution ROP area.**
- UEZ Pipe & cable ducting structure for fuel distribution ROP area.**

UG Structures for water supply and disposal.

- UGA Structures for raw water supply, cleaning and treatment.**
Raw / city / ground water pumping station incl. mech. cleaning and transport structures.
- UGB Structure for raw water system.**
Structure for storage tank / pond / pumping station.
- UGC Structure for raw and de-mineralised water storage and forwarding.**
Structure for storage tanks / ponds / pumping stations.
- UGD Structure for water demineralization system.**
Structures for treatment plant, water storage tank / pond / pumping station incl. dosing, electrical and control systems.
- UGE Structure for neutralization system incl. dosing plant.**
Structures for neutralization tank / pit / dosing plant.
- UGF Structure for fire distinguishing water storage and forwarding.**
Structures for storage tank / pond / pumping station.
- UGG Potable water treatment plant, storage incl. pumping station.**
Structures for storage tank / pond / pumping station.
- UGH Structures for rainwater collecting and treatment system.**
Structures for surface water catchment pond incl. sand trap and pumping station sand and residue disposal, rain and fire fighting retention pond.
- UGJ Structure for sand trap / settling pond.**

- UGK Structures for biological and chemical raw water treatment plant.**
Structure for settling pond incl. pumping station, dosing and outlet control equipment used in domestic water treatment.
- UGL Flocculator structure.**
Structure for flocculation plant incl. inlet and outlet channels used in process drains precleaning plant and sludge disposal / domestic waste water treatment.
- UGM Structures for filters, dosing, electrical and control systems.**
used in process drains precleaning plant and sludge disposal / domestic waste water treatment.
- UGN Structures for clean water storage incl. pumping station.**
used in process drains precleaning / treatment plant and sludge disposal / domestic waste water treatment.
- UGP Structure for sludge treatment system.**
incl. inlet and outlet channels.
- UGQ Sludge dewatering building.**
Structures for sludge press incl. auxiliaries / waste water collection and return ponds.
- UGR Sludge storage structure.**
used in process drains precleaning plant and sludge disposal / domestic waste water treatment.
- UGS Structure for operational / chemical waste water treatment incl. pumping station.**
collection pond incl. pumping station, cooler etc.
structures for (chemical) treatment plant and residue storage / disposal.
- UGT Calming pond incl. outlet regulation.**
used in process drains precleaning plant and sludge disposal / domestic waste water and rainwater treatment / disposal.
- UGU Oil separator pit incl. oil sludge collection.**
used in process drains precleaning plant and sludge disposal / domestic waste water and rainwater treatment / disposal.
- UGV Structure for sanitary waste water treatment incl. pumping station.**
Structure for septic tank incl. pumping station / BioGest plant / disposal.
- UGW Storm water evaporation pond.**
Sewage collecting pond and treatment incl. sludge pit, clean water return channels, pure water return / inlet structures.
- UGX Special structure (plant specific).**

STRUCTURES AND BUILDINGS FOR CCPP

UGY	Pipe bridge structure for water distribution / waste water collecting - ROP area.	UL	Structures for steam-, water-, gas cycles.
UGZ	Pipe & cable ducting structure for water distribution / waste water collecting - ROP area. Potable water system. Demineralized water system. Surface / storm water system. Sanitary water drainage. Oily water drainage. Boiler blowdown drainage.	ULA	Structure for feedwater system. Foundation / shelter / building for feedwater tank / pumping system.
UH	Structures for conventional heat generation.	ULC	Structure for condensate system. Foundation / shelter / building for condenser, condensate pumps, ADV, clean condensate collecting tank etc.
UHA	Structure for HRSG / HRSG housing. Foundation / pit / shelter / building.	ULD	Structure for condensate polishing plant. Foundation / shelter / building.
UHB	Foundations for HRSG auxiliaries. Foundation / pit / shelter / building.	ULF	Structure for exhaust steam pipes (steam turbine - air cooled condenser). Foundation / shelter / building.
UHG	Structure for exhaust gas ducts / housing (GT to HRSG). Foundation / pit / shelter / building. (if in HRSG scope of supply).	ULG	Structure for ACC (air cooled condenser) area. Foundation / support / shelter / building.
UHH	Structure for bypass stack / housing. Foundation / pit / shelter / building. (if in HRSG scope of supply).	ULH	Structure for air cooled condenser auxiliaries. Foundation / shelter / building.
UHL	Structure for boiler compressed air supply. Foundation / shelter / building.	ULJ	Structure for atmospheric drains vessel (ADV) (*LCM*). Foundation / shelter / building.
UHM	Structure for exhaust gas ducts (HRSG to stack). Foundation / pit / shelter / building.	ULK	Structure for clean drains pumps (*LCM*). Foundation / shelter / building.
UHN	Structure for exhaust gas stack (free standing). Foundation / pit.	ULS	Structure for auxiliary condensate collecting tank (*LCN*). Foundation / shelter / building.
UHQ	Structure for flue gas treatment system (mechanical). Foundation / pit / shelter / building.	ULT	Structure for auxiliary condensate pumps (*LCN*). Foundation / shelter / building.
UHR	Structure for flue gas treatment system (catalytic process). Foundation / pit / shelter / building.	ULV	Structure for chemical dosing system (*LFN*). Foundation / shelter / building.
UHS	Structure for flue gas treatment system (catalytic process). Foundation / pit / shelter / building.	ULW	Structure for sampling system (*QUL*). Foundation / shelter / building.
UHT	Structure for emission monitoring station. CEMS container.	ULX	Special structures (plant specific).
UHW	Structures for HRSG blowdown equipment. Foundation / shelter / building.	ULY	Bridge structures.
UHX	Structure for HRSG enclosure and housing. Foundation / shelter / building.	ULZ	Pipe ditch & cable duct structure for W/S cycle - outside power island.
UHY	Bridge structure.	UM	Structures for main machine sets.
UHZ	Ducting structures.	UMA	Steam turbine building / ST outdoor area incl. crane support structure. Table / floor construction, foundations for crane rails in ST outdoor area, foundations for ST auxiliaries. Complete ST foundations: *MPA*.

STRUCTURES AND BUILDINGS FOR CAPP

UMB Gas turbine building / GT outdoor area incl. crane support structure.

Table / floor construction, foundations for crane rails in GT outdoor area, foundations for GT auxiliaries.

Complete GT foundations: *MPA*.

UMC GT/ST building and outdoor area incl. crane support structure (single shaft).

Table / floor construction, foundations for crane rails in GT / ST outdoor area, foundations for GT / ST auxiliaries.

Complete GT / ST foundations: *MPA*.

UMD Common GT/ST building and outdoor area incl. crane support structure(multi shaft).

Table / floor construction, foundations for crane rails in GT / ST outdoor area, foundations for GT / ST auxiliaries.

Complete GT / ST foundations: *MPA*.

UMF Structure for exhaust gas duct / housing (GT to HRSG).

Foundation / shelter / building.
(if in GT scope of supply).

UMH Structure for bypass stack / housing.

Foundation / shelter / building (if in GT scope of supply).

UMN Structure for annex to *UMA*, *UMC* or *UMD*.

(only if annex to UMA, UMC or UMD).

UMP Annex for circulating (main cooling) and service (secondary cooling) water pipes.

(structures for main and secondary cooling water pipes in *UMA*, *UMC*, *UMD*).

UMS Air storage structure (GT combustion air reservoir plant).

cavern / building.

UMT Annex for components of blast furnace gas supply.

(only if annex to *UMB*, *UMC* or *UMD*).

UMV Connecting structures between main machine set halls.

(subways, gangways, covered paths etc.).

UMX Special structure (plant specific).

UMY Pipe bridge & cable rack structure for main machine set systems ROP area.

(not *ULY*, *UNY*).

UMZ Pipe ditch & cable duct structure for main machine set system - ROP area.

UN Structures for process energy / media supply for external users.

UNA Structure for process steam station.

e.g. for metering and control station, distribution; foundation / shelter / building.

UNB Structure for process steam condensate return.

e.g. storage tank, forwarding station; foundation / shelter / building.

UNC Structure / annex for hot water pumping station.

(e.g. for district heating); foundation / shelter / building.

UND Structure / annex for process heat transfer.

(e.g. heater / heater-condenser for hot water processing); foundation / shelter / building.

UNE Structure / foundation for hot water pond.

Foundation / shelter / building.

UNF Containment for hot water pond.

Foundation / shelter / building.

UNG Structures for chilled water supply.

UNX Special structure (plant specific).

UNY Pipe bridge & cable rack structure for process steam/hot water - ROP area (not *ULY*).

UNZ Pipe ditch & cable duct structure for process steam/hot water ROP area (not *ULY*).

UP Structures for cooling water systems (water intake and cleaning).

UPA Circulating (main cooling) water intake structure.

(intake head, shaft etc.); underwater structures.

UPB Service (secondary cooling) water intake structure.

(intake head, shaft etc.); underwater structures.

UPC Circulating (main cooling) water intake structure.

channel / pipe.

UPD Service (secondary cooling) water intake structure.

channel / pipe.

UPE Circulating (main cooling) water source.

Intake underwater and shore / bank structures.

STRUCTURES AND BUILDINGS FOR CCPP

UPF Service (secondary cooling) water source.

Intake underwater and shore / bank structures.

UPG Shore structure and embankment.

Shore structure and bank works.

UPH Circulating (main cooling) water treatment structures.

for mechanical cleaning, chemical dosing equipment, settling pond for main cooling water.

UPJ Service (secondary cooling) water treatment structures.

for mechanical cleaning, chemical dosing equipment, settling pond for secondary cooling water.

UPN Circulating (main cooling) water feed inlet culvert.

from mechanical cleaning (*UPE*) to pump house (*UQA*).

UPP Service (secondary cooling) water feed culvert.

from mechanical cleaning (*UPF*) to pump house (*UQB*).

UPQ Biocide treatment building.

Annex building / structure for chlorine dosing plant incl. storage.

UPR Structure for electro-chlorination plant.

Foundation / shelter / building.

UPS Flushing water system.

channel from mechanical cleaning to return.

UPT Screen wash water cleaning structure.

UPU Building, space / pit for dirt container.

Foundation / shelter / building.

UPV Structure for waste storage and disposal.

Foundation / pit.

UPW Structure. storage space for insulating closure slabs.

Foundation.

UPX Special structure (plant specific).

UPY Cooling water pipe bridge - between pumphouse and turbine building.

UPZ Cooling water pipe/duct/channel - between pumphouse and turbine building.

UQ Structures for cooling water systems (forwarding and outfall).

UQA Circulating (main cooling) water pump station (direct cooling).

incl. mech. cleaning structures, direct cooling Underwater / pit / shelter / building.

UQB Service (secondary cooling) water pump station.

incl. mech. cleaning structures, direct cooling Underwater / pit / shelter / building.

UQC Circulating (main cooling) water pump station.

excl. mechanical cleaning structure, cooling tower Underwater / pit / shelter / building.

UQD Service (secondary cooling) water pump station.

excl. mechanical cleaning structure, cooling tower Underwater / pit / shelter / building.

UQG Cooling water overflow structure, surge tank / chamber.

Underwater / pit.

UQH Screen wash water discharge culvert / channel.

Underwater / channel.

UQJ Cooling water seal pit, aeration, incl. outfall.

Underwater / pit.

UQK Cooling water de-aeration structure.

Underwater / pit.

UQL Service (secondary cooling) water surge pond.

Underwater / pit.

UQM Service (secondary cooling) water collecting pond.

Underwater / pit.

UQN Circulating (main cooling) water return channel.

Underwater / channel.

UQP Service (secondary cooling) water return channel.

Underwater / channel.

UQQ Circulating (main cooling) water outfall / return structure.

Underwater / pit.

UQR Service (secondary cooling) water outfall / return structure.

Underwater / pit.

UQS Circulating (main cooling) water discharge structure.

Underwater.

STRUCTURES AND BUILDINGS FOR CCPP

UQT	Service (secondary cooling) water discharge culvert. Underwater.	US	Structures for ancillary systems.
UQU	Circulating (main cooling) water spillway structure incl. aeration structure. Underwater / bank / shore structure.	USA	Structure for central HVAC system and compressed air plant. Foundation / shelter / building.
UQV	Structure for artificial circulating (main cooling) water aeration. Underwater / pit.	USB	Structure for space heating systems. e.g. boiler house / foundations / shelters for space heating units etc.
UQW	Routing structure for circulating (main cooling) water discharge. Underwater / bank / shore structure.	USC	Structure for stationary compressed air supply system (not *UTF*). Foundation / shelter / building.
UQX	Special structure (plant specific).	USG	Fire extinguishing water pumping station. Foundation / shelter / building.
UQY	Main / secondary cooling water return/discharge pipe bridge.	USH	Structure for waterway facilities. Underwater / pit / shore- / bank-structure / foundation / shelter etc.
UQZ	Ducting structure.	USS	Combined workshop and storage building. Shelter / building.
UR	Structures for circulating (cooling) water systems (recirculation).	UST	Workshop building. Shelter / building.
URA	Cooling tower structure for circulating (main cooling) water. Foundation / pit / basin / structure.	USU	Storage building, covered storage area, warehouse. Shelter / building.
URB	Cooling tower structure for service (secondary cooling) water. Foundation / pit / basin / structure.	USV	Laboratory building. Building.
URG	Cooling tower connecting structure. Foundation / channel.	USX	Special structure.
URH	Cooling tower outlet structure. Foundation / pit / channel.	USY	Bridge structure.
URJ	Cooling tower outfall culvert. Foundation / channel.	USZ	Ducting structure.
URK	Cooling tower return structure. Foundation / pit / channel.	UT	Structures for auxiliary systems.
URL	Cooling tower return culvert. Foundation / pit / channel.	UTA	Common / central structure for auxiliary systems (process-related). Foundation / shelter / building.
URM	Circulating (main cooling) water distribution structure. Foundation / pit / channel.	UTB	Structure for central water chilling plant. Foundation / shelter / building.
URN	Cooling tower bypass structure. Foundation / pit / channel.	UTF	Structure for central compressed and control air supply (*QE*, *QF*). Foundation / shelter / building.
URP	Cooling tower blowdown structure. Foundation / pit / channel.	UTG	Structure for central gas supply (not fuel gas). Foundation / shelter / building. e.g. hydrogen / nitrogen bottle / tank stored.
URQ	Cooling tower blowdown culvert. Foundation / pit / channel.	UTH	Auxiliary steam generator building. Foundation / shelter / building.
URR	Structure for closed cooling water system. e.g. foundation / support for air blast inter-coolers.	UTJ	Structure for auxiliary steam generator stack (not *UHN*). Foundations.
URX	Special structure (plant specific).	UTM	Structures for hydrogen production plant. Foundation / shelter / building.
URY	Bridge structures.		
URZ	Ducting structure.		

STRUCTURES AND BUILDINGS FOR CCPP

UTS	Structure for chemicals storage. Foundation / pit / shelter / building.	UXH	Grounding (PI), (ROP). Foundation / shelter / building.
UTT	Containment in the chemicals unloading area. Foundation / pit / shelter / building.	UXP	Structure for seawater supply. Seawater intake head channel / pipe, structure and embankment, supply channel from mechanical cleaning.
UTX	Special structure.	UXQ	Structure for seawater cleaning and forwarding. Mechanical cleaning and chemical dosing plant. structures for chlorination plant, pump station.
UTY	Bridge structure.	UXX	Special structure for bridge and duct (ROP). eg. Structure for bridge and duct (PI), (ROP). Foundation / shelter / building.
UTZ	Ducting structure.	UXY	Bridge structure.
UU	Shaft structures.	UXZ	Ducting structure.
UUG	Deep well. Shaft / pit.	UY	General service structures.
UUX	Special structure.	UYA	Office and staff amenities building. Building.
UUY	Bridge structure.	UYB	Staff amenities building. Building.
UUZ	Ducting structure.	UYC	Administration building. Building.
UV	Structures for chemical flue gas treatment incl. residues removal (for *HR*. *HS*. *HT*).	UYD	Canteen. Building.
UVE	Structure for ammonia unloading and storage. Foundation / shelter / building.	UYE	Gate house. Building.
UVM	Structure for catalyst element storage and handling. Shelter / building.	UYF	Security gate house, guard tower. Building.
UVX	Special structure (plant specific).	UYG	Information center. Building.
UVY	Bridge structure.	UYH	Training facilities. Building.
UVZ	Ducting structure.	UYJ	Medical center. Building.
UX	Structures for external systems (power plant-specific).	UYN	Railway engine shed. Foundation / shelter / building.
UXA	Structure for seawater desalination plant (in general). Foundation / shelter / building.	UYP	Fire station. Foundation / shelter / building.
UXB	Structures for brine storage and conditioning system. incl. blowdown pit and pumping station.	UYQ	Garage. Foundation / shelter / building.
UXC	Structure for seawater desalination plant auxiliaries. Foundation / shelter / building.	UYR	Automobile workshop. Vehicle repair shop; Foundation / shelter / building.
UXD	Structure for storage and forwarding of desalinated sea water. Foundation / pit / shelter / building.	UYS	Petrol station. Foundation / shelter / building.
UXE	Structure for desalinated seawater storage tanks. Foundation / pit.	UYX	Special structure (plant specific).
UXF	Structure for desalinated seawater pump station. Foundation / shelter / building.	UYZ	Bridge structures.
UXG	Cathodic protection (PI). Foundation / shelter / building.		Ducting structures.

STRUCTURES AND BUILDINGS FOR CCPP

UZ Structures for transport, traffic, fencing, gardens and other purposes.

UZA Structure for roads, parking and fences (infrastructure).

Road construction, parking, fences, hard standing area.

UZC Open areas and yards (general).

e.g. helicopter landing area, deposit area, storage area, scrap storage area.

UZD Parking areas, incl. associated structures.

Road construction / shelter.

UZE Railway structure incl. access railway and internal rail system.

Track construction.

UZF Lifting gear structure.

Track construction / foundation.

UZJ Fencing and gates.

Foundation.

UZK Structure for landscaping, gardening and outdoor work.

Green areas / landscaping / fountains etc.

UZL Noise abatement structures.

UZM Protective structures against external impact.

e.g. security walls etc.

UZN Structure for shore protection and jetty.

UZP Structure for bank / shoreline stabilization.

e.g. fender wall etc.

UZQ Structure for river regulation.

Hydraulic engineering.

UZR Jetty, quay.

Hydraulic engineering.

UZS Breakwater.

Hydraulic engineering.

UZT Open land, building lots, site owner rights

Building site preparation, power plant site, recreation area, meteorological station etc.

UZU Site security structure.

Structure for site stabilization / slope protection.

UZV Site preparation (PI/ROP).

Site surveying, soil investigation, site dewatering structures, soil improvement, piling.

UZW Structure for independent buildings and residential area.

Building.

UZX Special structures (plant specific).

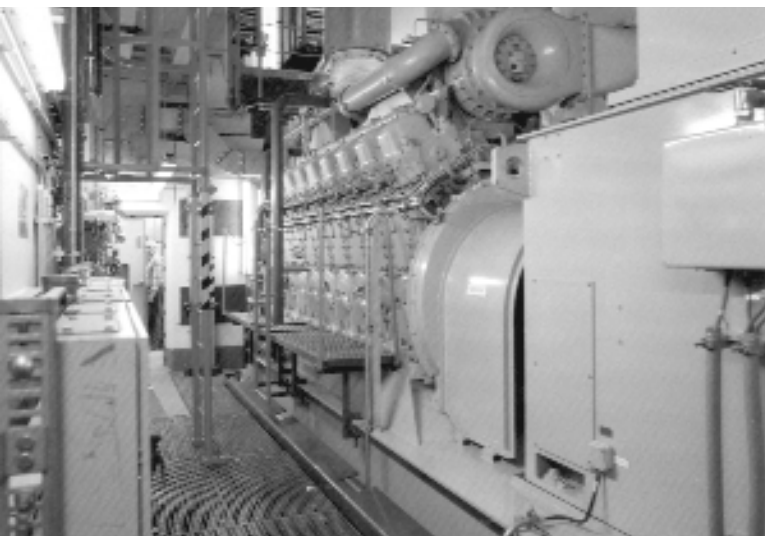
UZY Bridge structures.

UZZ Ducting structures.



HEAVY MACHINERY (NOT MAIN MACHINE SETS)

- XA Steam turbine plant.**
- XB Gas turbine plant.**
- XE Hydraulic turbine plant.**
- XJ Diesel engine plant.**
- XK Generator plant.**
- XL Electro-motive plant (incl. motor generator).**
- XP Common installations for heavy machinery.**
- XR Gas engine plant.**
- XV Lubricant supply system.**
- XW Sealing fluid supply system.**
- XX Fluid supply system for control and protection equipment.**
- XY Control and protection equipment.**



XA Steam turbine plant.

XAA HP turbine.

from incl. steam admission (main stop valve) or combined main stop and control valve
to incl. automatic/non-automatic extraction and exhaust nozzles and
to incl. the interfaces with other turbine internal systems.

XAB IP turbine.

from incl. crossover line, incl. control element or
from incl. intercept valve
to incl. automatic/non-automatic extraction and exhaust nozzles and
to incl. the interfaces with other turbine internal systems.

XAC LP turbine.

from incl. crossover line, incl. control element or
from incl. intercept valve or steam inlet nozzle (in reheat system without intercept valves)
to incl. automatic/non-automatic extraction and exhaust nozzles and
to incl. the interfaces with other turbine internal systems.

XAD Bearings.

XAG Condensing system.

from incl. condenser neck or inlet nozzle
to incl. condenser outlet nozzle, incl. connected flash tanks, incl. instrumentation equipment associated with condenser.

XAH Motive water system (if separate from *XAJ*).

from excl. outlet of other system
to excl. water-operated air ejector inlet.

XAJ Air removal system.

from excl. condenser outlet
to excl. atmosphere.

XAK Transmission gear between prime mover and driven machine, incl. turning gear.

XAL Drain and vent systems.

from incl. collector point or
from incl. final drain
to excl. discharge into other system.

XAM Leak-off steam system.

from excl. branch from seal leak-off
to excl. discharge into other system.

XAN Turbine bypass station, incl. desuperheating spray system.

from incl. bypass valve and
from incl. desuperheating spray valve
to incl. steam inlet to condenser.

XAV Lubricant supply system.

from incl. dedicated lubricant tank or common lubricant and control fluid tank or
from excl. branch off lubricant fluid supply system
to excl. user and
from excl. user.

XAW Sealing, heating and cooling steam system.

from excl. branch
to excl. casing nozzle of steam user and leak-off
to excl. condenser or
to incl. gland steam condenser or
to excl. heating/cooling steam user.

XAX Non-electric control and protection equipment, incl. fluid supply system.

XAY Electrical control and protection equipment.

XB Gas turbine plant.

XBA Turbine, compressor rotor with common casing.

from incl. compressor inlet
to incl. compressor outlet
from incl. turbine inlet
to incl. turbine outlet, incl. exhaust gas diffuser.

XBB Turbine casing and rotor.

from incl. turbine inlet
to incl. turbine outlet, incl. exhaust gas diffuser.

XBC Compressor casing and rotor.

from incl. compressor inlet
to incl. compressor outlet.

XBD Bearings.

XBE Cooling medium system.

XBH Cooling and sealing gas system.

from incl. extraction point
to excl. user and
from excl. user, incl. leak-off
to incl. inlet to other system.

XBJ Start-up unit.

XBK Transmission gear between prime mover and driven machine, incl. turning gear, barring gear.

XBL Intake air, cold gas system (open cycle).

from excl. atmosphere
to excl. combustion chamber or
to excl. compressor inlet or
to incl. exhaust gas heat exchanger, excl. compressor.

XBM Combustion chamber (gas heating, combustion).

from incl. cold gas, fuel inlet
to incl. hot gas outlet.

XBN Fuel supply system (liquid).

from excl. branch off main supply line or
from incl. temporary (day) tank
to excl. combustion chamber or
to excl. motive gas generating unit, i
ncl. fuel return system.

XBP Fuel supply system (gaseous).

from excl. branch off main supply line
to excl. combustion chamber or
to excl. motive gas generating unit.

XBQ Ignition fuel supply system (if separate).

from excl. branch off main supply line or
from incl. storage tank
to excl. combustion chamber or
to excl. motive gas generating unit.

XBR Exhaust gas system (open cycle).

from excl. combustion chamber or
from excl. exhaust gas diffuser
to excl. discharge into atmosphere, excl.
turbine, or
to excl. inlet to other system
(e.g. combustion air system).

XBS Storage system.

to excl. connection to main system and
from excl. connection to main system.

XBT Motive gas generator unit, incl. combustion chamber.

from incl. air/fuel inlet
to incl. motive gas outlet of motive gas
generating unit.

XBU Additive system.

from incl. supply
to incl. injection.

XBV Lubricant supply system.

from incl. dedicated lubricant tank or
common lubricant and control fluid tank or
from excl. branch off control fluid supply
system
to excl. user and from excl. user.

XBW Seal oil supply.

from incl. dedicated sealing oil tank or
from excl. sealing oil pump suction line
to excl. user and
from excl. user.

XBX Non-electric control and protection equipment, incl. fluid supply system.

XBY Electrical control and protection equipment.

XBZ Lubricant and control fluid treatment system.

XE Hydraulic turbine plant.

XE A Turbine (casing, shaft, runner etc.).

from excl. turbine inlet or
from excl. isolating valve
to incl. turbine outlet or
to excl. isolating valve, incl. draft tube.

XEB Isolating valve.

from incl. isolating valve inlet
to incl. isolating valve outlet.

XED Bearings.

XEG Stabilizing air system.

from incl. air compressor
to incl. outlet to turbine.

XEK Transmission gear between prime mover and driven machine.

XEL Water depression air supply system.

from incl. air compressor
to incl. outlet to turbine.

XES Shaft gland cooling water system.

XEV Lubricant supply system.

from incl. lubricant tank
to excl. user and
from excl. user.

XEW Sealing water supply system.

from incl. sealing water supply main
isolating valve
to excl. casing nozzle of sealing water user.

XEX Non-electric control and protection equipment, incl. fluid supply.

XEY Electrical control and protection equipment.

XJ Diesel engine plant.

XJA Engine.

from incl. fuel injection nozzle inlet or
from incl. air intake nozzles or
from incl. cooling water inlet nozzles
to incl. exhaust nozzle outlet or
to incl. cooling water nozzle outlet, incl.
engine- internal systems.

XJB Turbocharger, blower.

from incl. turbocharger or blower inlet
to incl. turbocharger or blower outlet.

- XJG Liquid cooling system.**
from excl. engine cooling water nozzle outlet or
from incl. turbocharger air cooling system inlet
to excl. engine cooling water nozzle inlet or
to incl. turbocharger air cooler outlet.
- XJH Air intercooling system.**
from incl. intercooler inlet to incl. intercooler outlet
to excl. outlet to other cooling systems.
- XJK Transmission gear between prime mover and driven machine.**
- XJN Fuel systems.**
from incl. temporary (day) tank or
from excl. branch off piping system
to excl. fuel injection nozzle inlet.
- XJP Start-up unit, (incl. flywheel).**
- XJQ Air intake system.**
from excl. atmosphere
to excl. turbocharger or
to excl. engine air intake nozzles.
- XJR Exhaust gas system.**
from excl. engine exhaust nozzle outlet
to excl. discharge to atmosphere.
- XJV Lubricant supply system.**
from incl. dedicated lubricant tank or
common lubricant and control fluid tank or
from excl. branch off control fluid supply system
to excl. user
from excl. user.
- XJW Sealing fluid supply system.**
- XJX Fluid supply system for control and protection equipment.**
- XJY Control and protection equipment.**
- XK Generator plant.**
- XKA Black Start Generator Sets.**
to incl. generator bushing.
- XKB Exciter set (use only if *XKC* is not sufficient for identification).**
- XKC Exciter set.**
- XKD Bearings.**
- XKF Stator/rotor liquid cooling system, incl. coolant supply system (Note: for cooling oil see *XKU*).**
Task: dissipate heat produced by
stator/rotor to coolant
from excl. stator/rotor outlet
to excl. stator/rotor inlet.
- XKG Stator/rotor gas cooling system, incl. coolant supply system (Note: for nitrogen cooling see *XKH*).**
Task: dissipate heat produced by
stator/rotor to coolant
from excl. stator/rotor outlet
to excl. stator/rotor inlet.
- XKH Stator/rotor nitrogen cooling system, incl. coolant supply system (Note: for other gas cooling see *XKG*).**
Task: dissipate heat produced by
stator/rotor to coolant
from excl. stator/rotor outlet
to excl. stator/rotor inlet.
- XKP Secondary cooling water system.**
from excl. cooling water system (*PC*, *PG*)
to excl. users incl. blowdown, discharge, venting
from excl. users
to excl. inlet into other system.
- XKQ Exhaust gas system (if separate from *XKG* and *XKH*).**
- XKU Stator/rotor cooling oil cooling system, incl. coolant supply system (Note: for other liquid cooling see *XKF*).**
Task: dissipate heat produced by
stator/rotor to coolant
from excl. stator/rotor outlet
to excl. stator/rotor inlet.
- XKV Lubricant supply system (if separate system for generator).**
- XKW Sealing fluid supply system (Sealing oil system, incl. supply and treatment).**
from excl. branch off sealing oil supply system
to excl. stator inlet and from excl. stator outlet
to excl. inlet to other system or in closed systems
from excl. stator outlet
to excl. stator inlet.
- XKX Fluid supply system for control and protection equipment.**
- XKY Control and protection equipment.**
- XL Electro-motive plant (incl. motor generator).**
- XLA Motor frame, motor generator frame, incl. stator, rotor and all integral cooling equipment.**
to incl. motor or generator bushing

- XLC Exciter set.**
XLD Bearings.
XLF Stator/rotor liquid cooling system, incl. coolant supply system (Note: for cooling oil see *XLU*).
 Task:
 dissipate heat produced by stator/rotor to coolant
 from excl. stator/rotor outlet
 to excl. stator/rotor inlet.
- XLG Stator/rotor gas cooling system, incl. coolant supply system (Note: for nitrogen cooling see *XLH*).**
 Task:
 dissipate heat produced by stator/rotor to coolant
 from excl. stator/rotor outlet
 to excl. stator/rotor inlet.
- XLH Stator/rotor nitrogen cooling system, incl. coolant supply system (Note: for other gas cooling see *XLG*).**
 Task:
 dissipate heat produced by stator/rotor to coolant
 from excl. stator/rotor outlet
 to excl. stator/rotor inlet.
- XLQ Exhaust gas system (if separate from *XLG* and *XLH*).**
- XLU Stator/rotor cooling oil cooling system, incl. coolant supply system (Note: for other liquid cooling see *XLF*).**
 Task:
 dissipate heat produced by stator/rotor to coolant
 from excl. stator/rotor outlet
 to excl. stator/rotor inlet.
- XLV Lubricant supply system (if separate system for electro-motive units).**
- XLW Sealing fluid supply system (Sealing oil system, incl. supply and treatment).**
 from excl. branch off sealing oil supply system
 to excl. stator inlet and
 from excl. stator outlet
 to excl. inlet to other system or in closed systems
 from excl. stator outlet
 to excl. stator inlet.
- XLX Fluid supply system for control and protection equipment.**
XLX Control and protection equipment.

XP Common installations for heavy machinery.

- XPA Foundations.**
XPB Sheating.
XPG Frame, support structure.
XPR Forced cooling system.
XPS Drying and layup system.

XR Gas engine plant.

- XRA Gas engine plant.**
 (free for use).
- XRБ Gas engine plant.**
 (free for use).
- XRC Gas engine plant.**
 (free for use).
- XRD Gas engine plant.**
 (free for use).
- XRE Gas engine plant.**
 (free for use).
- XRF Gas engine plant.**
 (free for use).
- XRG Gas engine plant.**
 (free for use).
- XRH Gas engine plant.**
 (free for use).
- XRJ Gas engine plant.**
 (free for use).
- XRK Gas engine plant.**
 (free for use).
- XRL Gas engine plant.**
 (free for use).
- XRM Gas engine plant.**
 (free for use).
- XRN Gas engine plant.**
 (free for use).
- XRP Gas engine plant.**
 (free for use).
- XRQ Gas engine plant.**
 (free for use).
- XRR Gas engine plant.**
 (free for use).
- XRS Gas engine plant.**
 (free for use).
- XRT Gas engine plant.**
 (free for use).
- XRU Gas engine plant.**
 (free for use).
- XRV Lubricant supply system.**
- XRW Sealing fluid supply system.**
- XRX Fluid supply system for control and protection equipment.**
- XRY Control and protection equipment.**

XV Lubricant supply system.

- XVA** Lubricant supply system.
(free for use).
- XVB** Lubricant supply system.
(free for use).
- XVC** Lubricant supply system.
(free for use).
- XVD** Lubricant supply system.
(free for use).
- XVE** Lubricant supply system.
(free for use).
- XVF** Lubricant supply system.
(free for use).
- XVG** Lubricant supply system.
(free for use).
- XVH** Lubricant supply system.
(free for use).
- XVJ** Lubricant supply system.
(free for use).
- XVK** Lubricant supply system.
(free for use).
- XVL** Lubricant supply system.
(free for use).
- XVM** Lubricant supply system.
(free for use).
- XVN** Lubricant supply system.
(free for use).
- XVP** Lubricant supply system.
(free for use).
- XVQ** Lubricant supply system.
(free for use).
- XVR** Lubricant supply system.
(free for use).
- XVS** Lubricant supply system.
(free for use).
- XVT** Lubricant supply system.
(free for use).
- XVU** Lubricant supply system.
(free for use).

XW Sealing fluid supply system.

- XWA** Sealing fluid supply system.
(free for use).
- XWB** Sealing fluid supply system.
(free for use).
- XWC** Sealing fluid supply system.
(free for use).
- XWD** Sealing fluid supply system.
(free for use).
- XWE** Sealing fluid supply system.
(free for use).
- XWF** Sealing fluid supply system.
(free for use).

- XWG** Sealing fluid supply system.
(free for use).
- XWH** Sealing fluid supply system.
(free for use).
- XWJ** Sealing fluid supply system.
(free for use).
- XWK** Sealing fluid supply system.
(free for use).
- XWL** Sealing fluid supply system.
(free for use).
- XWM** Sealing fluid supply system.
(free for use).
- XWN** Sealing fluid supply system.
(free for use).
- XWP** Sealing fluid supply system.
(free for use).
- XWQ** Sealing fluid supply system.
(free for use).
- XWR** Sealing fluid supply system.
(free for use).
- XWS** Sealing fluid supply system.
(free for use).
- XWT** Sealing fluid supply system.
(free for use).
- XWU** Sealing fluid supply system.
(free for use).

XX Fluid supply system for control and protection equipment.

- XXA** Fluid supply system for control and protection equipment.
(free for use).
- XXB** Fluid supply system for control and protection equipment.
(free for use).
- XXC** Fluid supply system for control and protection equipment.
(free for use).
- XXD** Fluid supply system for control and protection equipment.
(free for use).
- XXE** Fluid supply system for control and protection equipment.
(free for use).
- XXF** Fluid supply system for control and protection equipment.
(free for use).
- XXG** Fluid supply system for control and protection equipment.
(free for use).
- XXH** Fluid supply system for control and protection equipment.
(free for use).

XXJ Fluid supply system for control and protection equipment.

(free for use).

XXK Fluid supply system for control and protection equipment.

(free for use).

XXL Fluid supply system for control and protection equipment.

(free for use).

XXM Fluid supply system for control and protection equipment.

(free for use).

XXN Fluid supply system for control and protection equipment.

(free for use).

XXP Fluid supply system for control and protection equipment.

(free for use).

XXQ Fluid supply system for control and protection equipment.

(free for use).

XXR Fluid supply system for control and protection equipment.

(free for use).

XXS Fluid supply system for control and protection equipment.

(free for use).

XXT Fluid supply system for control and protection equipment.

(free for use).

XXU Fluid supply system for control and protection equipment.

(free for use).

XY Control and protection equipment.

XYA Control and protection equipment.
(free for use).

XYB Control and protection equipment.
(free for use).

XYC Control and protection equipment.
(free for use).

XYD Control and protection equipment.
(free for use).

XYE Control and protection equipment.
(free for use).

XYF Control and protection equipment.
(free for use).

XYG Control and protection equipment.
(free for use).

XYH Control and protection equipment.
(free for use).

XYJ Control and protection equipment.
(free for use).

XYK Control and protection equipment.
(free for use).

XYL Control and protection equipment.
(free for use).

XYM Control and protection equipment.
(free for use).

XYN Control and protection equipment.
(free for use).

XYP Control and protection equipment.
(free for use).

XYQ Control and protection equipment.
(free for use).

XYR Control and protection equipment.
(free for use).

XYS Control and protection equipment.
(free for use).

XYT Control and protection equipment.
(free for use).

XYU Control and protection equipment.
(free for use).



WORKSHOP AND OFFICE EQUIPMENT

ZA Workshop and office equipment.

ZB Workshop and office equipment.

ZC Workshop and office equipment.

ZM Workshop and office equipment.

ZZ Workshop and office equipment.

ZA Workshop and office equipment.
(free for use).

ZB Workshop and office equipment.
(free for use).

ZC Workshop and office equipment.
(free for use).

ZM Workshop and office equipment.
(free for use).

ZMB Mobile cleaning equipment.

ZZ Workshop and office equipment.
(free for use).



4. KKS EQUIPMENT UNIT KEY (A₁ A₂)

A Mechanical equipment

AA Valves, dampers, etc., incl. actuators, also manual; rupturedisk equipment
 AB Isolating elements, air locks
 AC Heat exchangers, heat transfer surfaces
 AD -blocked-
 AE Turning, driving, lifting and slewing gear (also manipulators)
 AF Continuous conveyors, feeders (escalators)
 AG Generator units
 AH Heating, cooling and air conditioning units
 AJ Size reduction equipment, only as part of process
 AK Compacting and packaging equipment, only as part of process
 AL -blocked-
 AM Mixers, agitators
 AN Compressor units, fans
 AP Pump units
 AQ -blocked-
 AR -blocked-
 AS Adjusting and tensioning equipment for non-electrical variables (to be applied only, if the actuator forms itself a constructive unit with another equipment unit)
 AT Cleaning, drying, filtering and separating equipment, excl. *BT*
 AU Braking, gearbox, coupling equipment, non-electrical converters
 AV Combustion equipment
 AW Stationary tooling, treatment equipment

AX Test and monitoring equipment for plant maintenance
 AY -blocked-
 AZ -blocked-

B Mechanical equipment

BA -blocked-
 BB Storage equipment (tanks)
 BC -blocked-
 BD -blocked-
 BE Shafts (for erection and maintenance only)
 BF Foundations
 BG -blocked-
 BH -blocked-
 BJ -blocked-
 BK -blocked-
 BL -blocked-
 BM -blocked-
 BN Jet pumps, ejectors, injectors
 BP Flow restrictors, limiters, orifices (not metering orifices)
 BQ Hangers, supports, racks, piping penetrations
 BR Piping, ductwork, chutes, compensators
 BS Silencers
 BT Flue gas catalytic converter modules
 BU Insulation, sheathing
 BV -blocked-
 BW -blocked-
 BX -blocked-
 BY -blocked-
 BZ -blocked-

**C Direct measuring circuits
(Data character A2
following DIN 19227,
part 1, September 1973
edition, table 1, initial
letter)**

CA -blocked-
CB -blocked-
CC -blocked-
CD Density
CE Electrical variables (e.g.
current, voltage, power,
electr. frequency)
CF Flow, rate
CG Distance, length, position,
direction of rotation
CH Manual input as manually
operated sensor (e.g. fire
detector)
CJ -blocked-
CK Time
CL Level (also for dividing line)
CM Moisture, humidity
CN -blocked-
CP Pressure
CQ Quality variables (analysis,
material properties), other than
CD, *CM*, *CV*
CR Radiation variables
CS Velocity, speed, frequency
(mechanical), acceleration
CT Temperature
CU Combined and other variables
CV Viscosity
CW Weight, mass
CX Neutron flux
CY Vibration, expansion
CZ -blocked-

**D Closed loop control
circuits (Data character
A2 following DIN 19227,
part 1, September 1973
edition, table 1, initial
letter)**

DA -blocked-
DB -blocked-
DC -blocked-
DD Density
DE Electrical variables (e.g.
current, voltage, power,
electr. frequency)
DF Flow, rate
DG Distance, length, position,
direction of rotation
DH -blocked-
DJ -blocked-
DK Time
DL Level (also for dividing line)
DM Moisture, humidity
DN -blocked-
DP Pressure
DQ Quality variables (analysis,
material properties), other than
DD, *DM*, *DV*
DR Radiation variables
DS Velocity, speed, frequency
(mechanical), acceleration
DT Temperature
DU Combined and other variables
DV Viscosity
DW Weight, mass
DX Neutron flux
DY Vibration, expansion
DZ -blocked-

**E Analog and binary signal
conditioning**

EA Open loop control
(free for use)
EB Open loop control
(free for use)
EC Open loop control
(free for use)
ED Open loop control
(free for use)
EE Open loop control
(free for use)
EF -blocked-
EG Alarm, annunciation
(free for use)
EH Alarm, annunciation
(free for use)
EJ Alarm, annunciation
(free for use)
EK Alarm, annunciation
(free for use)
EL -blocked-
EM Process computer (free for use)
EN Process computer (free for use)
EP Process computer (free for use)
EQ Process computer (free for use)
ER Reactor protection
ES -blocked-
ET -blocked-
EU Combined analog and binary
signal conditioning
EV -blocked-
EW Protection (free for use)
EX Protection (free for use)
EY Protection (free for use)
EZ Protection (free for use)

**F Indirect measuring circuits
(Data character A2
following DIN 19227,
part 1, September 1973
edition, table 1, initial
letter)**

FA	-blocked-
FB	-blocked-
FC	-blocked-
FD	Density
FE	Electrical variables (e.g. electr. efficiency, power)
FF	Flow, rate
FG	Distance, length, position, direction of rotation
FH	-blocked-
FJ	-blocked-
FK	Time
FL	Level (also for dividing line)
FM	Moisture, humidity
FN	-blocked-
FP	Pressure
FQ	Quality variables (analysis, material properties), other than *FD*, *FM*, *FV*
FR	Radiation variables
FS	Velocity, speed, frequency (mechanical), acceleration
FT	Temperature
FU	Combined and other variables
FV	Viscosity
FW	Weight, mass
FX	Neutron flux
FY	Vibration, expansion
FZ	-blocked-

G Electrical equipment

GA	Junction boxes and cable/bus bar penetrations (free for use)
GB	Junction boxes and cable/bus bar penetrations (free for use)
GC	Junction boxes and cable/bus bar penetrations (free for use)
GD	Junction boxes and cable/bus bar penetrations (free for use)
GE	Junction boxes and cable/bus bar penetrations (free for use)
GF	Junction boxes and cable/bus bar penetrations (free for use)
GG	Junction boxes and cable/bus bar penetrations (free for use)
GH	Electrical and instrumentation and control installation units identified as per process system (e.g. cubicles, boxes)
GJ	-blocked-
GK	Information display and operator control equipment for process computers and automation systems
GL	-blocked-
GM	Junction boxes for light-current systems of national telecommunication services
GN	-blocked-
GP	Subdistribution/junction boxes for lighting
GQ	Subdistribution/junction boxes for power sockets
GR	DC generating equipment, batteries
GS	Switchgear equipment if not identified under process equipment
GT	Transformer equipment
GU	Converter equipment
GV	Structure-related earthing and lightning protection equipment, surge arrestors
GW	Cabinet power supply equipment
GX	Actuating equipment for electrical variables
GY	Junction boxes for light-current systems (not of national telecommunication services)
GZ	Hangers, supports and racks for electrical and instrumentation and control equipment

**H Subassemblies of main and heavy machinery
(only to be used in conjunction with *M* = Main machine sets and *X* = Heavy machinery)**

HA	Machine stationary assembly
HB	Machine rotating assembly
HC	-blocked-
HD	Bearing assembly
HE	-blocked-
HF	-blocked-
HG	-blocked-
HH	-blocked-
HJ	-blocked-
HK	-blocked-
HL	-blocked-
HM	-blocked-
HN	-blocked-
HP	-blocked-
HQ	-blocked-
HR	-blocked-
HS	-blocked-
HT	-blocked-
HU	-blocked-
HV	-blocked-
HW	-blocked-
HX	-blocked-
HY	-blocked-
HZ	-blocked-

5. KKS COMPONENT KEY (B₁ B₂)

Electrical components (To DIN 40719, part 2, September 1978 edition)

-A Assemblies and sub-assemblies	-V Tubes, semiconductors
-B Transducers for non-electrical to electrical variables and vice-versa	-W Transmission paths, waveguides, aerials
-C Capacitors	-X Terminals, plugs, sockets
-D Binary elements, delay devices, memory devices	-Y Electrical positioners, e.g. solenoids (not motors)
-E Special components	-Z Terminations, balancing equipment, filters, limiters, cable terminations
-F Protective devices	
-G Generators, power supplies	
-H Signalling devices	
-K Relays, contactors	
-L Inductors	
-M Motors	
-N Amplifiers, controllers	
-P Measuring instruments, testing equipment	
-Q Power switchgear	
-R Resistors	
-S Switches, selectors	
-T Transformers	
-U Modulators, convertors from electrical to other electrical variables	

A -blocked-

B -blocked-

C -blocked-

D -blocked-

E -blocked-

F -blocked-

G -blocked-

H -blocked-

J -blocked-

K Mechanical components

KA Gate valves, globe valves, dampers, cocks, rupture disks, orifices

KB Gates, doors, dam boards

KC Heat exchangers, coolers

KD Vessels/tanks, pools, surge tanks (fluid systems)

KE Turning, driving, lifting and slewing gear

KF Continuous conveyors, feeders

KG -blocked-

KH -blocked-

KJ Size reduction machines

KK Compacting, packaging machines

KL -blocked-

KM Mixers, agitators

KN Compressors, blowers, fans

KP Pumps

KQ -blocked-

KR -blocked-

KS -blocked-

KT Cleaning machines, dryers, separators, filters

KU -blocked-

KV Burners, grates

KW Stationary tooling and treatment machines for maintenance

KX -blocked-

KY -blocked-

KZ -blocked-

L -blocked-

M Mechanical components

MA -blocked-

MB Brakes

MC -blocked-

MD -blocked-

ME -blocked-

MF Foundations

MG Gearboxes

MH -blocked-

MJ -blocked-

MK Clutches, couplings

ML -blocked-

MM Engines, not electrical

MN -blocked-

MP -blocked-

MQ -blocked-

MR Piping components, ductwork components

MS Positioners, not electrical

MT Turbines

MU Transmission gear, non electrical, converters and boosters other than couplings and gearboxes

MV -blocked-

MW -blocked-

MX -blocked-

MY -blocked-

MZ -blocked-

N -blocked-

P -blocked-

Q Instrumentation and control components (non-electrical)

QA -blocked-

QB Sensors if not structurally integral with *QP*, metering orifices

QC -blocked-

QD -blocked-

QE -blocked-

QF -blocked-

QG -blocked-

QH Signalling devices

QJ -blocked-

QK -blocked-

QL -blocked-

QM -blocked-

QN Controllers, flybolt governor

QP Measuring instruments, testing equipment

QQ -blocked-

QR Instrument piping

QS Condensation chambers (datum reservoir) in measuring circuits

QT Thermowells and pockets for protection of sensors

QU -blocked-

QV -blocked-

QW -blocked-

QX -blocked-

QY -blocked-

QZ -blocked-

R -blocked-

S -blocked-

T -blocked-

U -blocked-

V -blocked-

W -blocked-

X Signal origins
(Subgroups are established as appropriate to the type of instrumentation and control system and are therefore subject to agreement between the parties to the project.)

XA Signal origins (free for use)
XB Signal origins (free for use)
XC Signal origins (free for use)
XD Signal origins (free for use)
XE Signal origins (free for use)
XF Signal origins (free for use)
XG Signal origins (free for use)
XH Signal origins (free for use)
XI Signal origins (free for use)
XK Signal origins (free for use)
XL Signal origins (free for use)
XM Signal origins (free for use)
XN Signal origins (free for use)
XP Signal origins (free for use)
XQ Signal origins (free for use)
XR Signal origins (free for use)
XS Signal origins (free for use)
XT Signal origins (free for use)
XU Signal origins (free for use)
XV Signal origins (free for use)
XW Signal origins (free for use)
XX Signal origins (free for use)
XY Signal origins (free for use)
XZ Signal origins (free for use)

Y Signal applications
(Subgroups are established as appropriate to the type of instrumentation and control system and are therefore subject to agreement between the parties to the project.)

YA Signal applications (free for use)
YB Signal applications (free for use)
YC Signal applications (free for use)
YD Signal applications (free for use)
YE Signal applications (free for use)
YF Signal applications (free for use)
YG Signal applications (free for use)
YH Signal applications (free for use)
YJ Signal applications (free for use)
YK Signal applications (free for use)
YL Signal applications (free for use)
YM Signal applications (free for use)
YN Signal applications (free for use)
YP Signal applications (free for use)
YQ Signal applications (free for use)
YR Signal applications (free for use)
YS Signal applications (free for use)
YT Signal applications (free for use)
YU Signal applications (free for use)
YV Signal applications (free for use)
YW Signal applications (free for use)
YX Signal applications (free for use)
YY Signal applications (free for use)
YZ Signal applications (free for use)

Z Gated signals
(Subgroups are established as appropriate to the type of instrumentation and control system and are therefore subject to agreement between the parties to the project.)

ZA Gated signals (free for use)
ZB Gated signals (free for use)
ZC Gated signals (free for use)
ZD Gated signals (free for use)
ZE Gated signals (free for use)
ZF Gated signals (free for use)
ZG Gated signals (free for use)
ZH Gated signals (free for use)
ZJ Gated signals (free for use)
ZK Gated signals (free for use)
ZL Gated signals (free for use)
ZM Gated signals (free for use)
ZN Gated signals (free for use)
ZP Gated signals (free for use)
ZQ Gated signals (free for use)
ZR Gated signals (free for use)
ZS Gated signals (free for use)
ZT Gated signals (free for use)
ZU Gated signals (free for use)
ZV Gated signals (free for use)
ZW Gated signals (free for use)
ZX Gated signals (free for use)
ZY Gated signals (free for use)
ZZ Gated signals (free for use)